

Appendix D

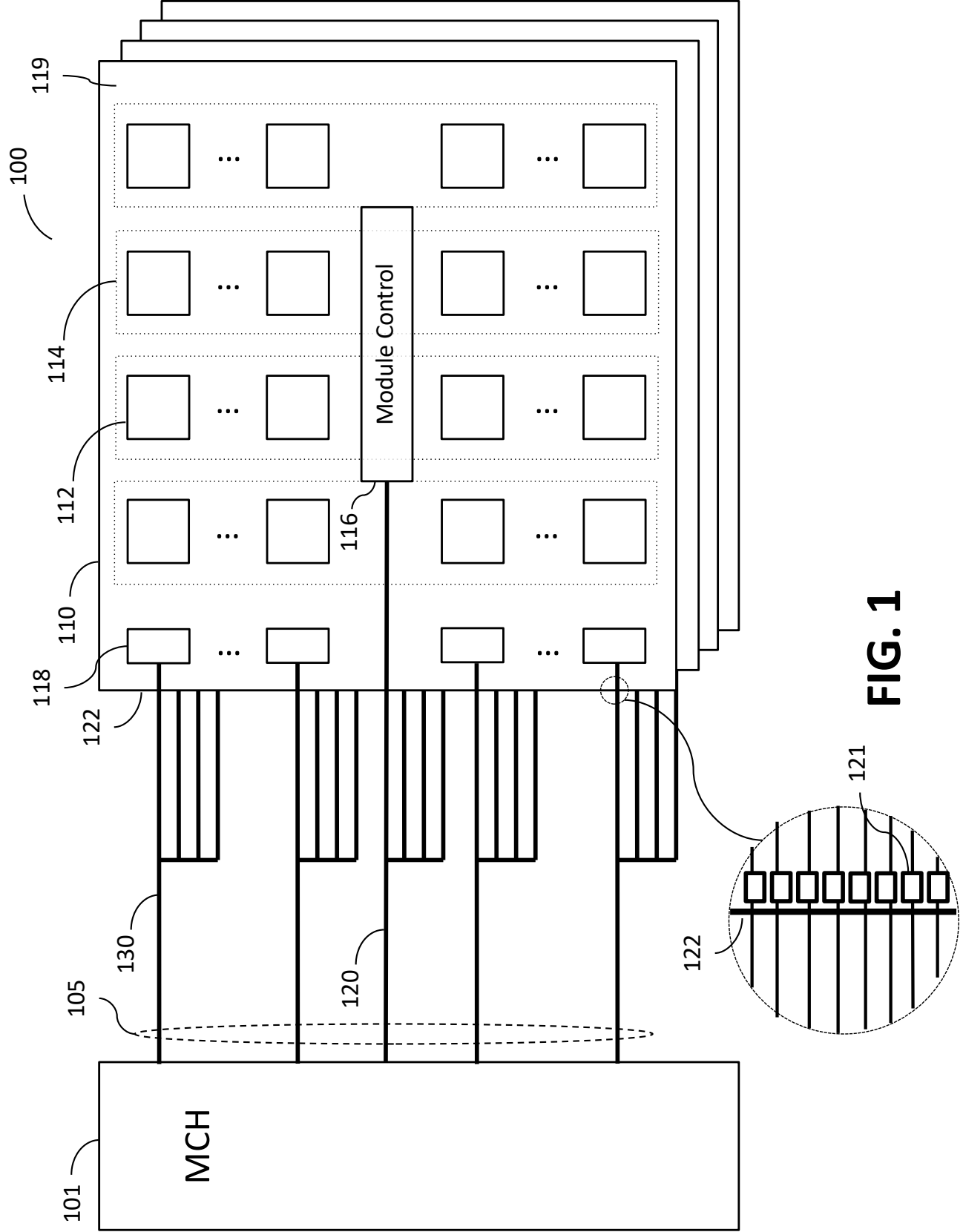


FIG. 1

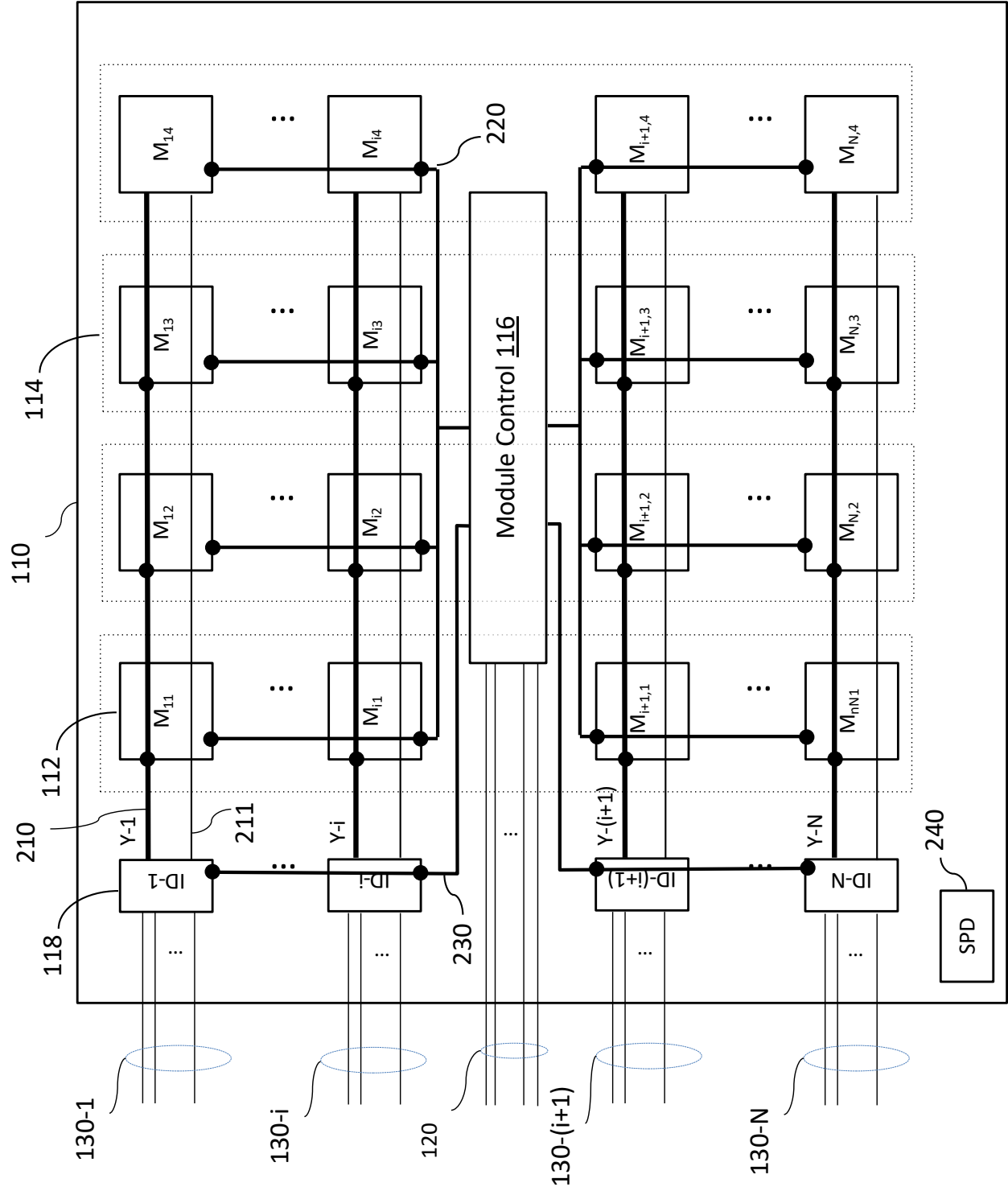


FIG. 2A

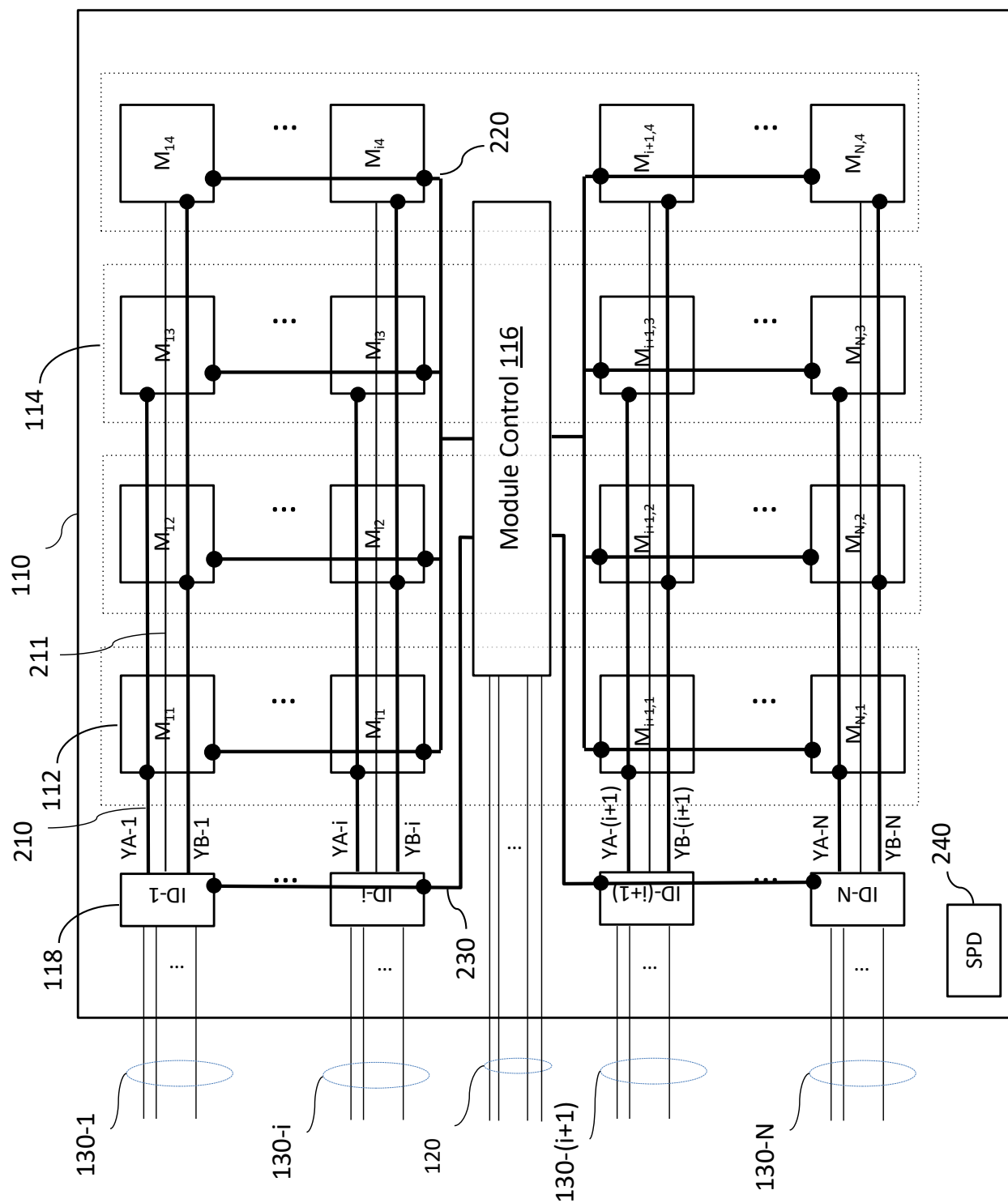


FIG. 2B

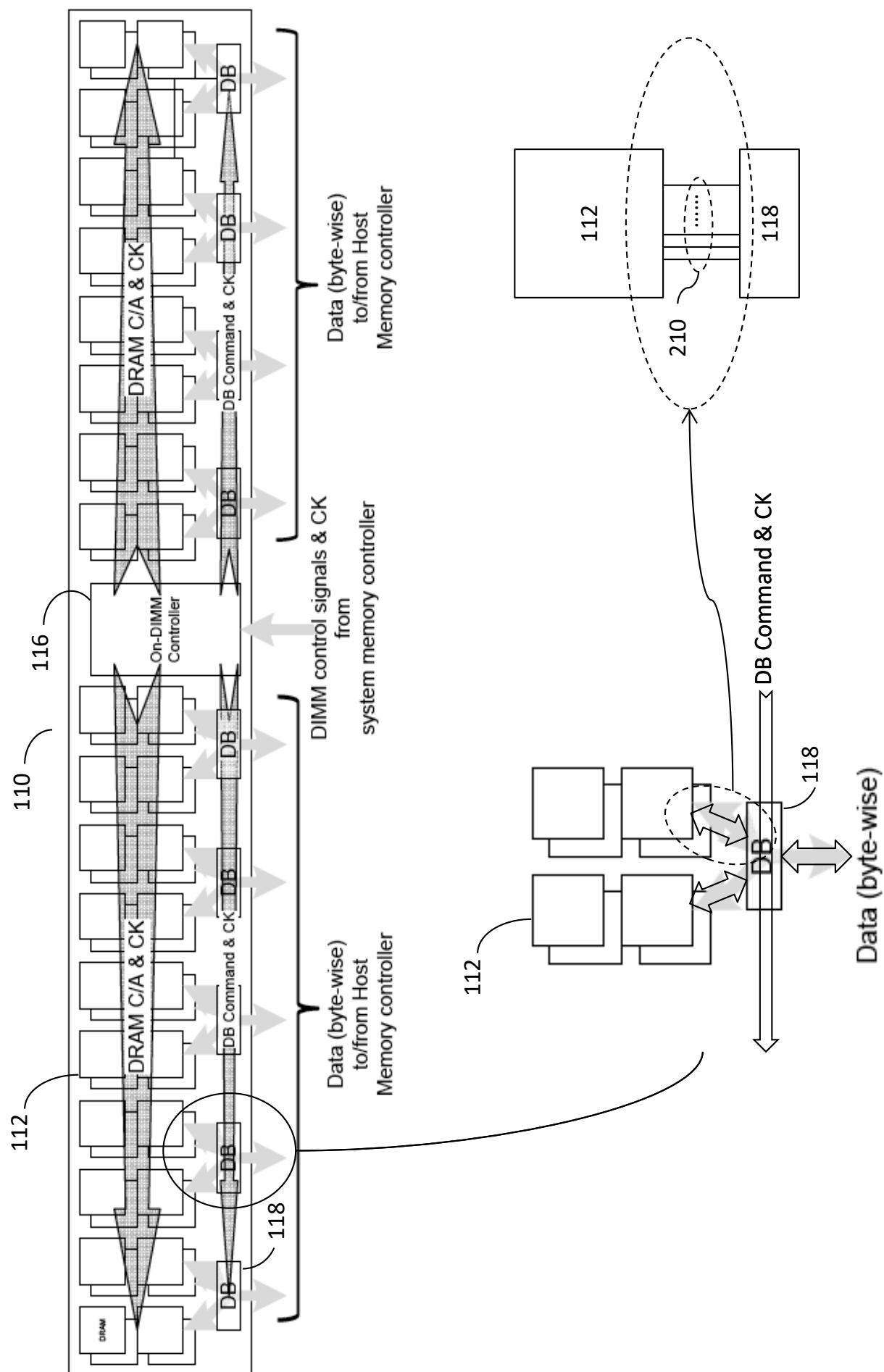


FIG. 2C

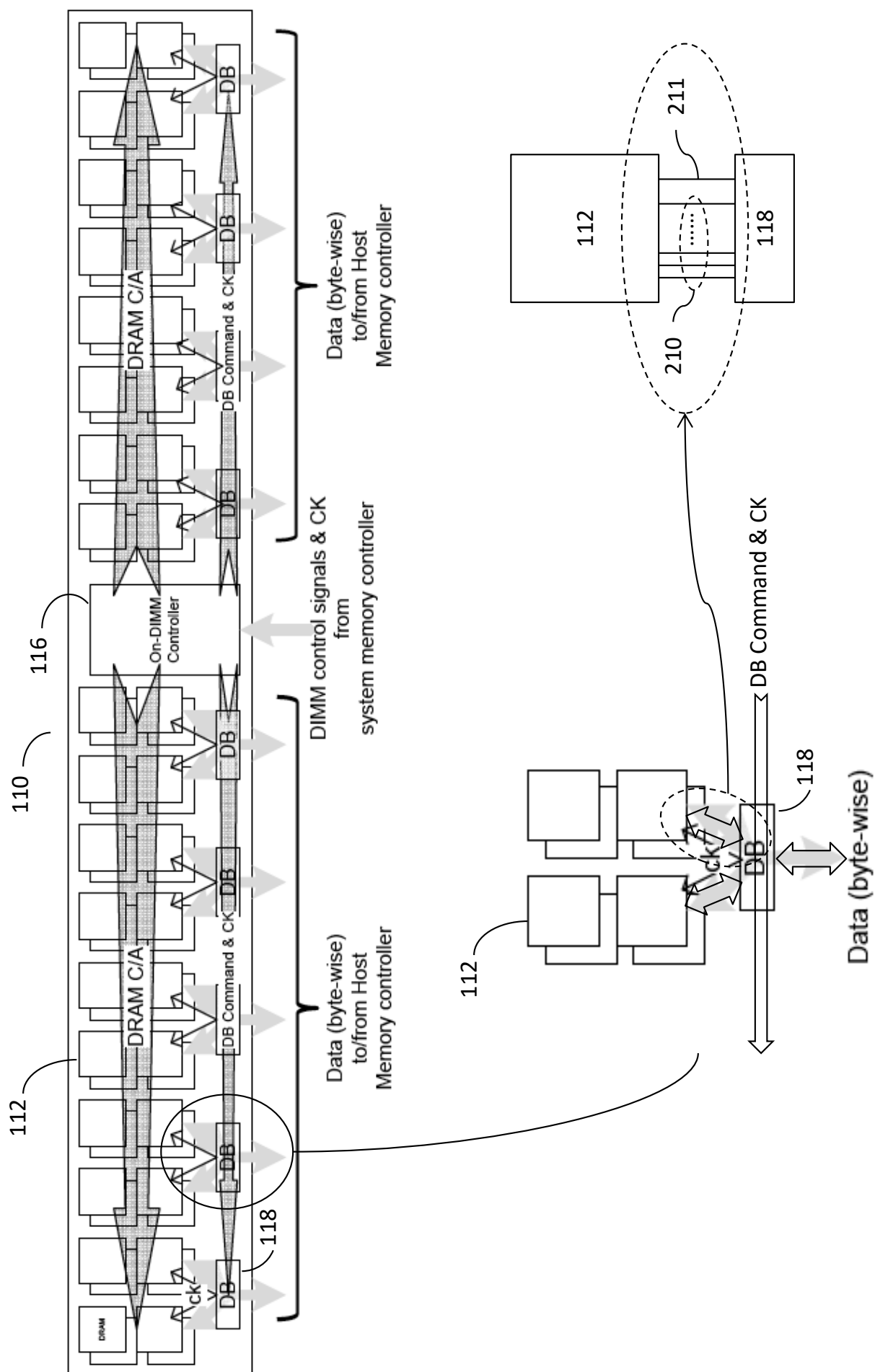


FIG. 2D

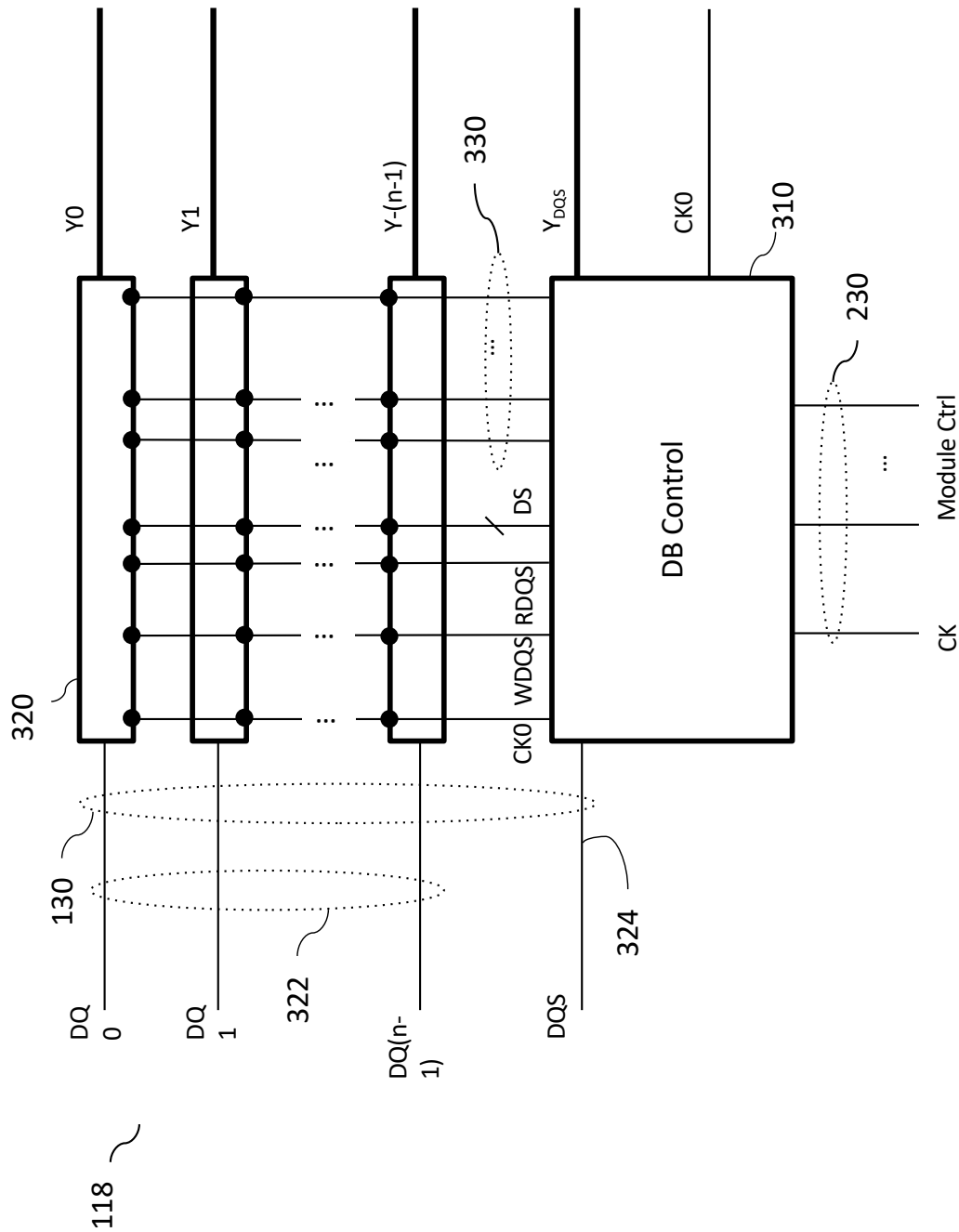


FIG. 3

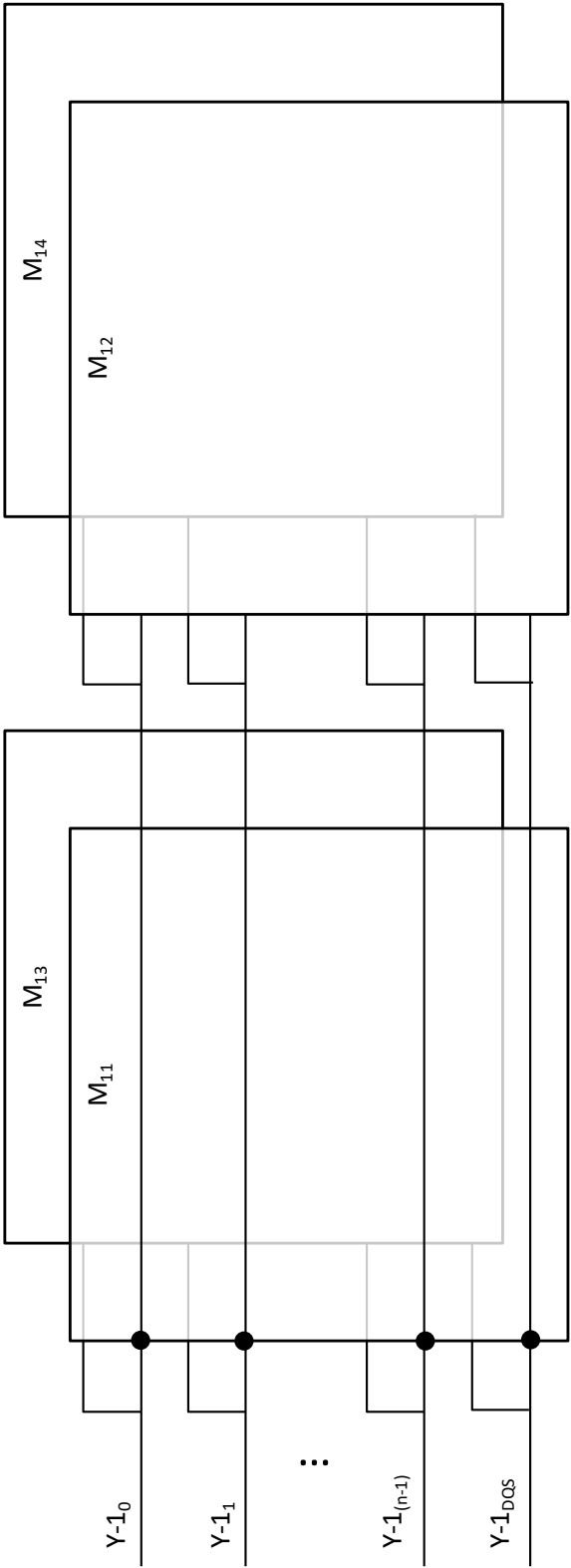


FIG. 4A

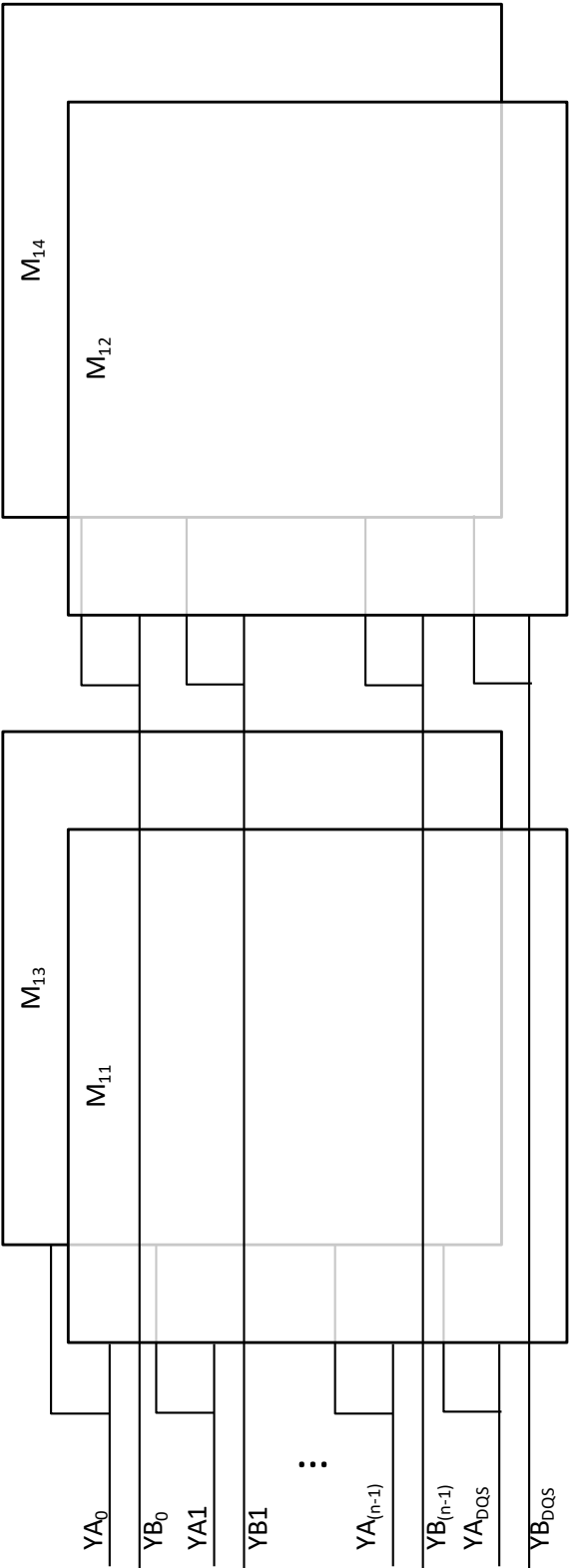


FIG. 4B

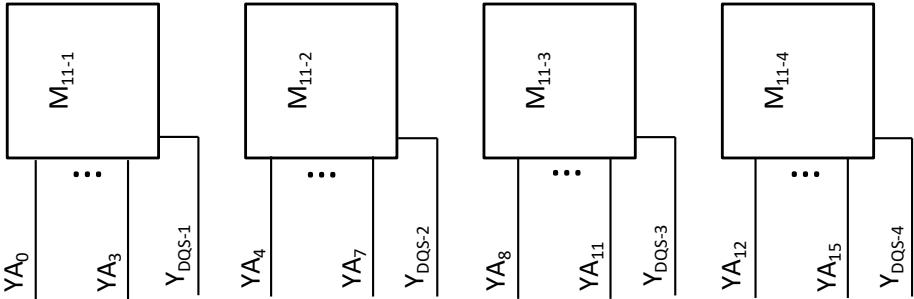


FIG. 5B

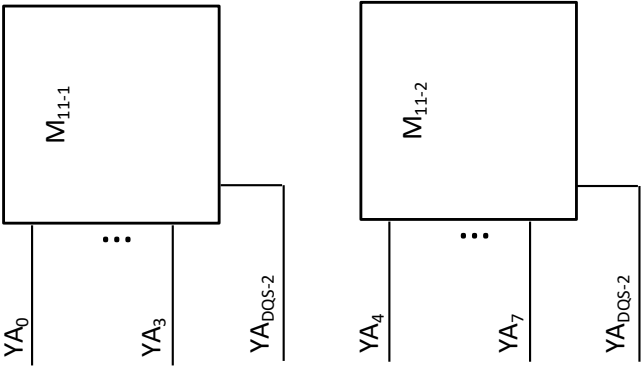


FIG. 5A

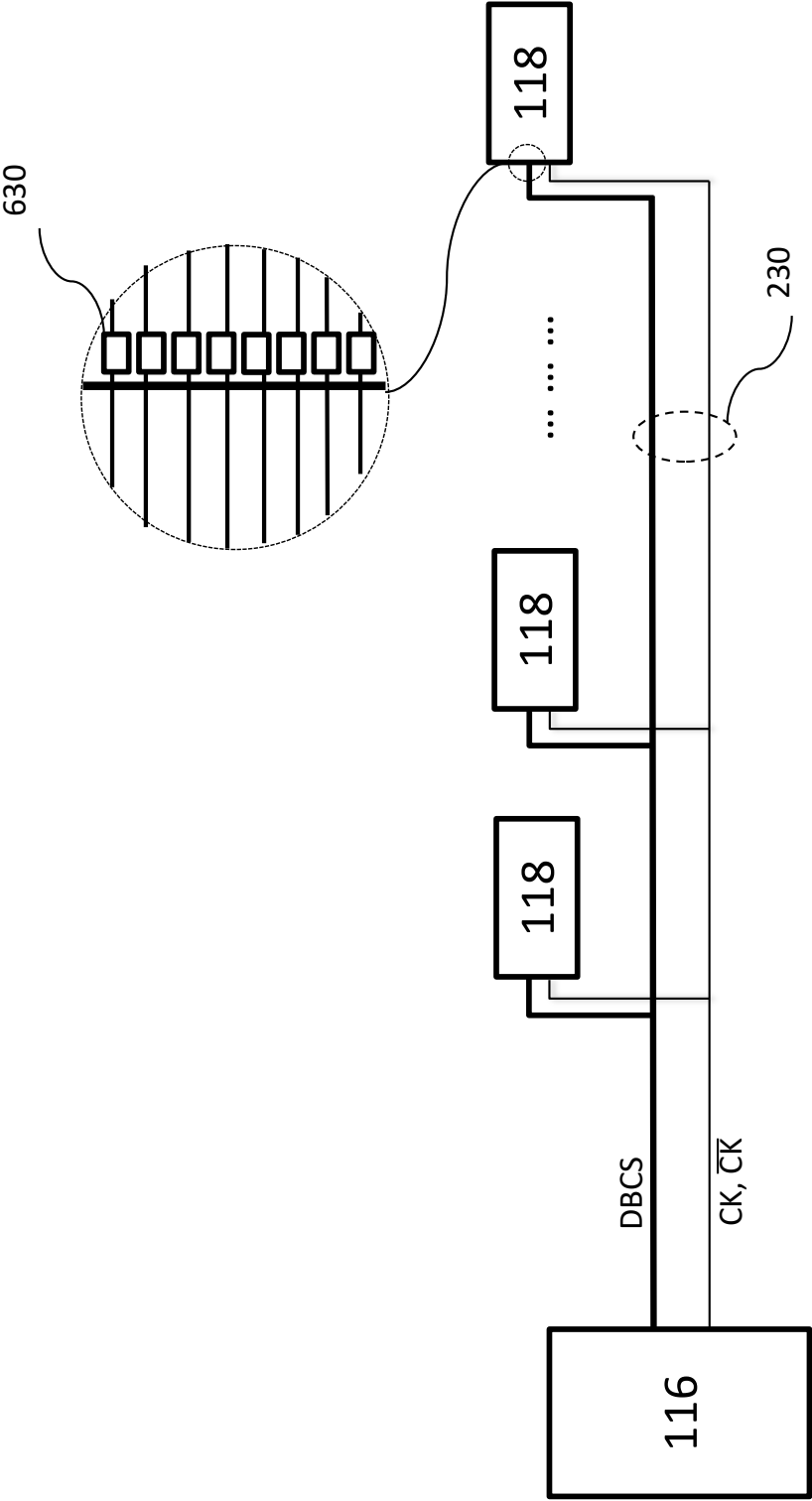


FIG. 6A

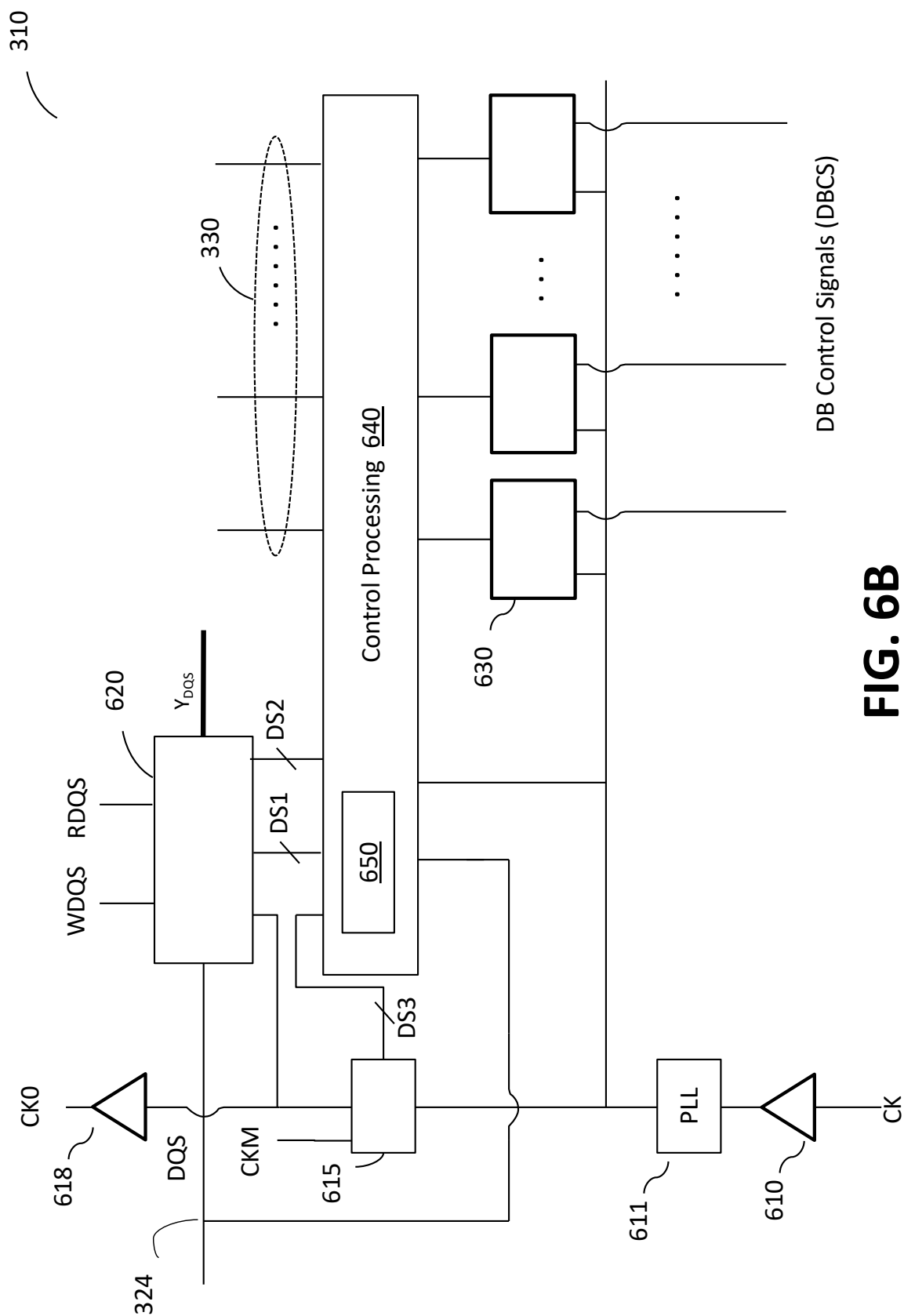


FIG. 6B

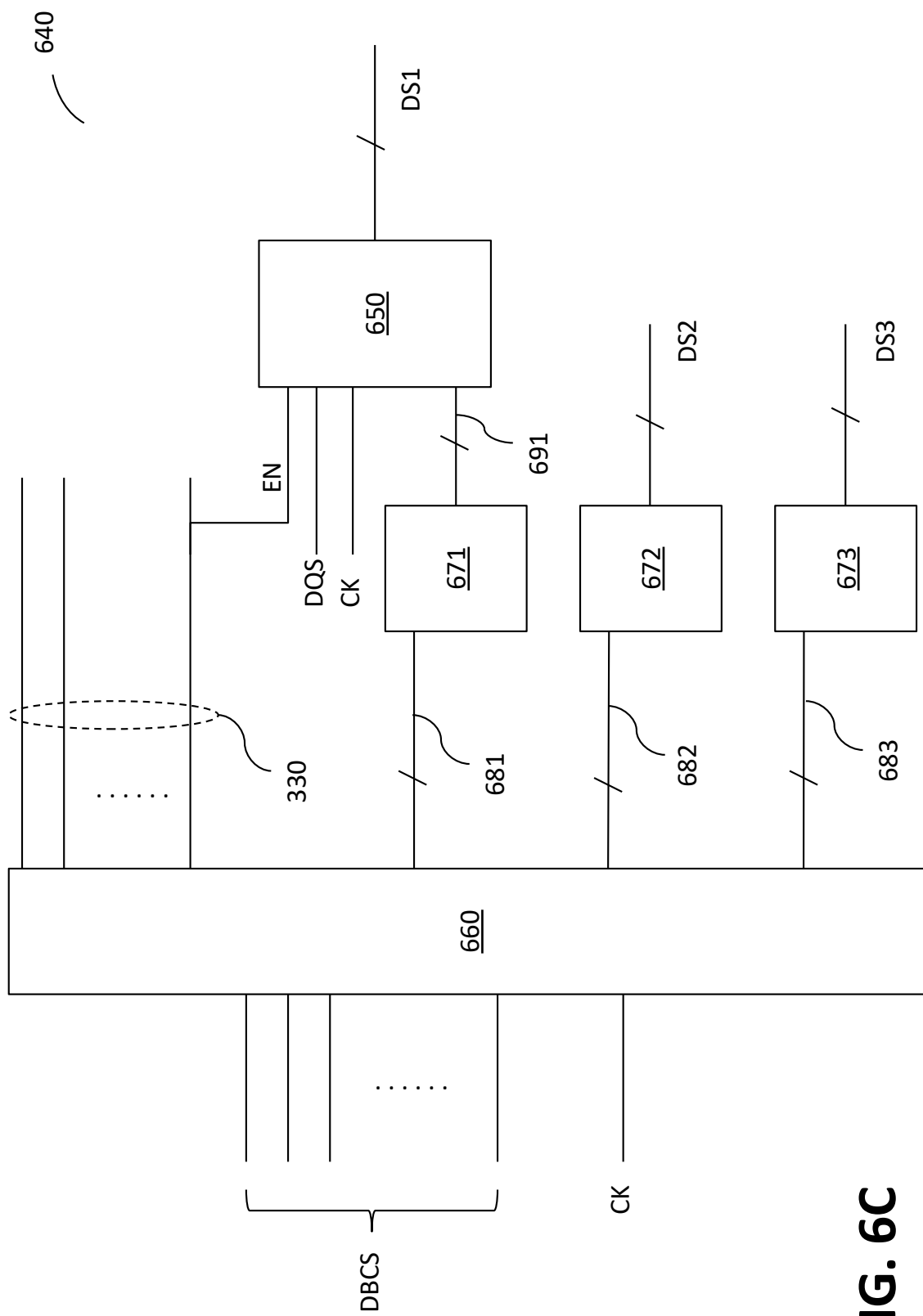


FIG. 6C

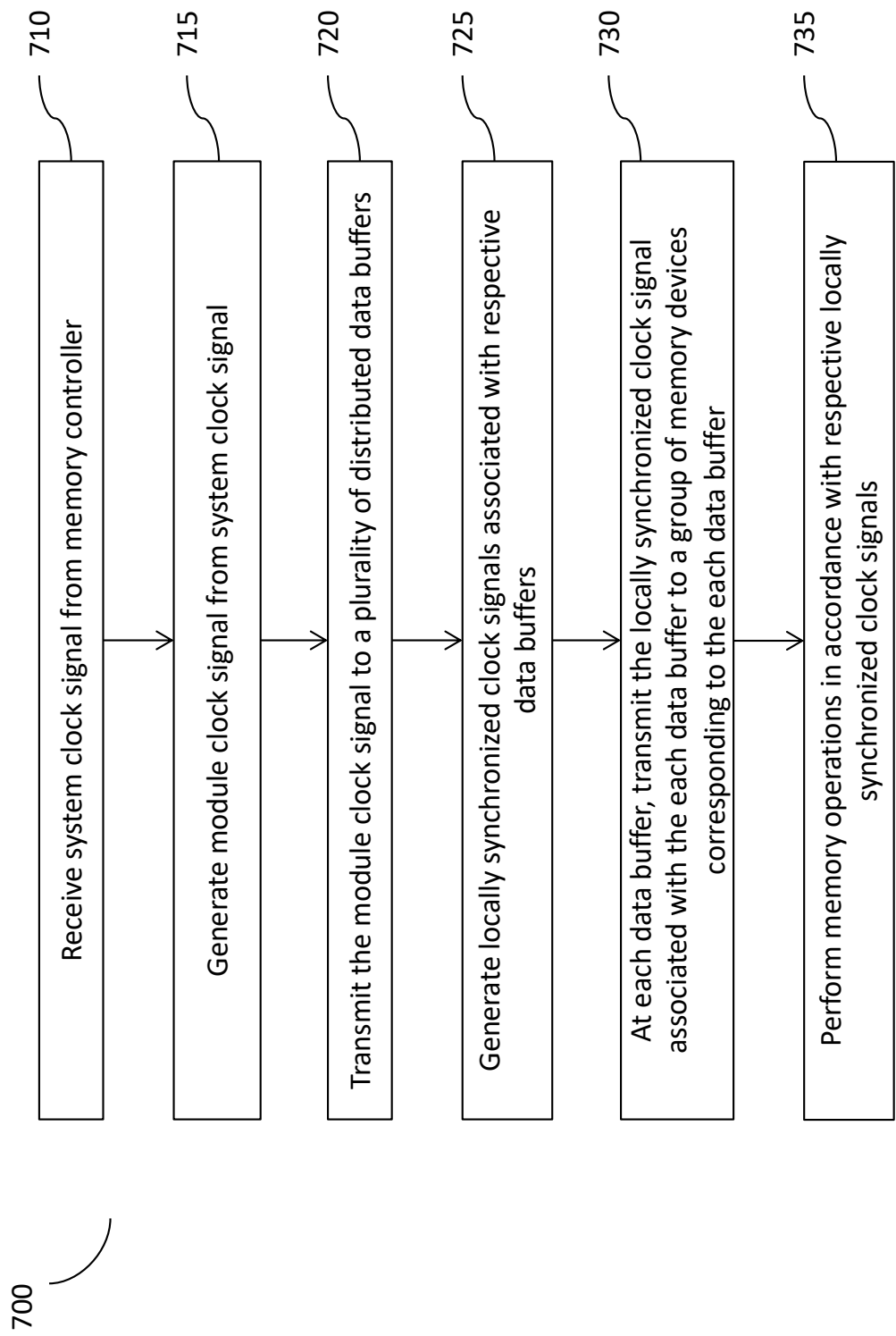


FIG. 7A

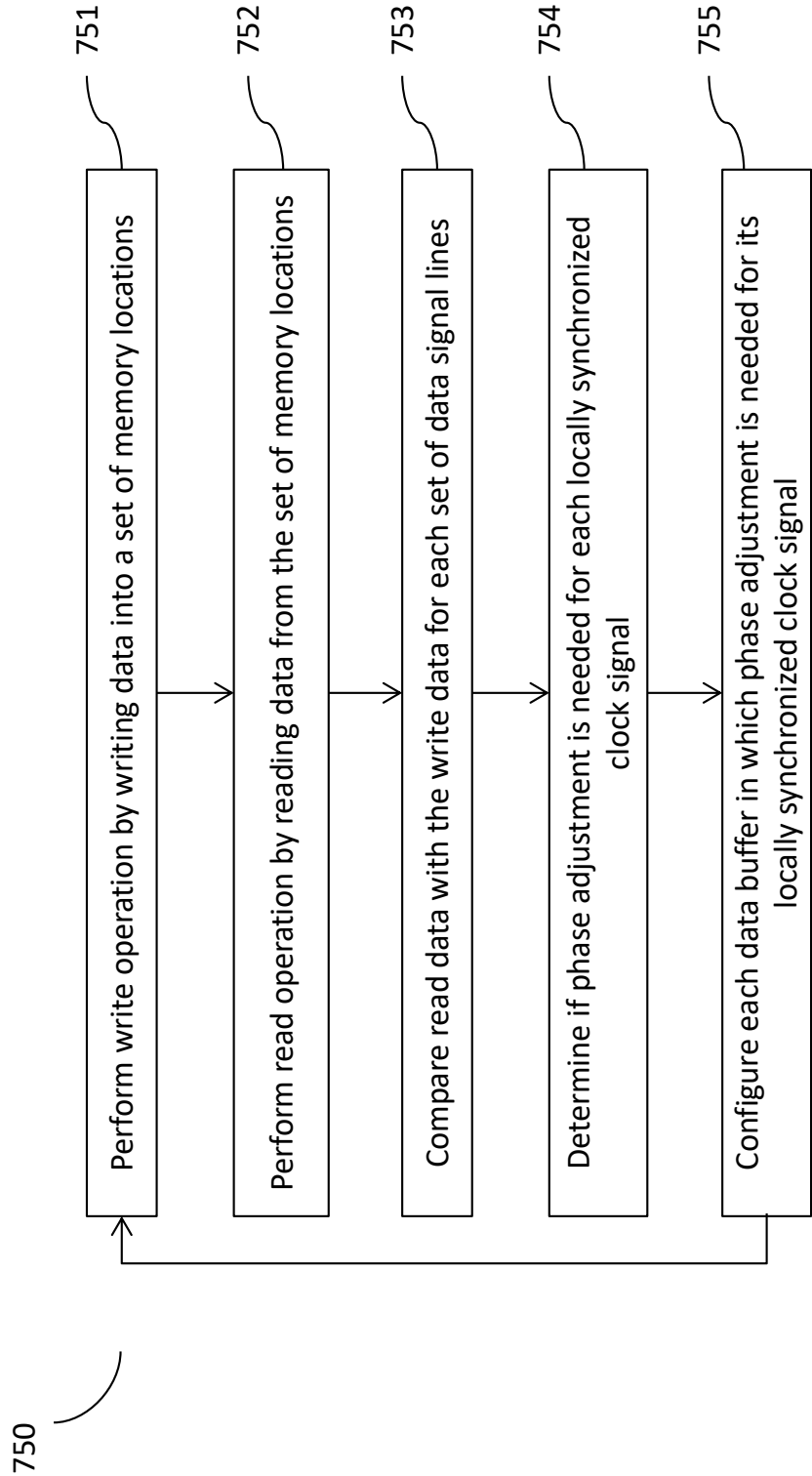


FIG. 7B

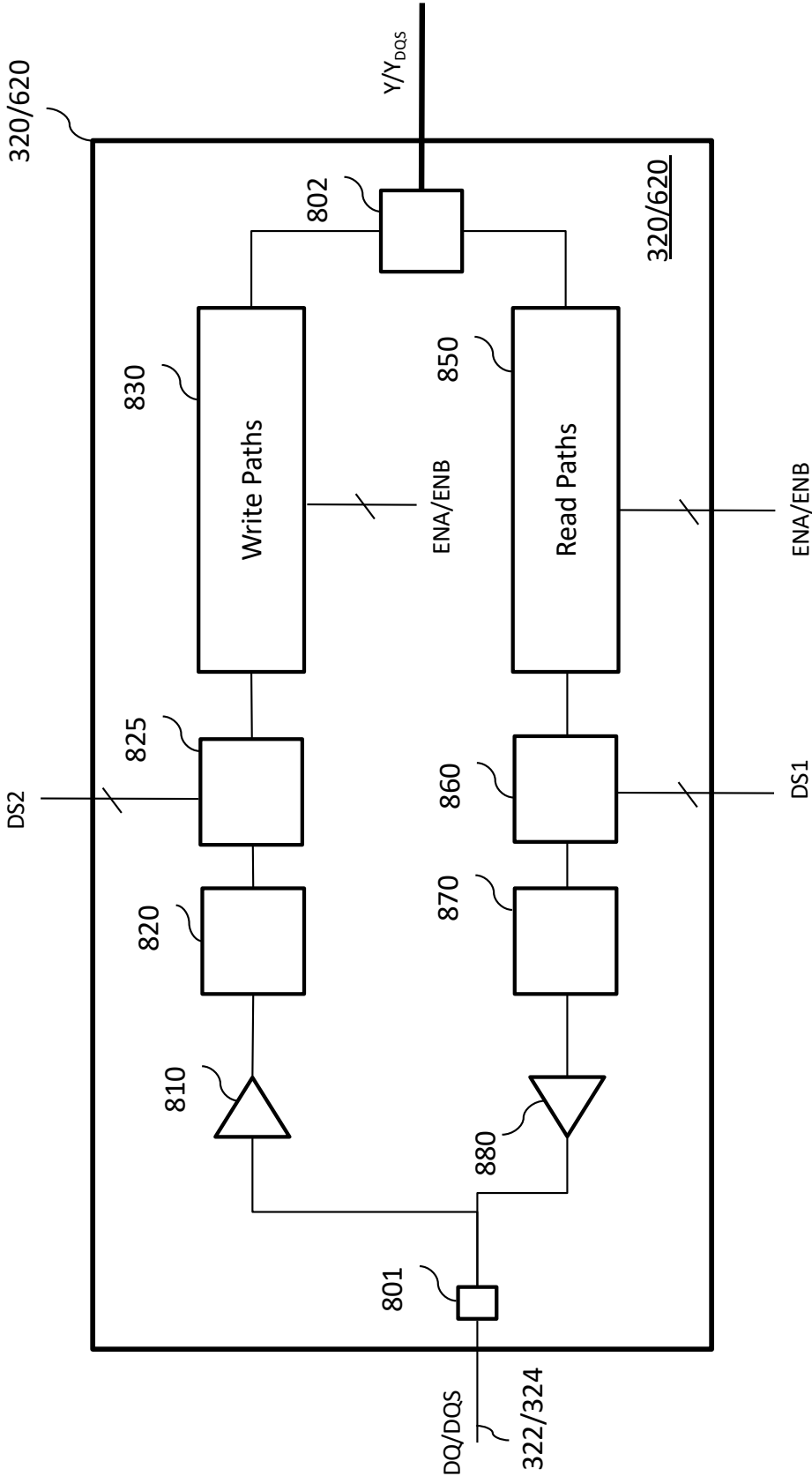


FIG. 8

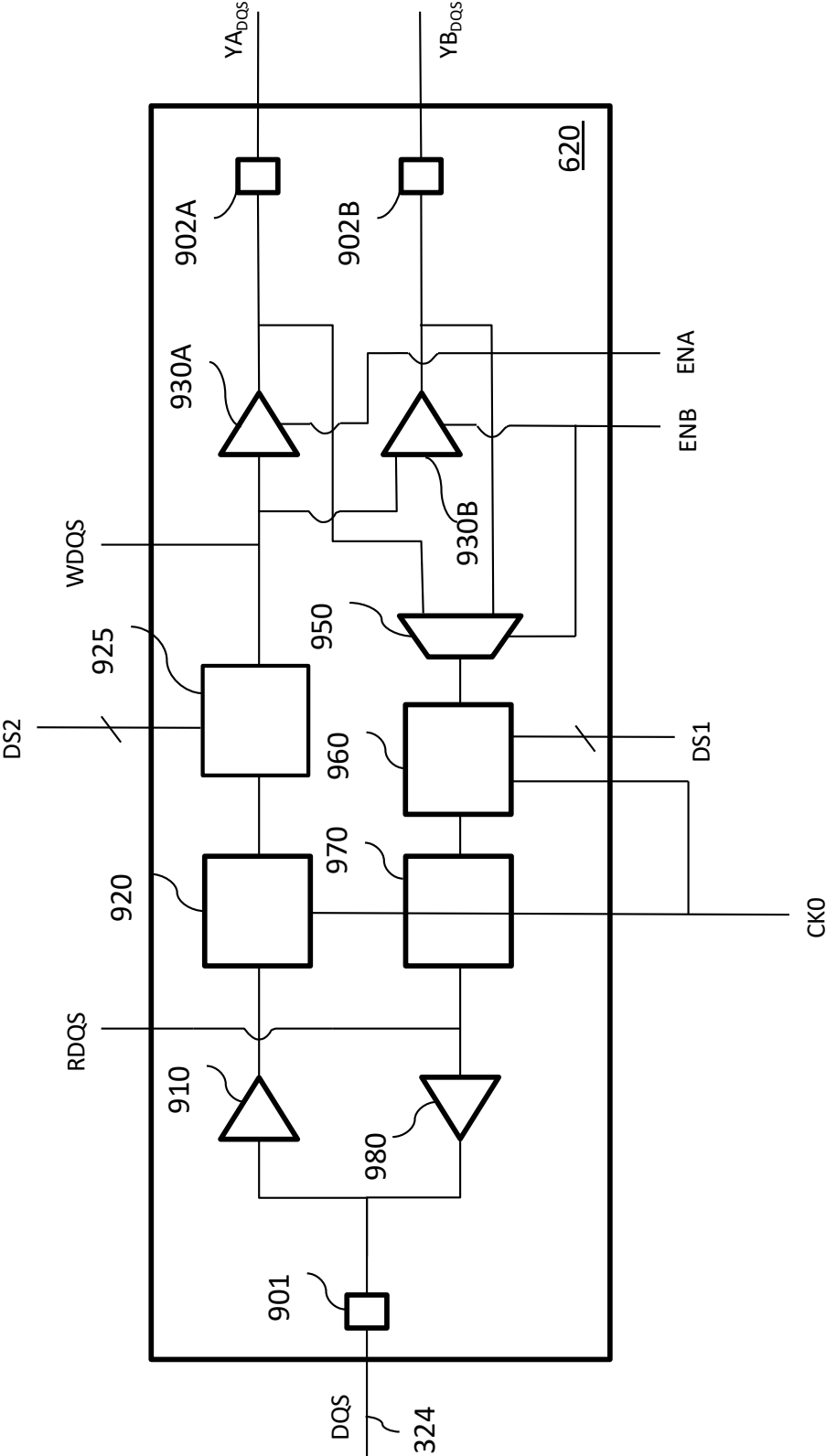


FIG. 9

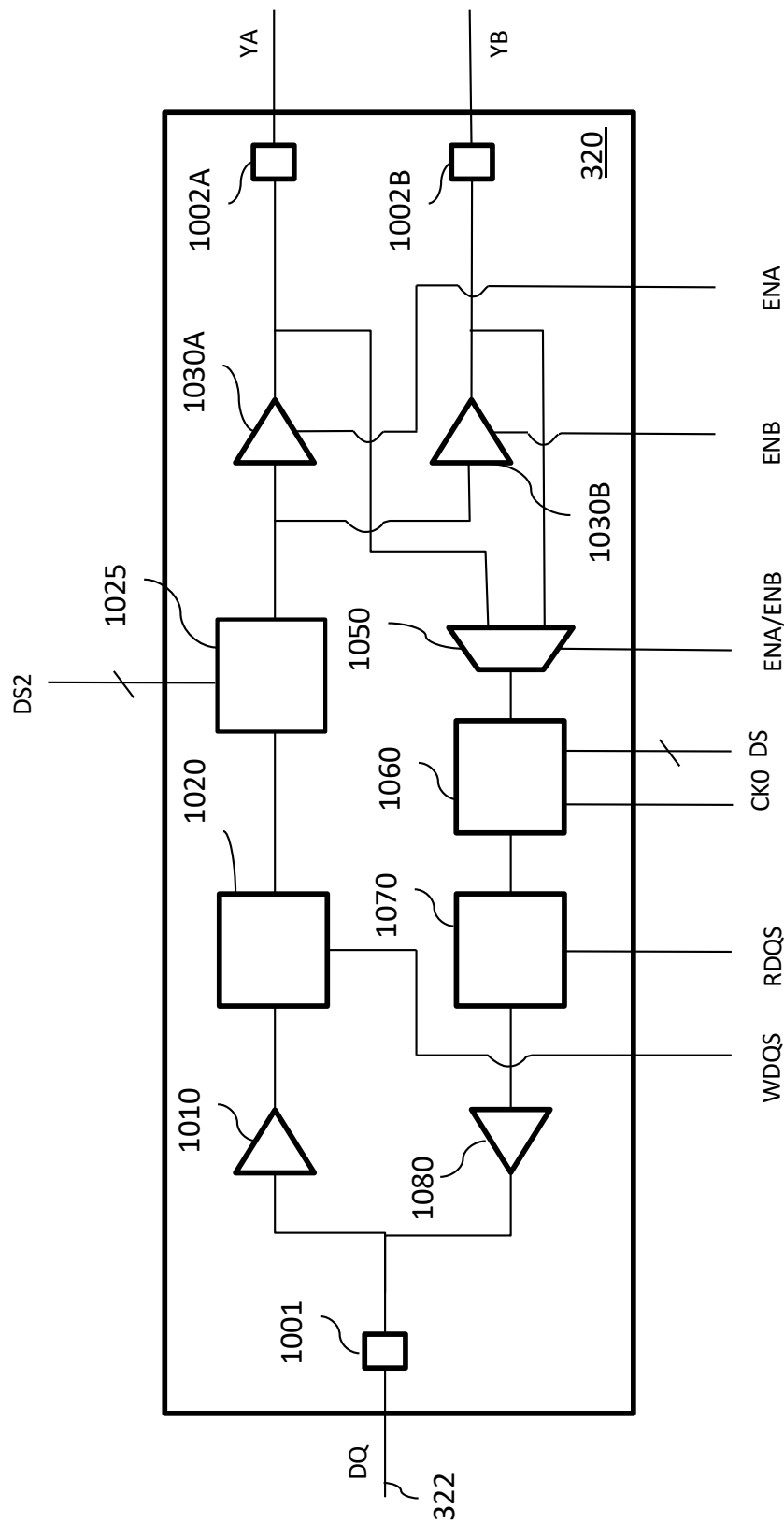


FIG. 10

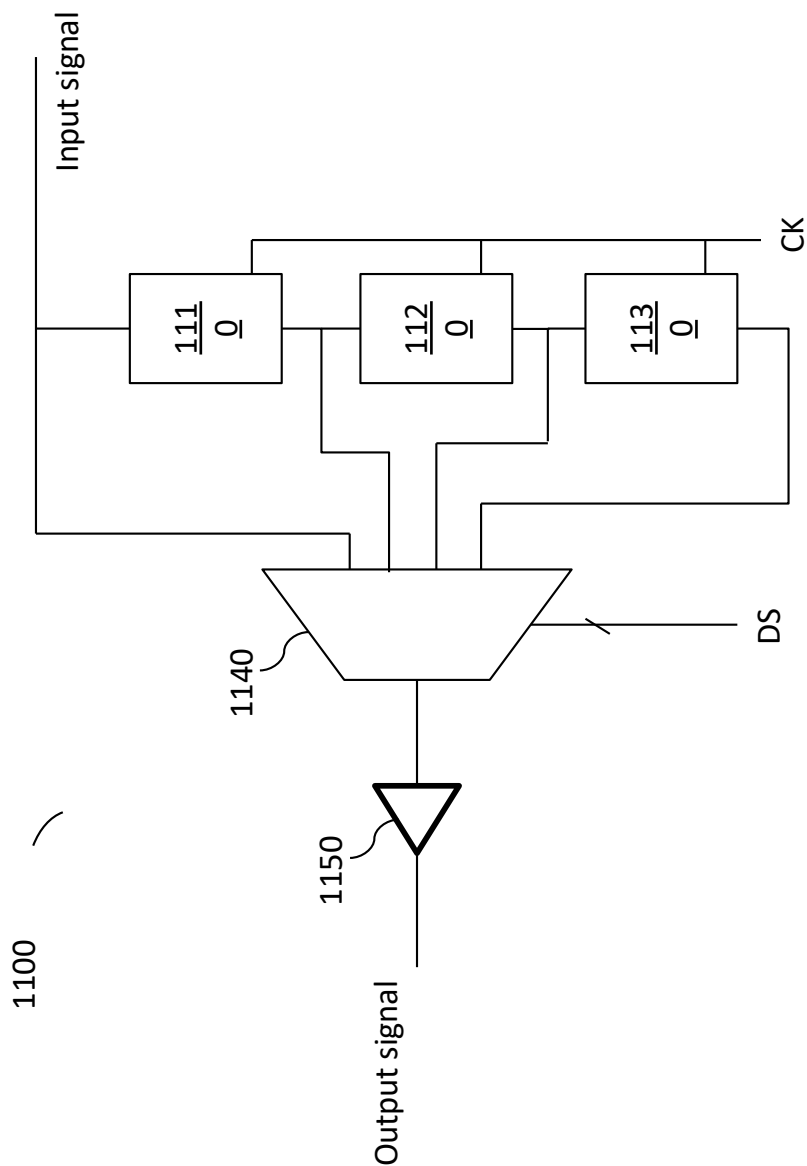


FIG. 11

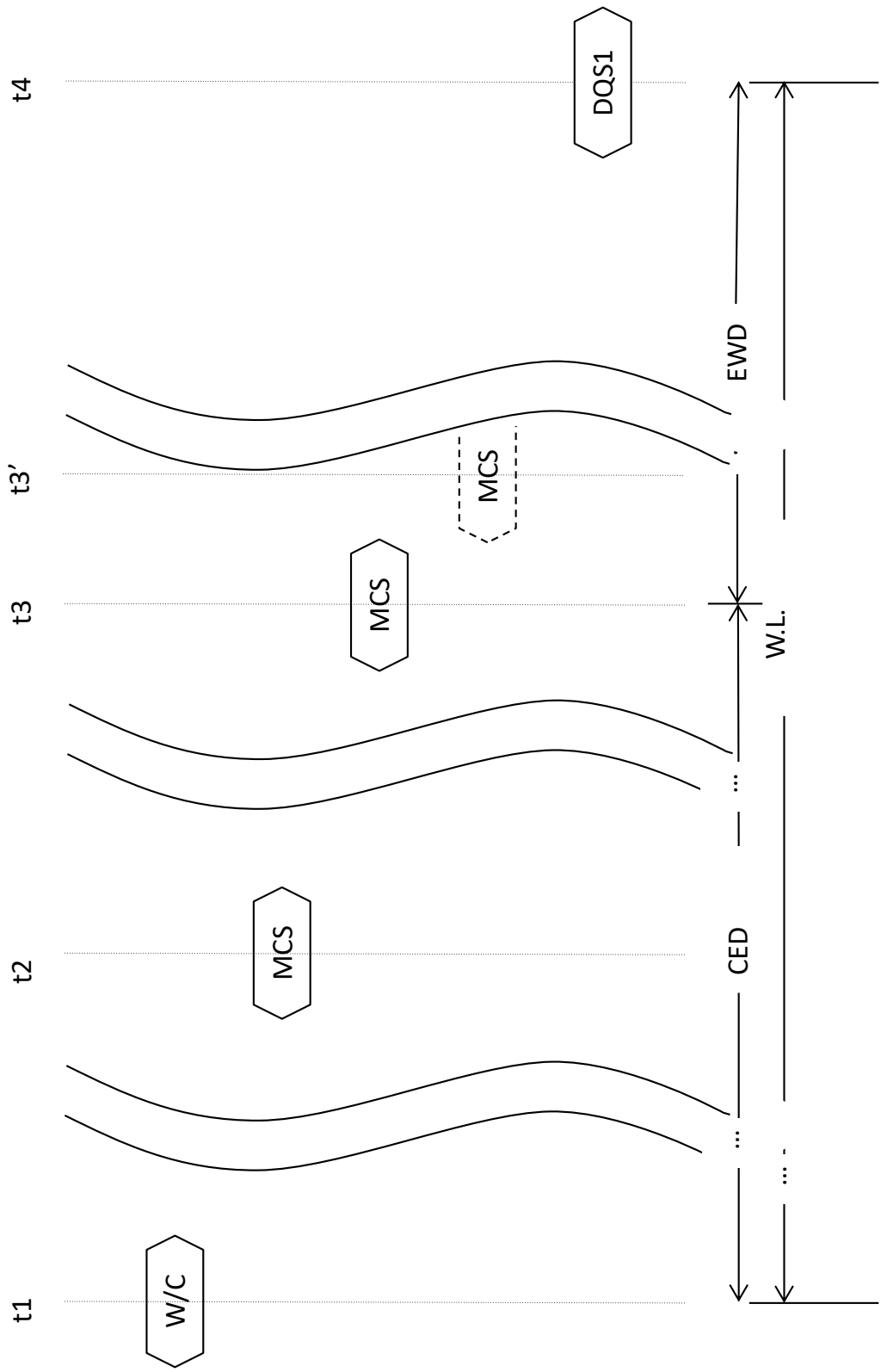


FIG. 12A

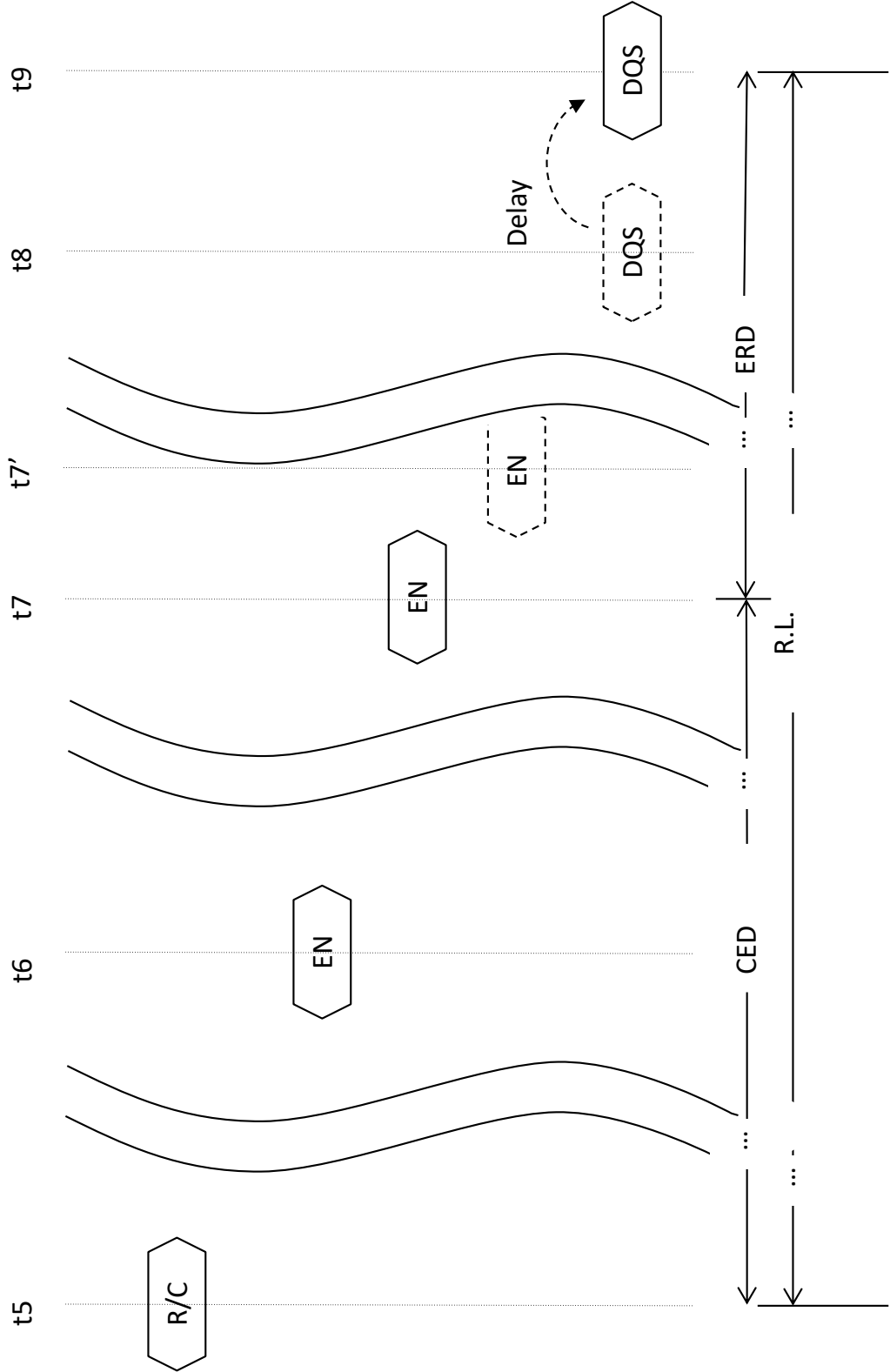


FIG. 12B

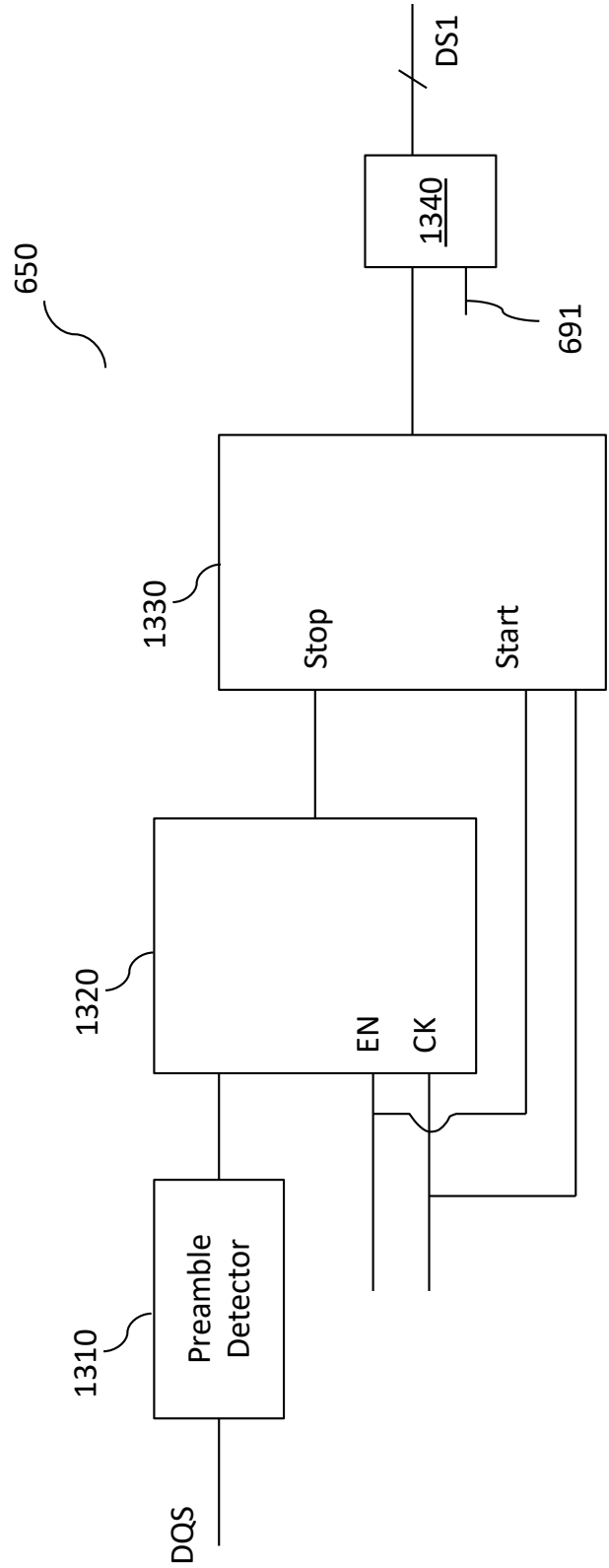


FIG. 13

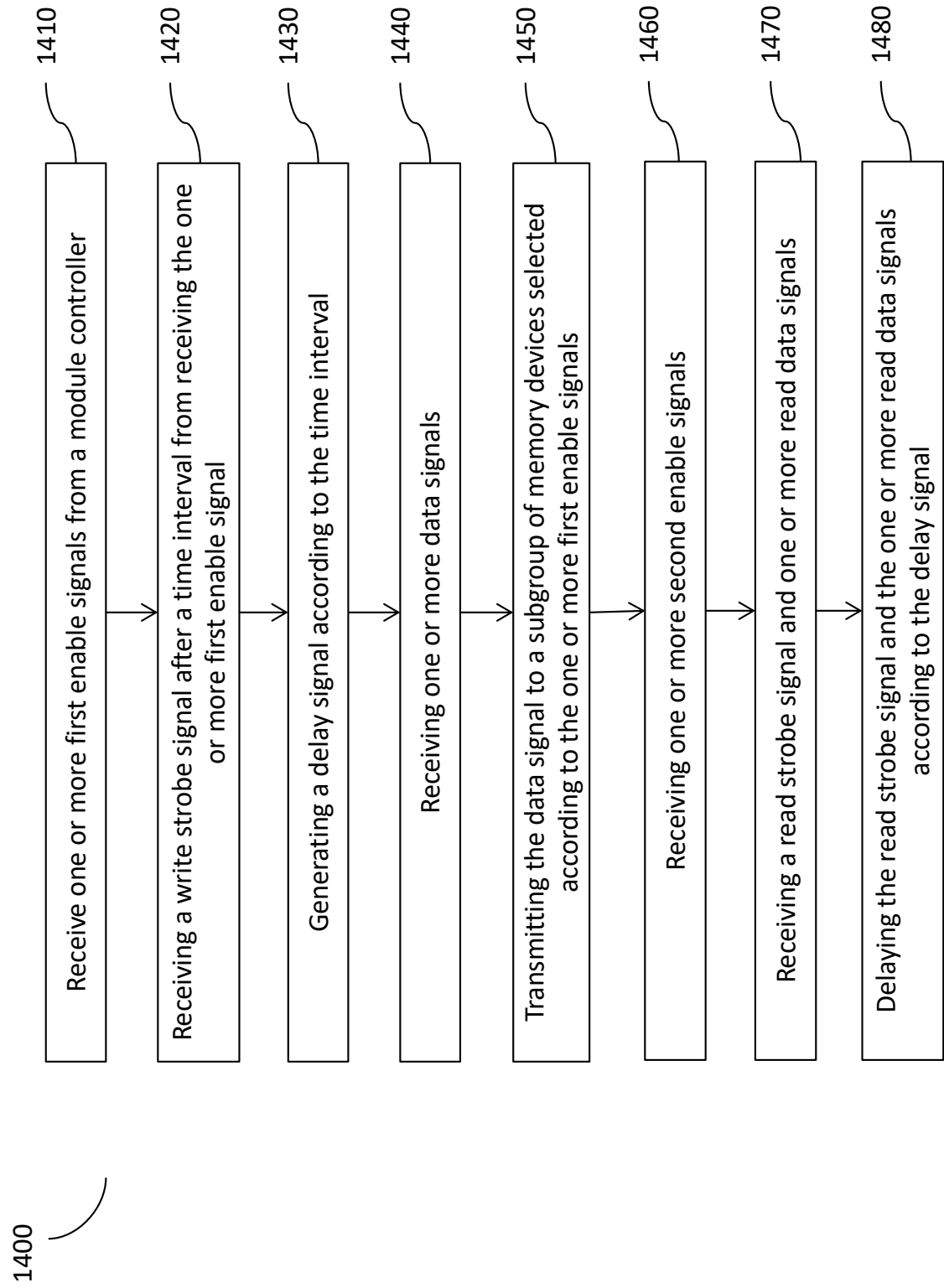


FIG. 14

Placeholder Sheet for Supplemental Content

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Document Date: 11/01/2024

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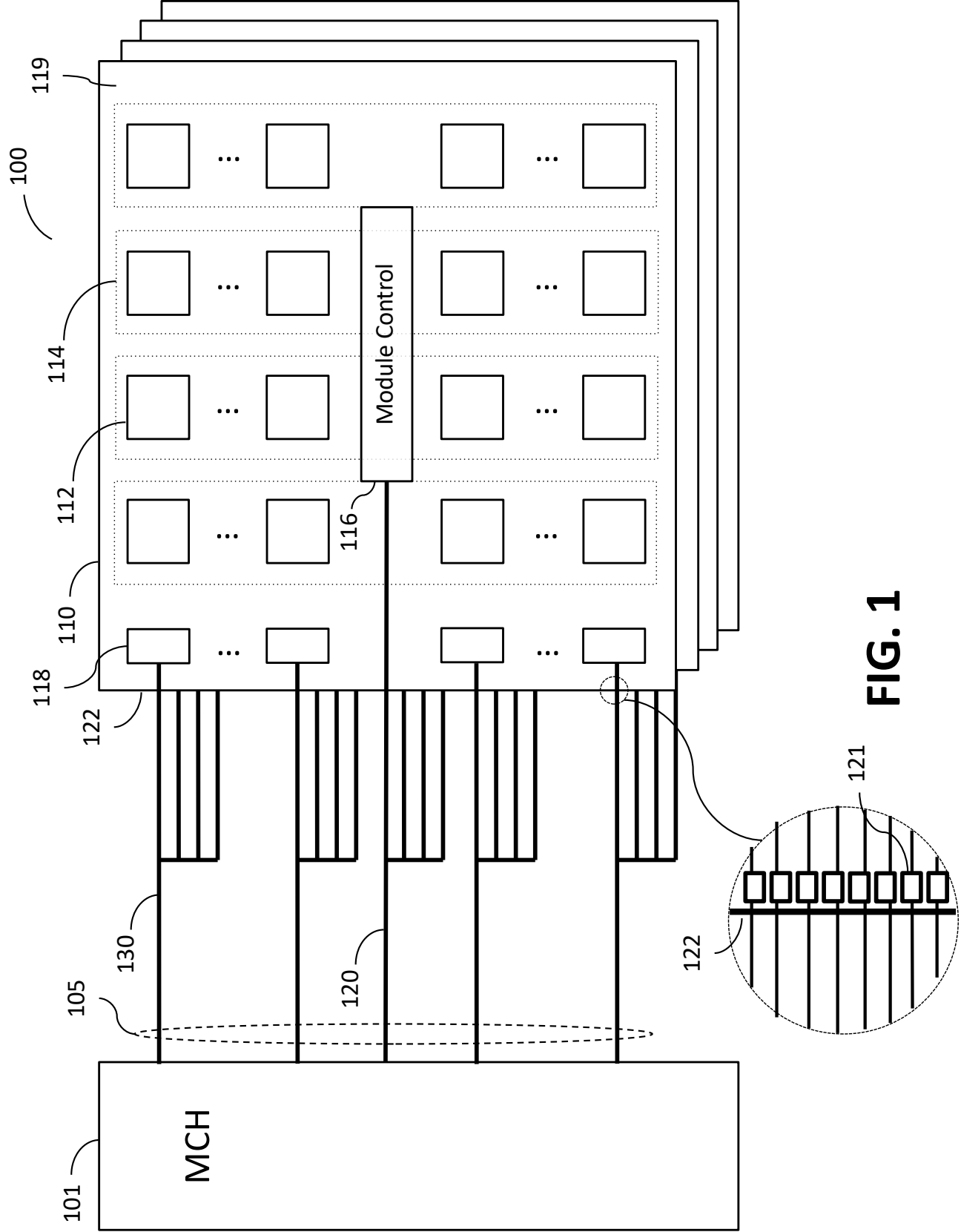


FIG. 1

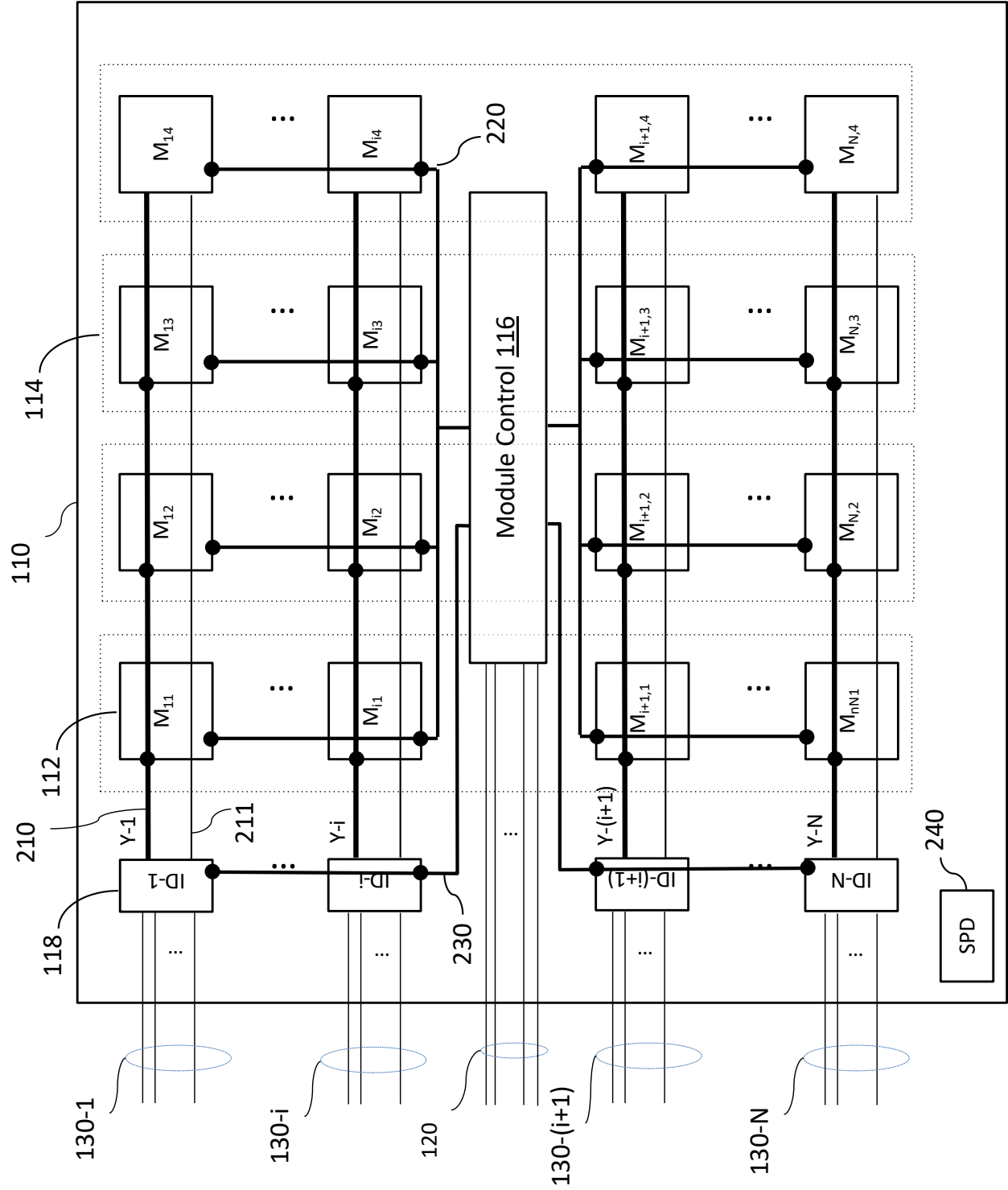


FIG. 2A

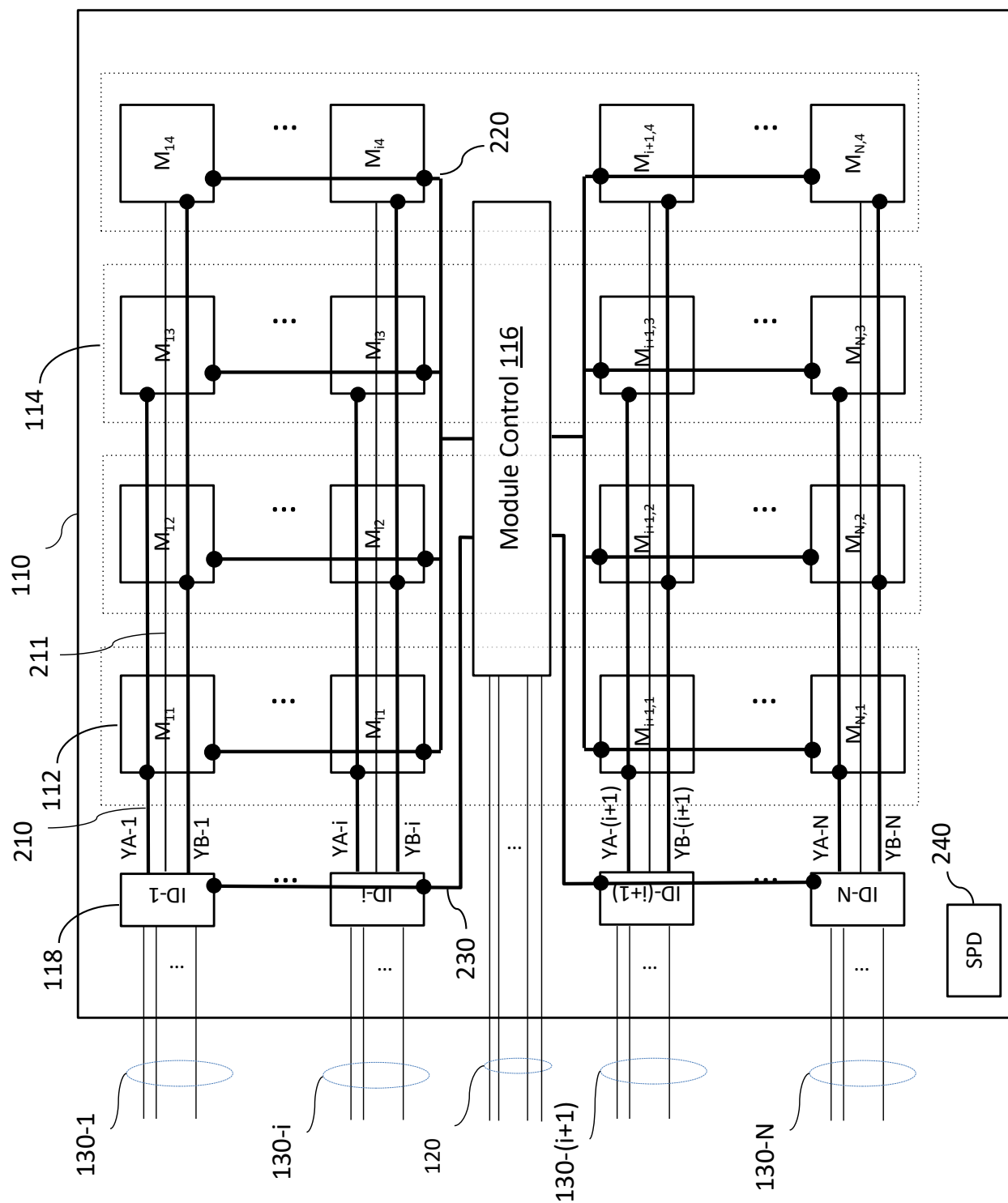


FIG. 2B

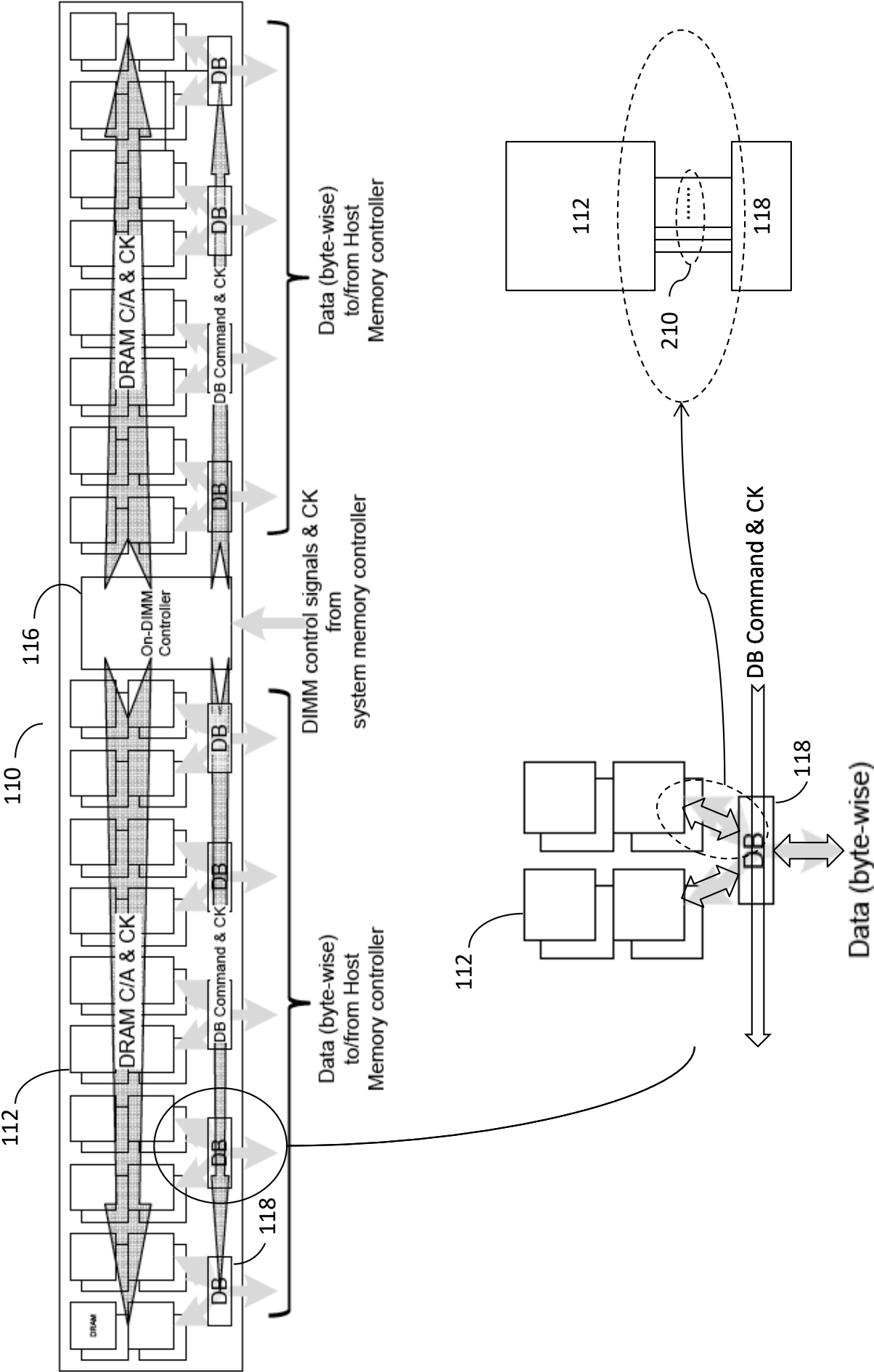


FIG. 2C

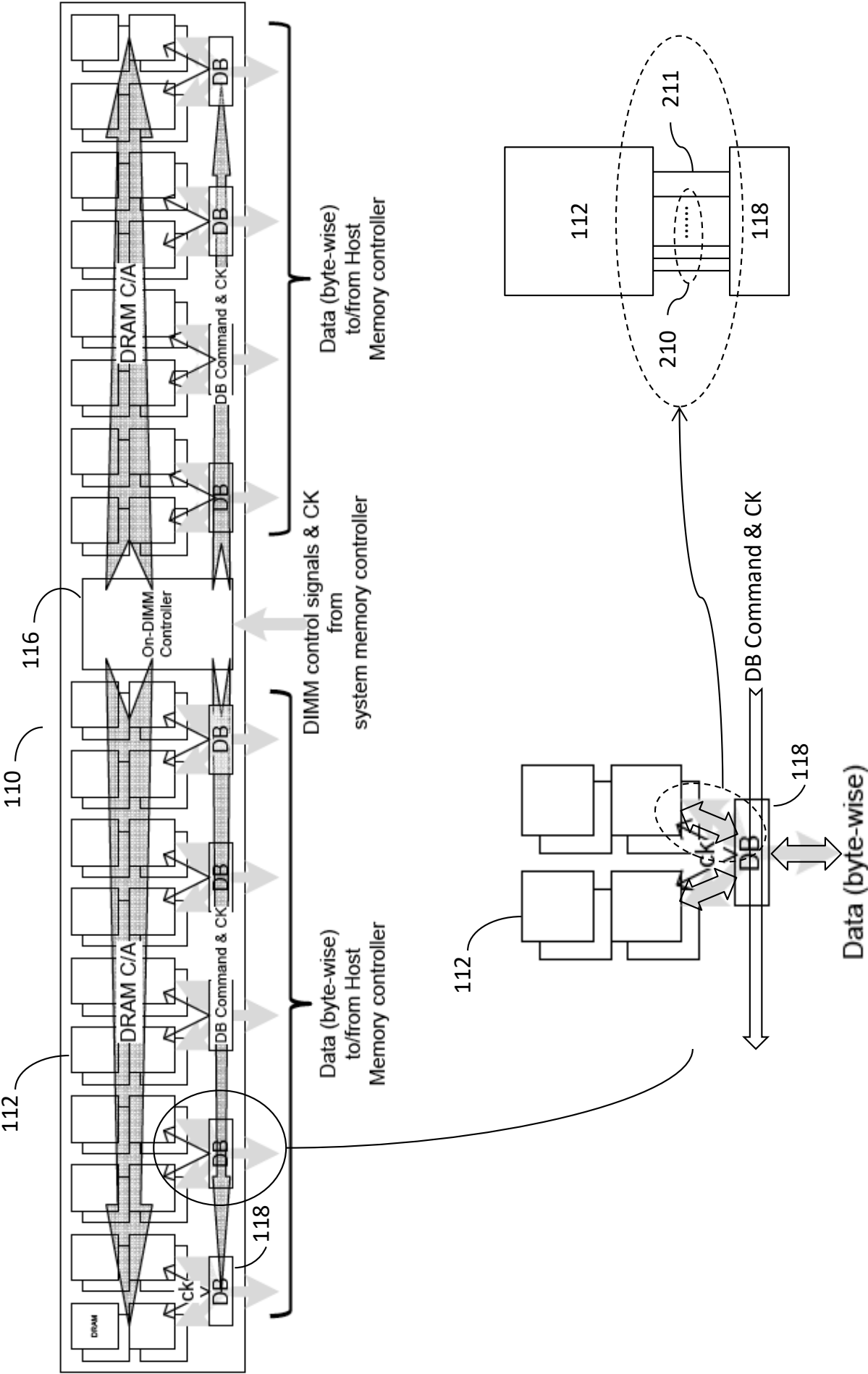


FIG. 2D

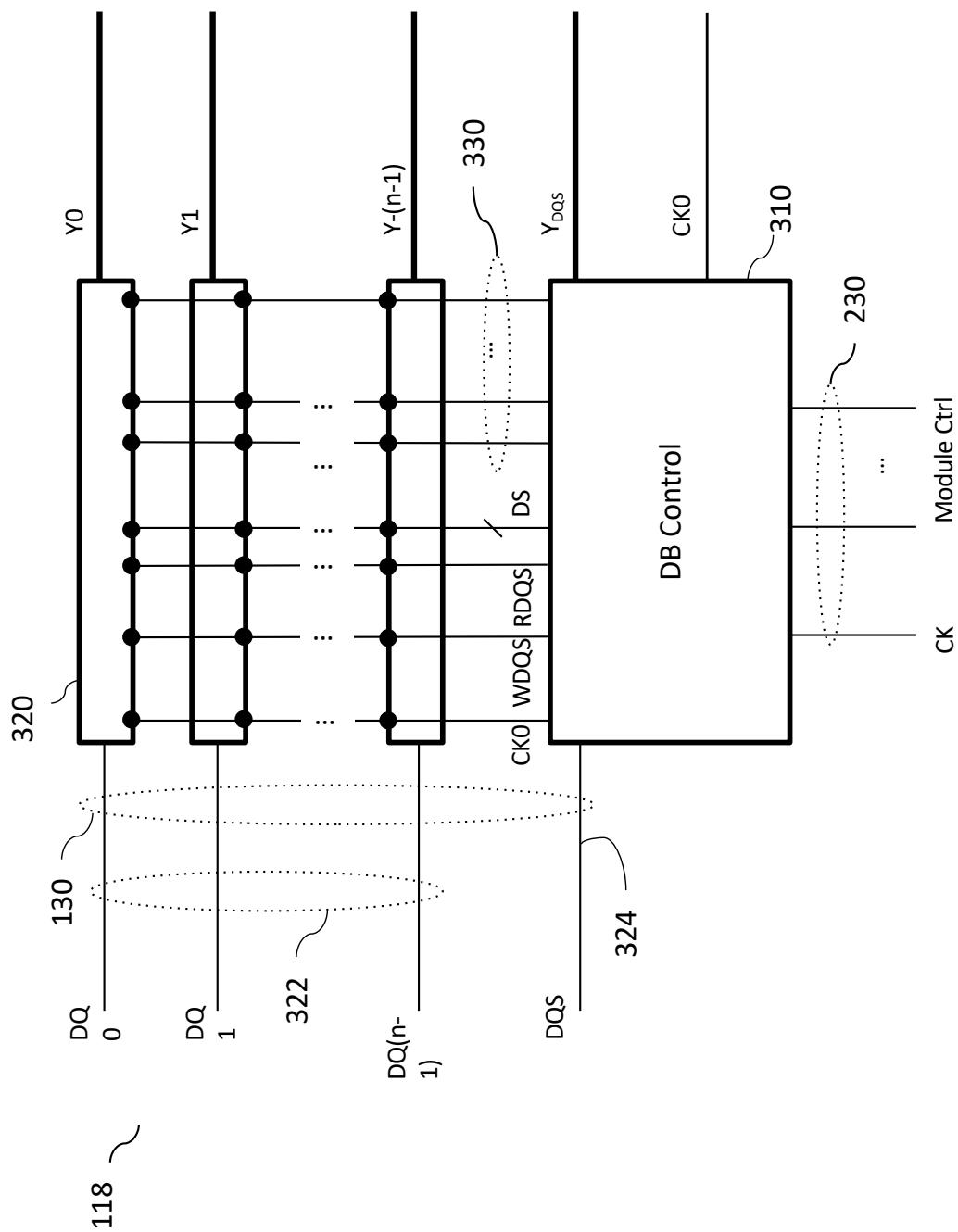


FIG. 3

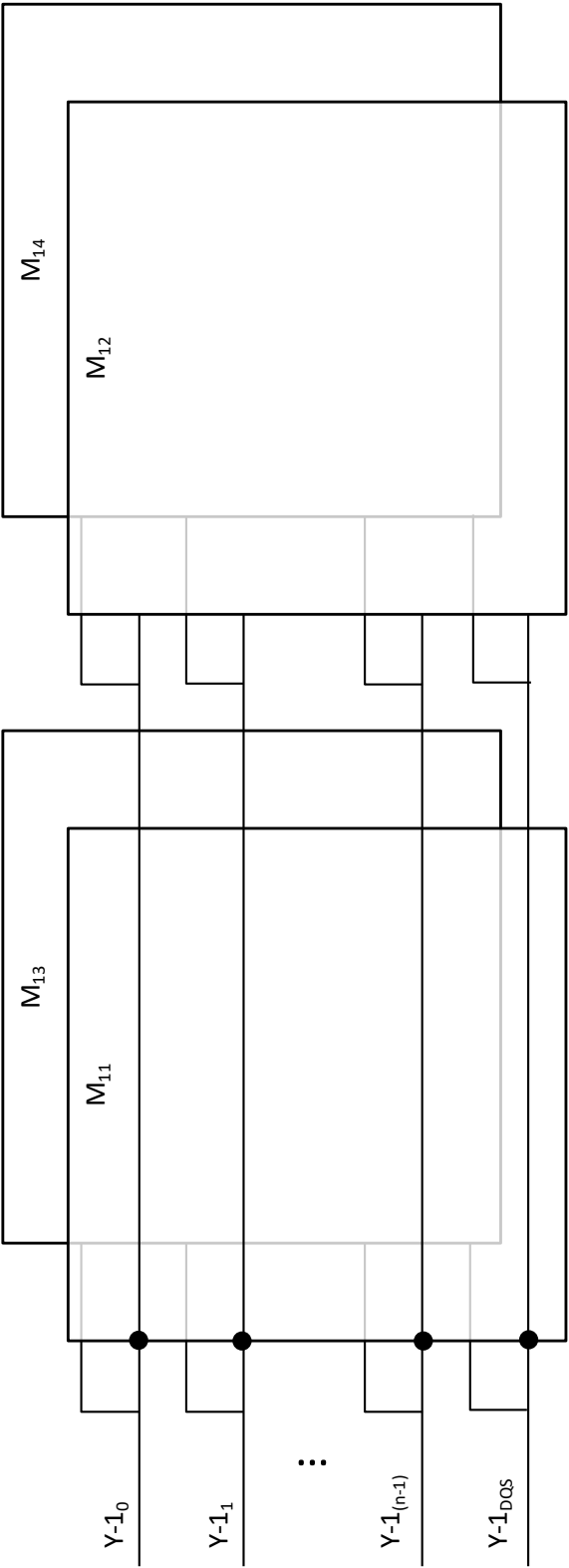


FIG. 4A

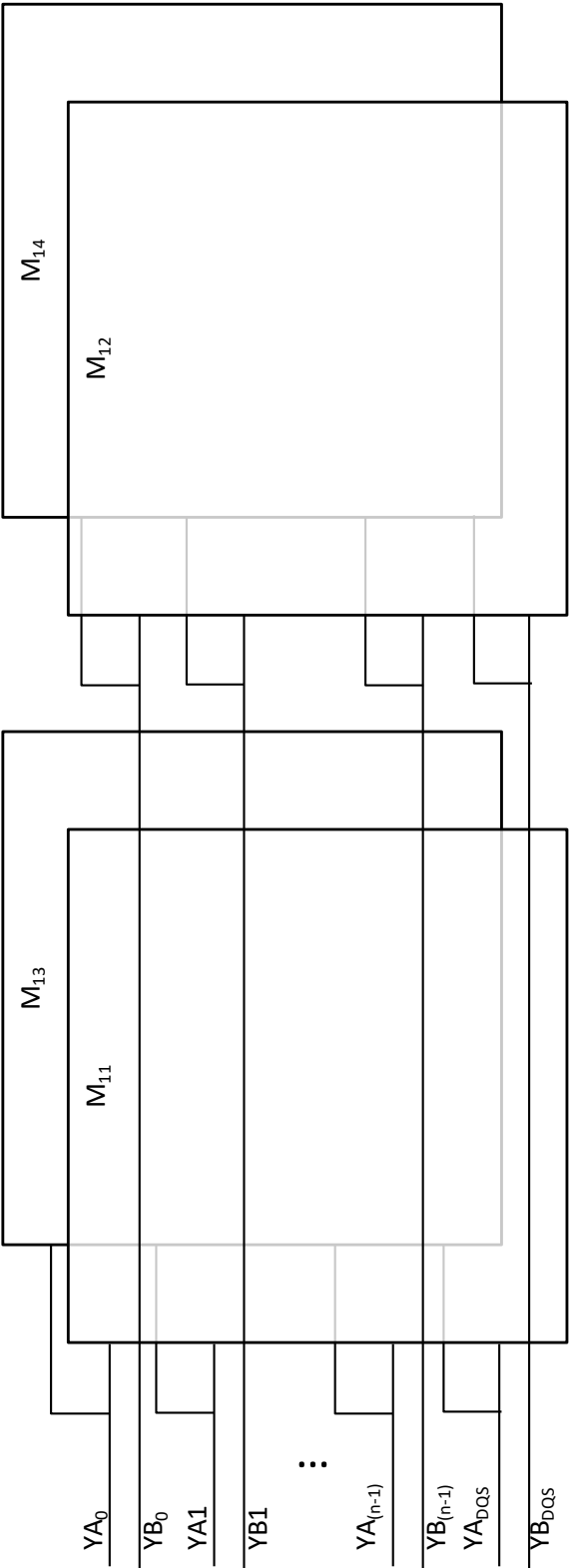


FIG. 4B

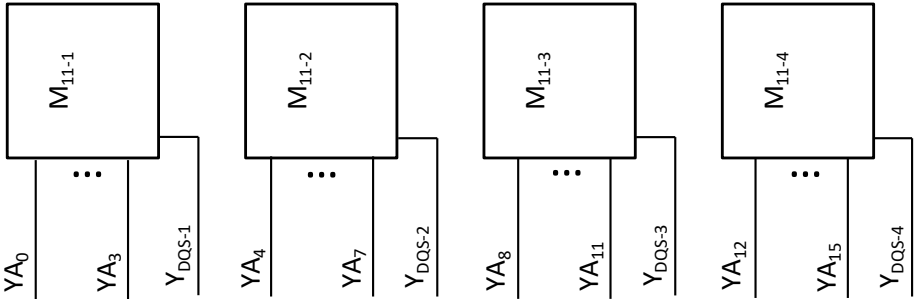


FIG. 5B

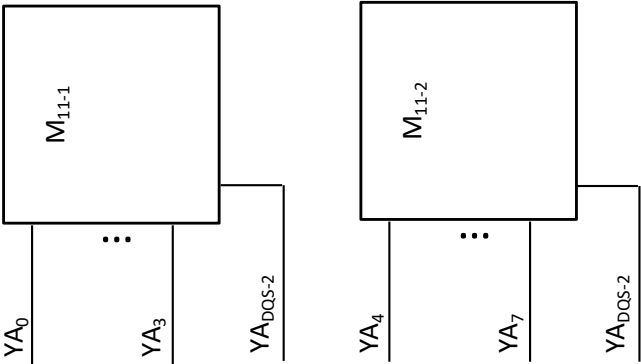


FIG. 5A

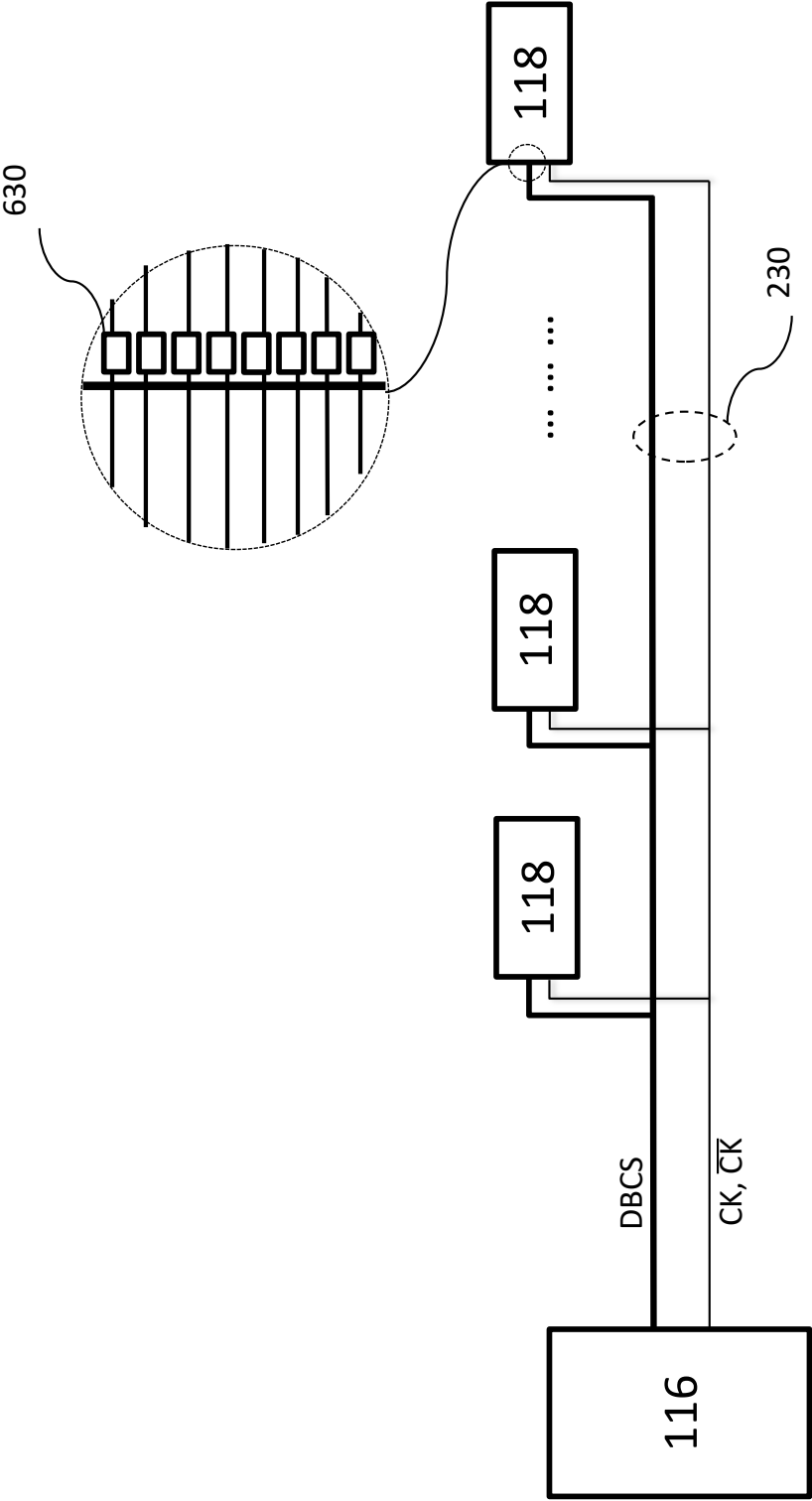


FIG. 6A

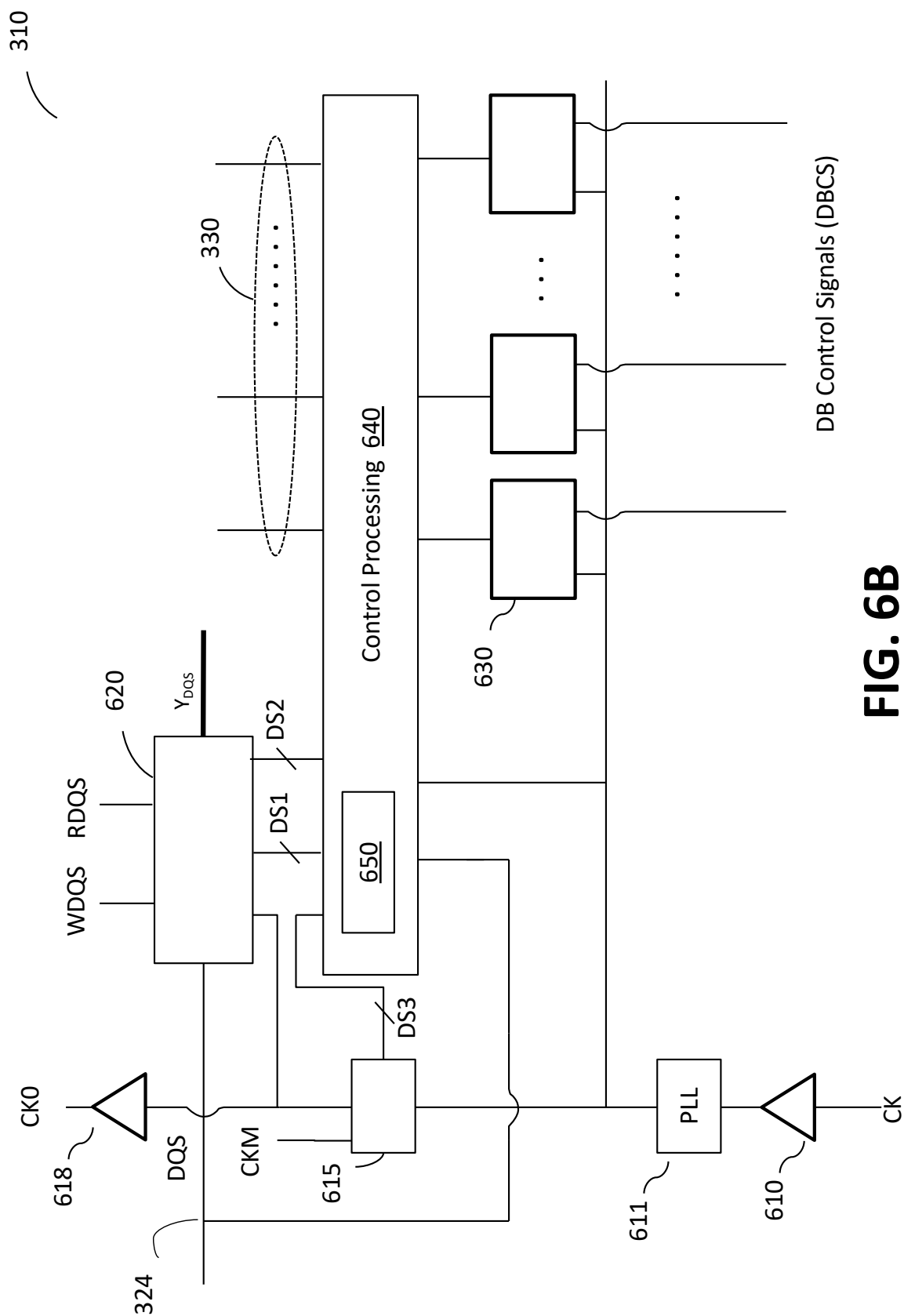


FIG. 6B

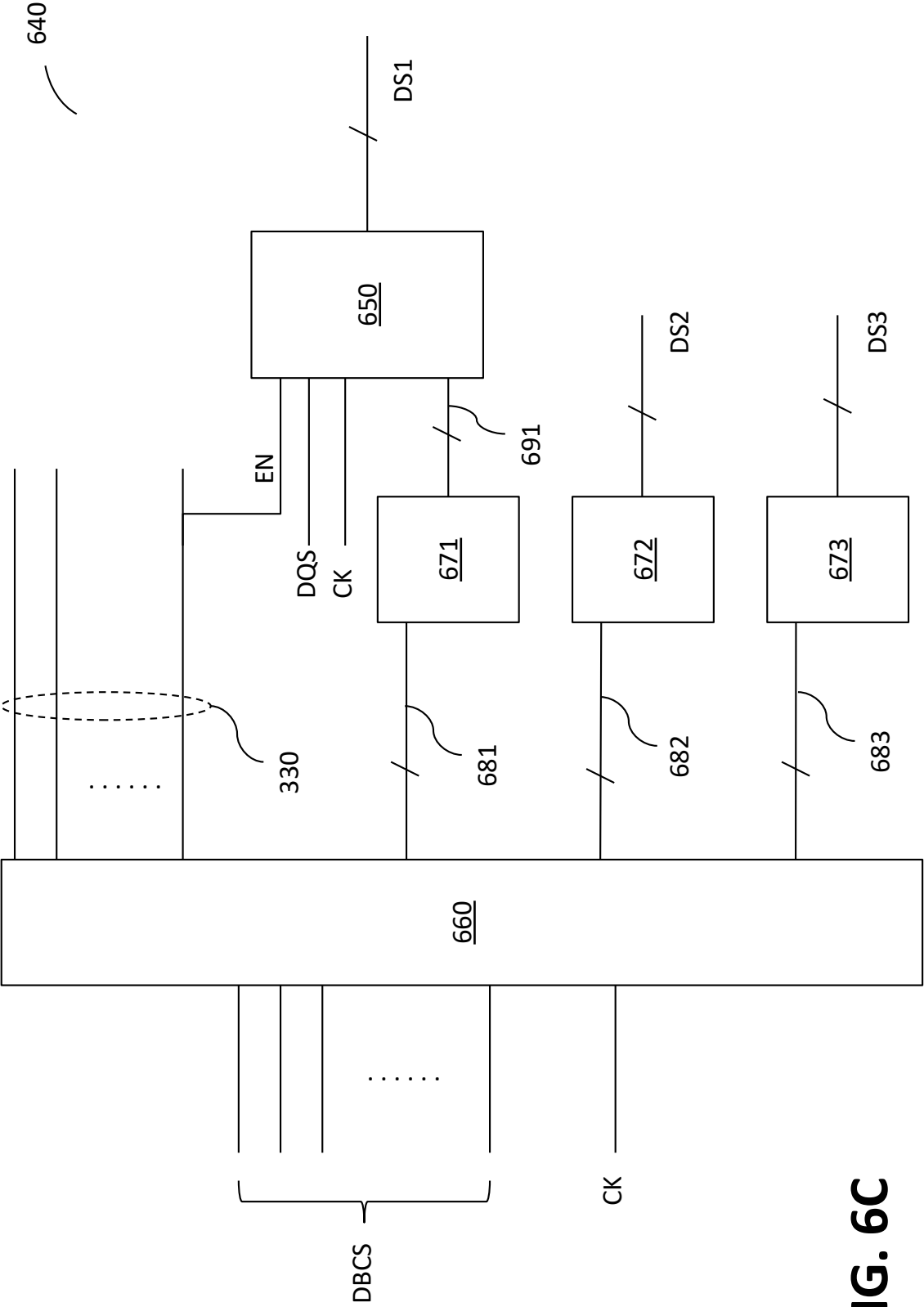


FIG. 6C

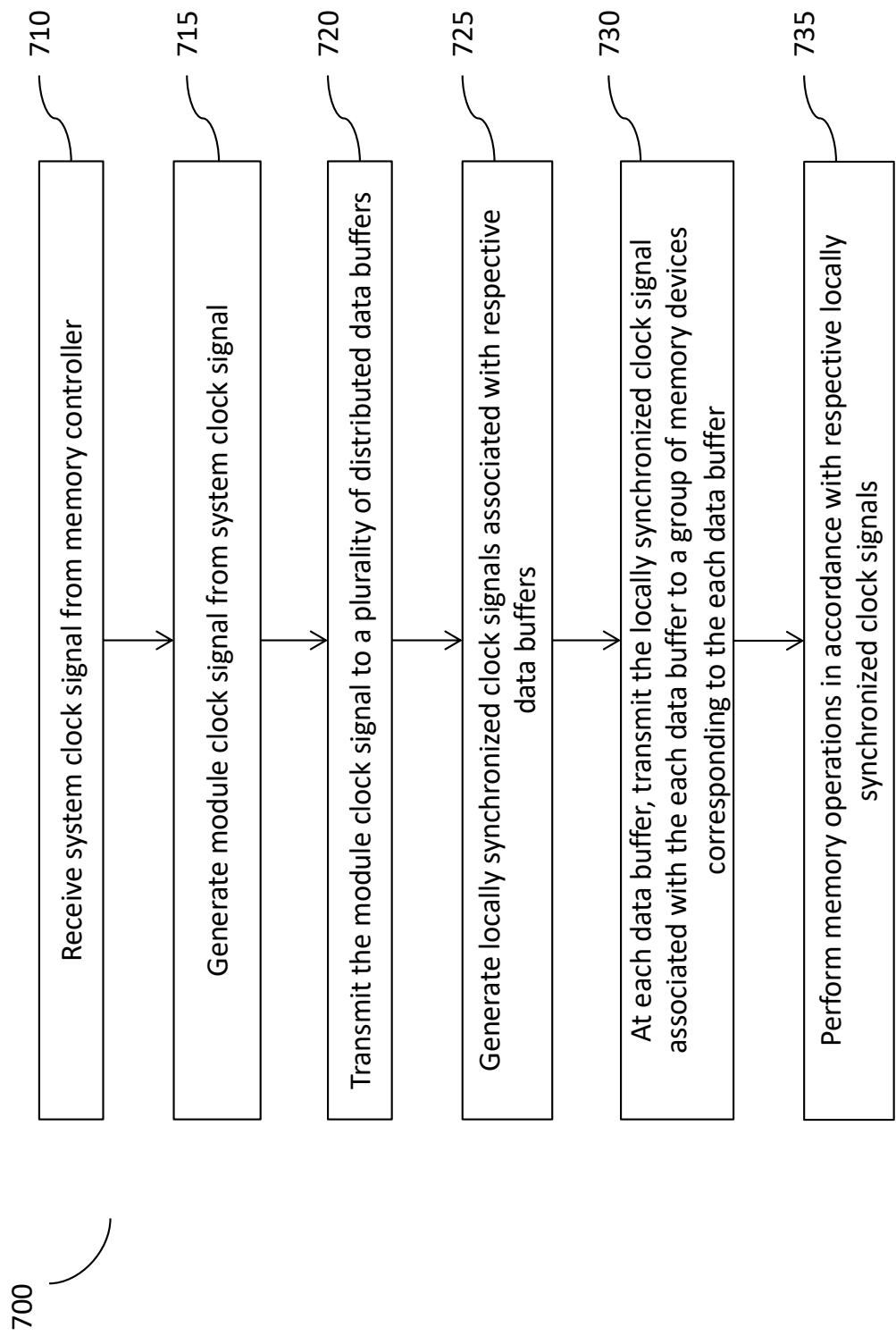


FIG. 7A

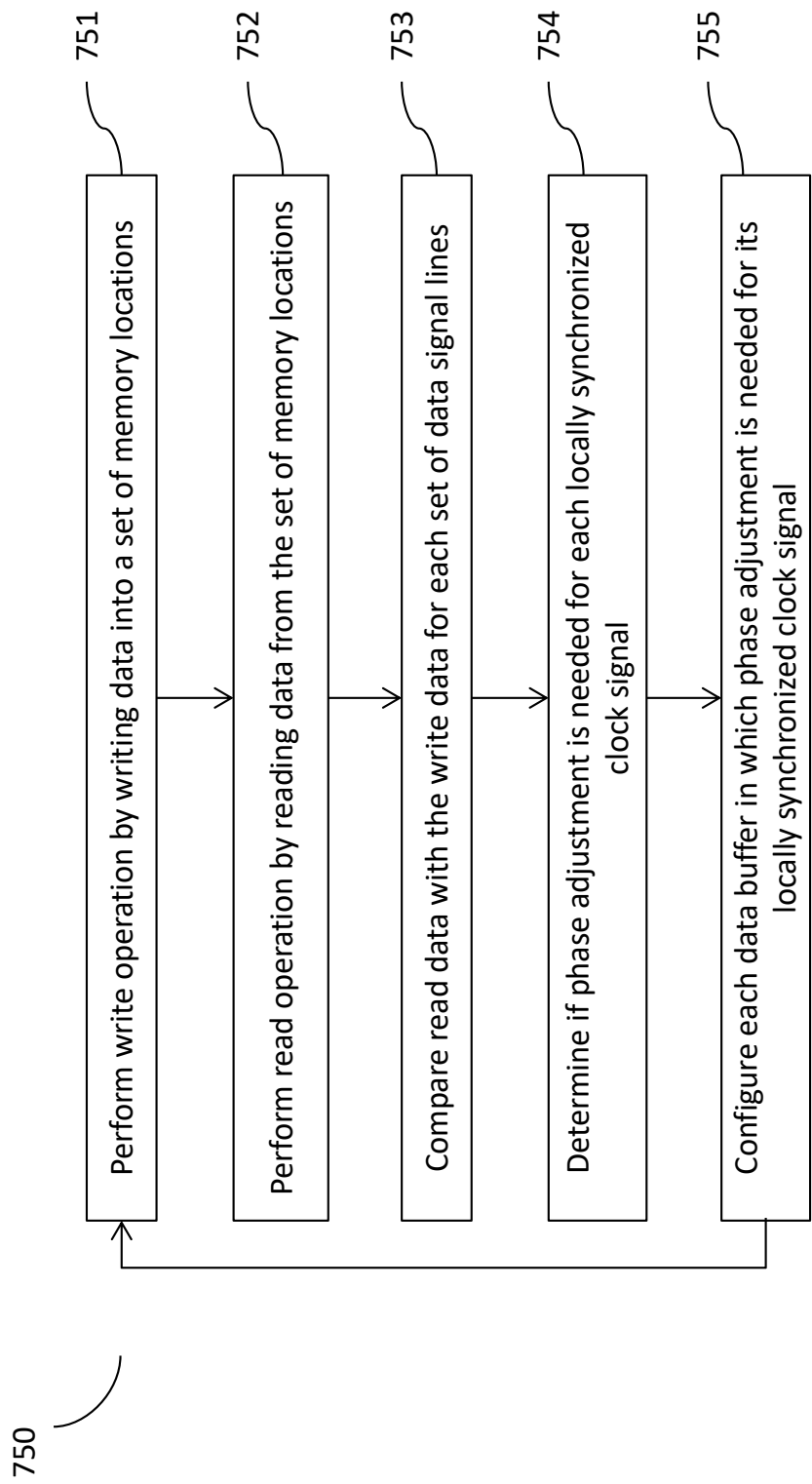


FIG. 7B

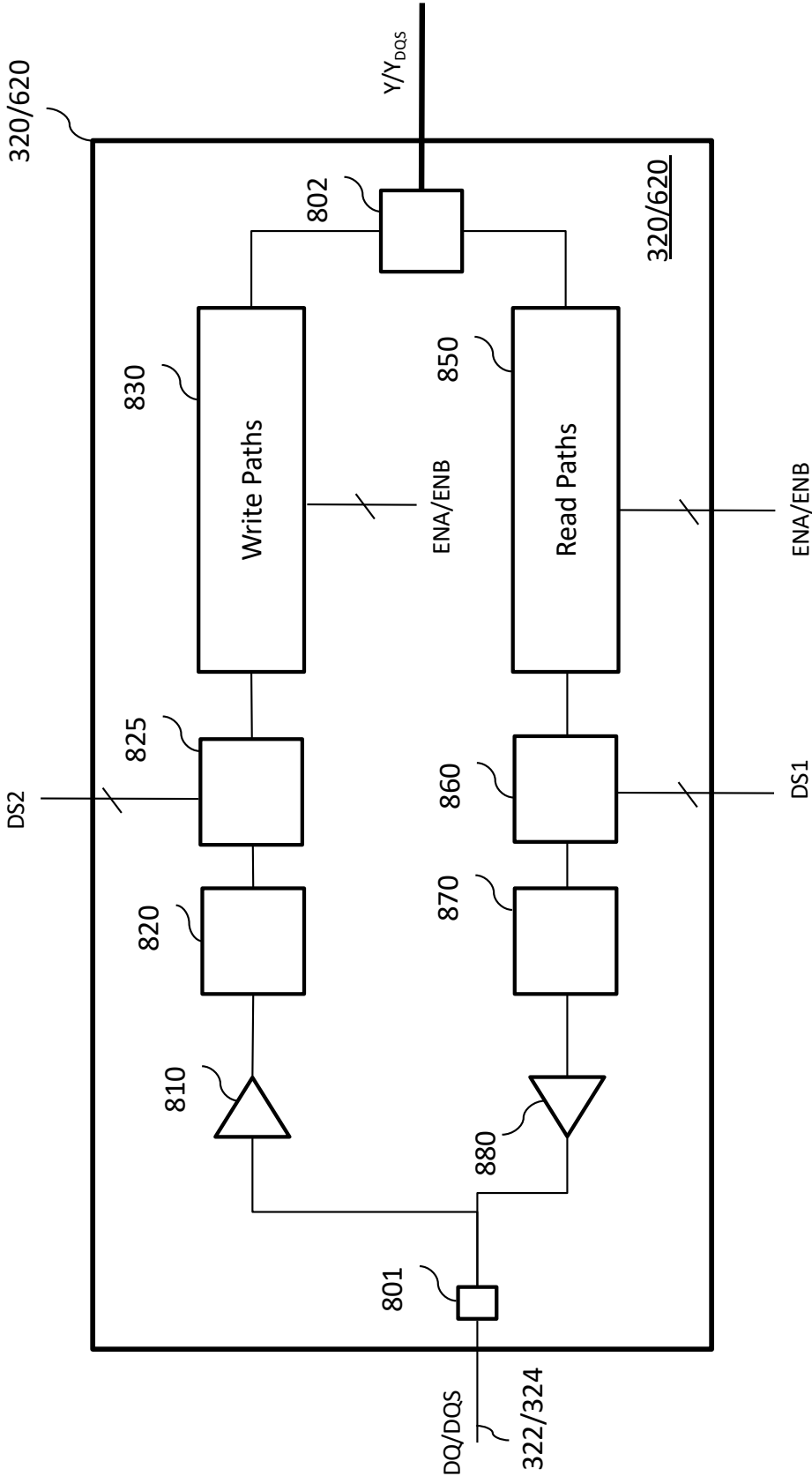


FIG. 8

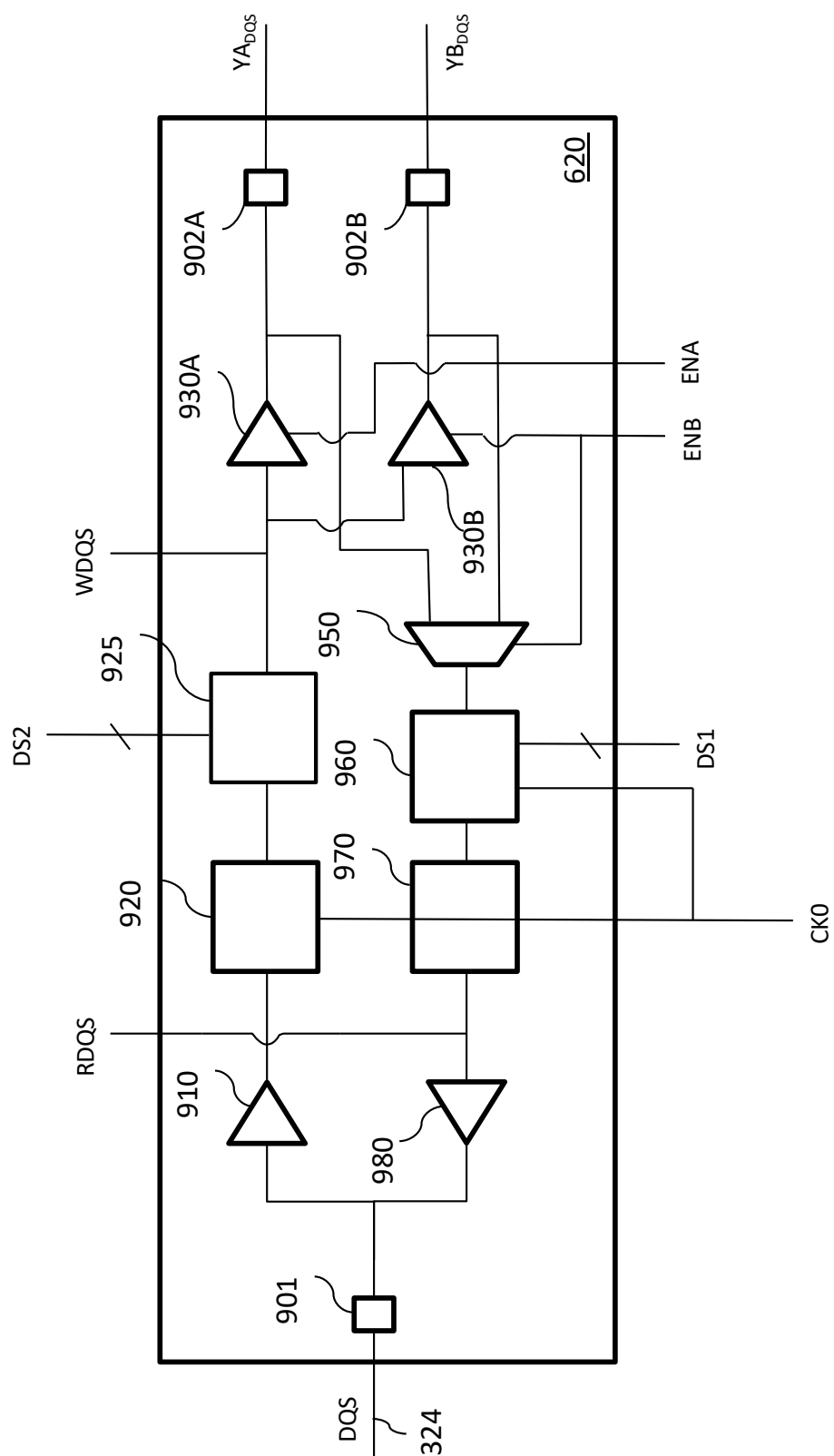


FIG. 9

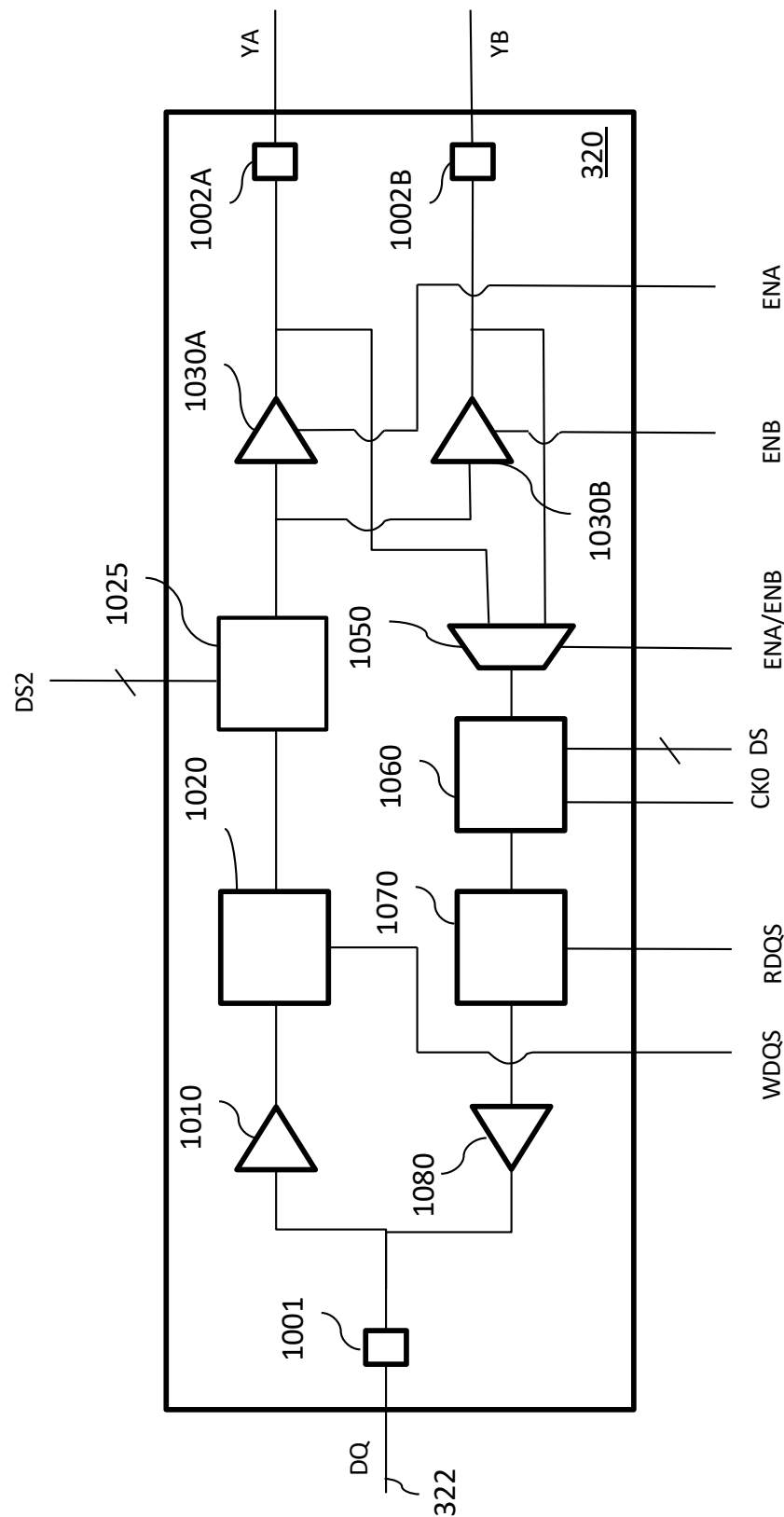


FIG. 10

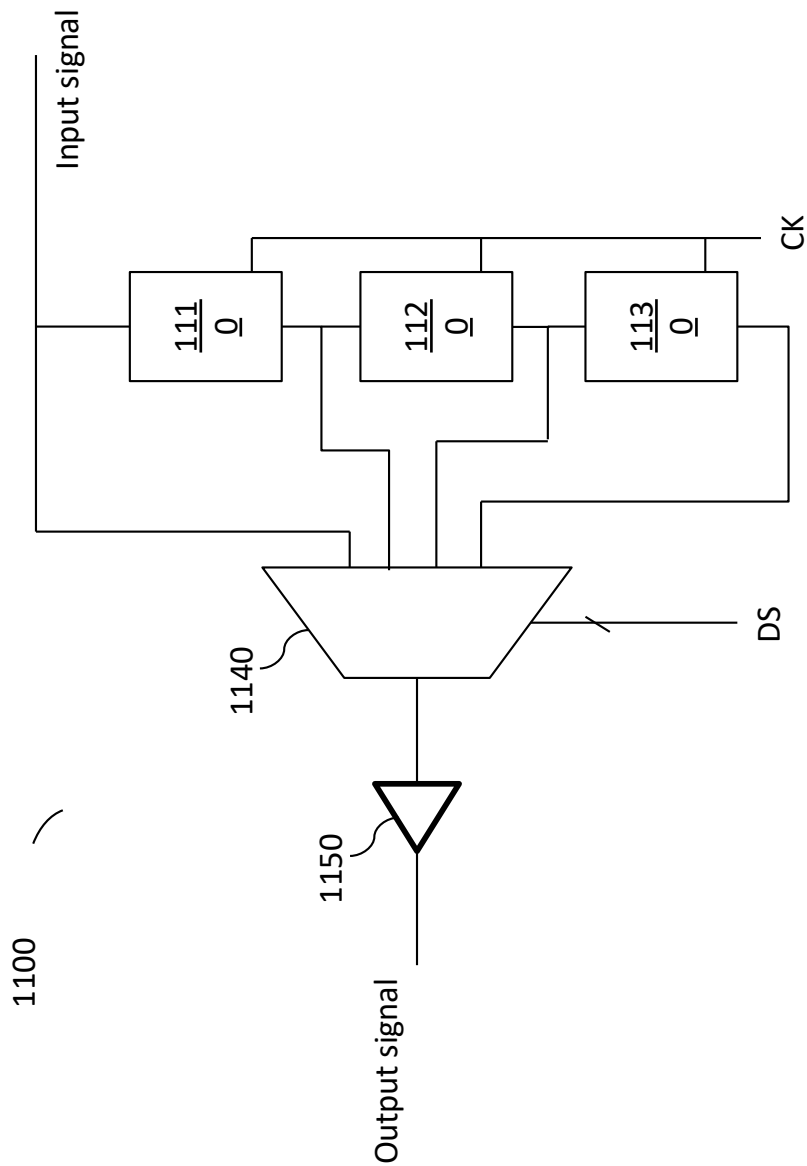


FIG. 11

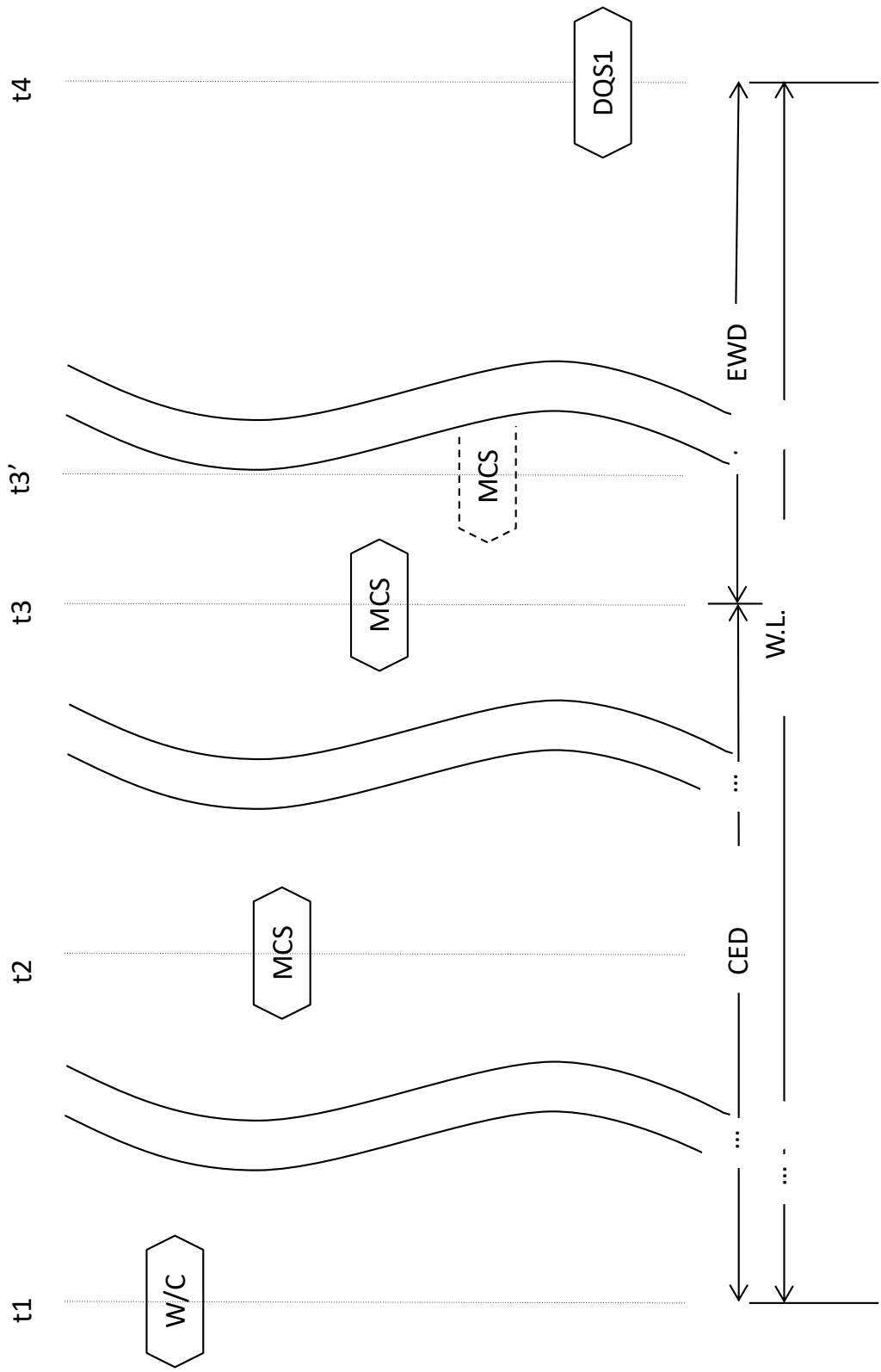


FIG. 12A

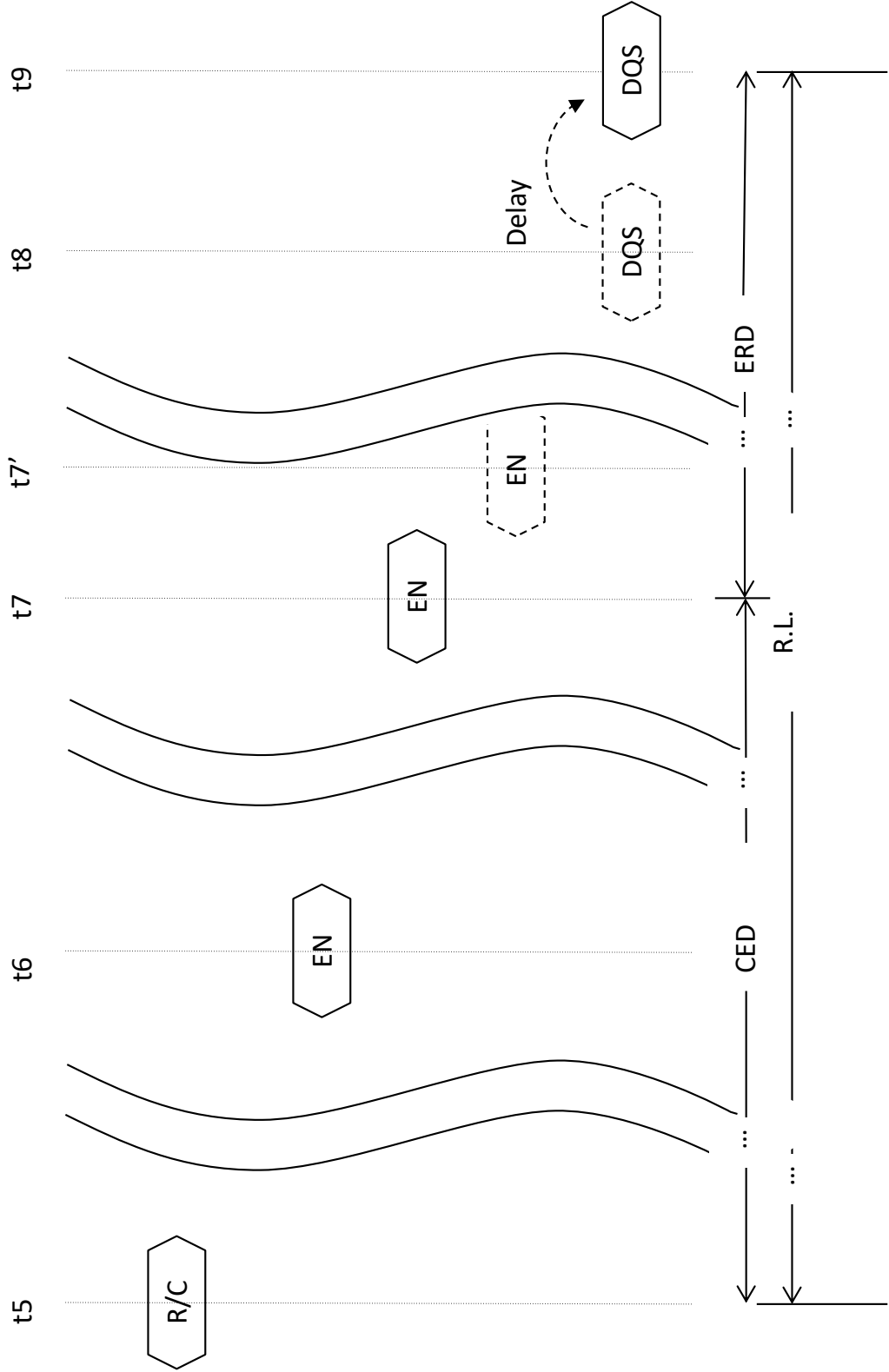


FIG. 12B

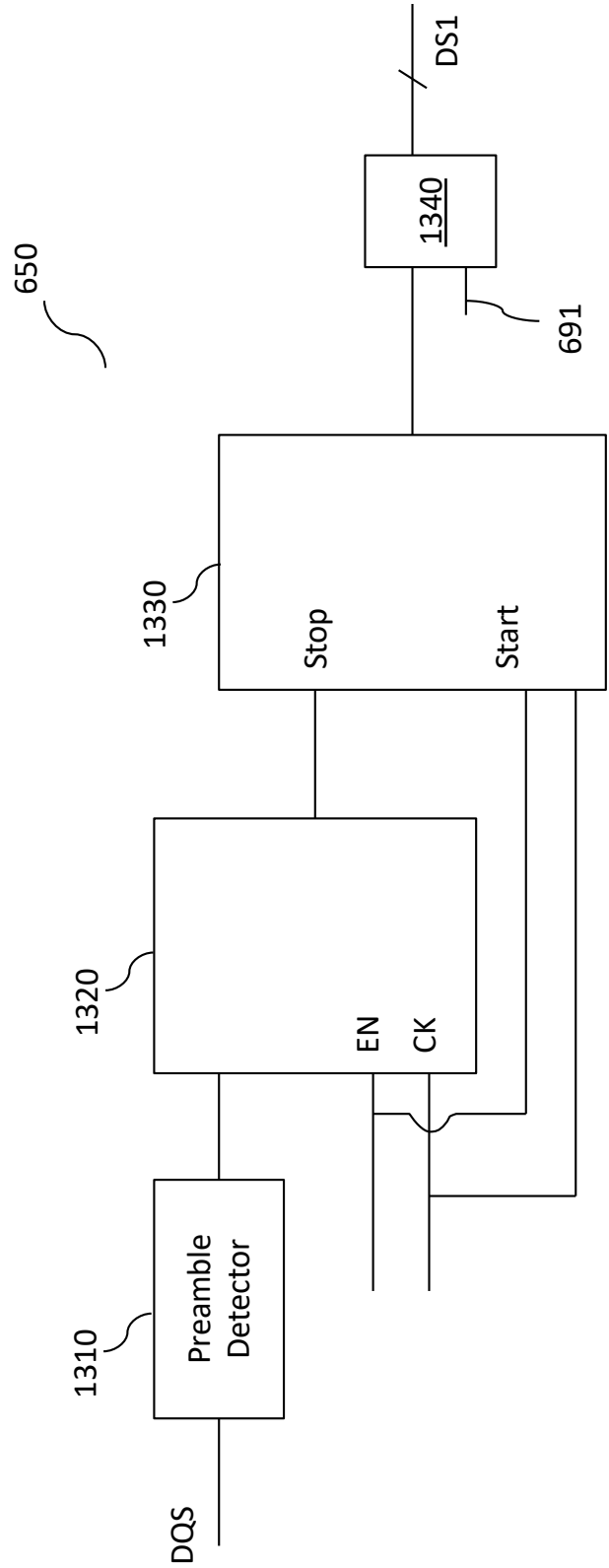


FIG. 13

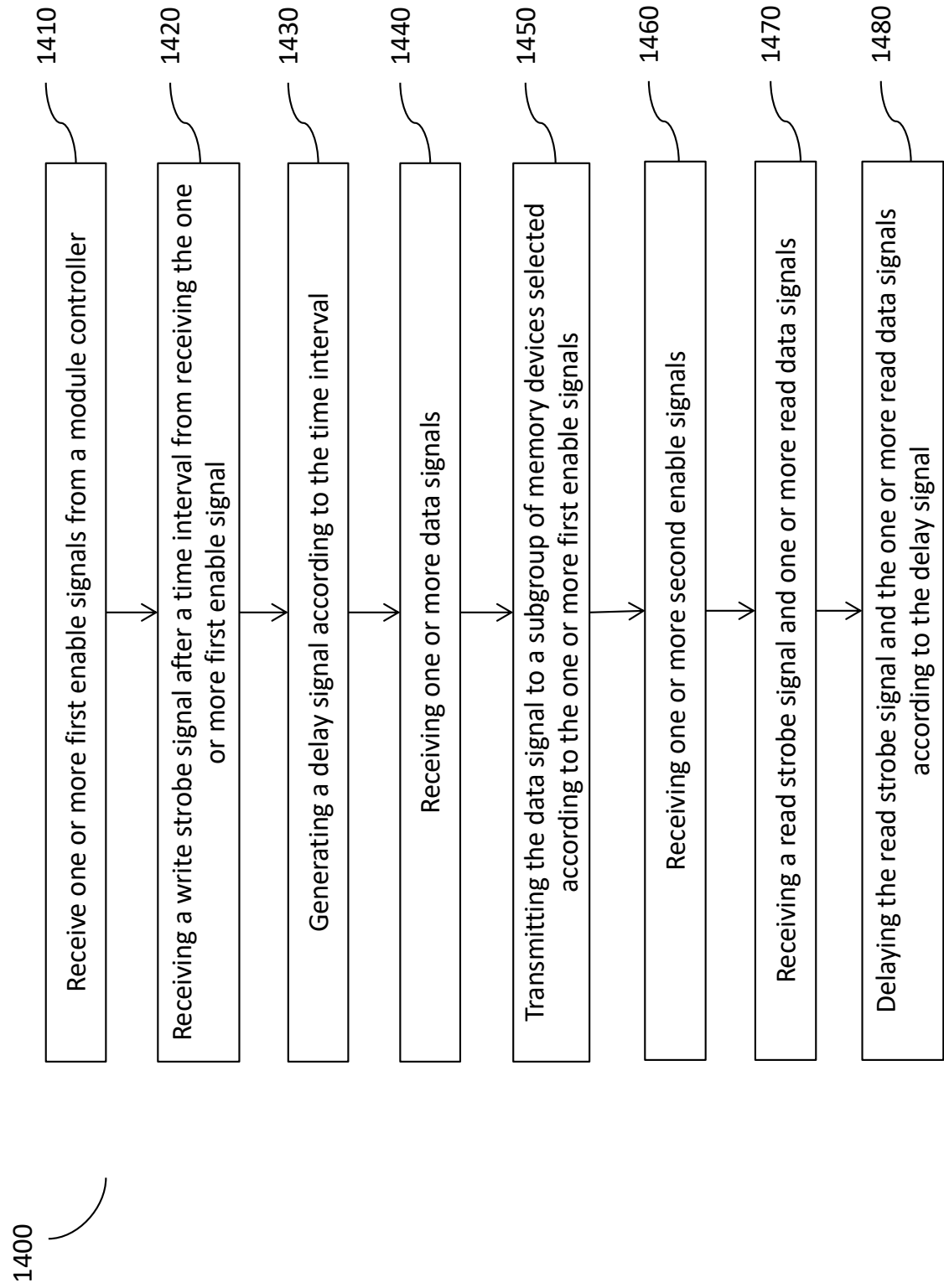


FIG. 14

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application a continuation of U.S. Patent Application Ser. No. 18/059,958, filed on November 29, 2022, entitled “Memory Module with Static and Dynamic Timing Adjustment in Data Buffers,” which is a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021, entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory

module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules.

The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 is a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 is a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer

circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical

communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to

latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer

control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory

devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected

subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such

data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the

memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very

limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the

clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not

be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or

ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines $Y-1_0$, $Y-1_1$, ..., $Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup

in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YA-1₀, YA-1₁, ..., YA-1_n and module strobe line YA-1_{DQS} and memory devices M₁₂ and M₁₄, which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YB-1₀, YB-1₁, ..., YB-1_n and module strobe line YB-1_{DQS}. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M₁₁₋₁ and M₁₁₋₂ may be used in place of the memory device M₁₁. Each of the two memory devices M₁₁₋₁ and M₁₁₋₂ has a data width of 4, and together they act like a memory device M₁₁ of a data width of 8. Thus, memory device M₁₁₋₁ is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory circuit M₁₁₋₂ is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M₁₁₋₁ to M₁₁₋₄ may be used as the memory device M₁₁. Each of the four memory devices M₁₁₋₁ to M₁₁₋₄ has a data width of 4, and together they act like a memory device M₁₁ of a data width of 16. Thus, memory device M₁₁₋₁ is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory device M₁₁₋₂ is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain

embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118 based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines $Y_{A_{DQS}}$ and $Y_{B_{DQS}}$. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read

paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 681 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 691 to the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the data path enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock

signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment.

As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS} , or YA/YA_{DQS} and YB/YB_{DQS} . The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825 controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830 that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line YA_{DQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line YB_{DQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line YA_{DQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay

circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line YB_{DQS}. The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line YA_{DQ}, a third DQ pin 1002B that is coupled to a corresponding module DQ line YB_{DQ}. The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line YA_{DQ}. The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second

write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line YB_{DQ}. The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having

programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different

cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the

one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of

memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide connections between the memory module and the system bus;

memory devices mounted on the PCB and organized in a plurality of groups, a respective group of the memory devices corresponding to a respective set of the plurality of sets of data/strobe signal lines;

circuits mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines;

generate a module clock signal and module C/A signals in response to the system clock and input C/A signals;

generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and

output the module C/A signals to the memory devices; and

output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the memory devices, wherein each memory device in the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines.

2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.

3. The memory module of claim 2, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in a plurality of groups, and circuits coupled to the memory devices. The circuits including circuitry configurable to receive from the memory controller a system clock and input control and address (C/A) signals, and to generate a module clock signal and module C/A signals in response to the system clock and input C/A signals. The circuits further include circuitry configurable to generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and to output the plurality of local clock signals to respective groups of the memory devices. A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices.



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NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Jamie Zheng

PATENT CENTER # 67819552

FILING DATE -

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun Lee

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 8

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073-20US04_ADS.pdf	9	Application Data Sheet	2174 KB
NL073-20US04_Appl_Transmittal.pdf	1	Transmittal of New Application	98 KB
NL073-20US03_Oath_Assign.pdf	2	Oath or Declaration filed	73 KB
NL073-20US04_373-Stmnt.pdf	1	Assignee showing of ownership per 37 CFR 3.73	58 KB
POA_Netlist_General.pdf	1	Power of Attorney	148 KB
NL073-20US04_Drawings.pdf	22	Drawings-only black and white line drawings	269 KB
NL073-	32	Application body structured	78 KB

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NL073- 20US04_Specification.pdf	32	Auxiliary PDF of Application	304 KB
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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Inventor Information:

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Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
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Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
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Inventor 2					Remove	
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	Jayesh		Bhakta			
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Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

☐ An Address is being provided for the correspondence information of this application.

Customer Number	79141		
Email Address	jzheng@usclaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		
Attorney Docket Number	NL073-20US04	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	6B

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending		Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
	Continuation of	18/059958	2022-11-29		
Prior Application Status	Patented		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/059958	Continuation of	17/141978	2021-01-05	11513955	2022-11-29
Prior Application Status	Patented		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/141978	Continuation of	16/432700	2019-06-05	10884923	2021-01-05
Prior Application Status	Patented		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
16/432700	Continuation of	14/445035	2014-07-28	10324841	2019-06-18
Prior Application Status	Expired		Remove		
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14/445035	Claims benefit of provisional	61/859215	2013-07-27		
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This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Remove
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

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Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

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A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

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☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information For Applicant:			
Address 1	111 Academy Way, Suite 100		
Address 2			
City	Irvine	State/Province	CA
Country	US	Postal Code	92617
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

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If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

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City			State/Province	
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This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Jamie J. Zheng/		Date (YYYY-MM-DD)	2024-11-01	
First Name	Jamie J.	Last Name	Zheng	Registration Number	51167

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US04
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

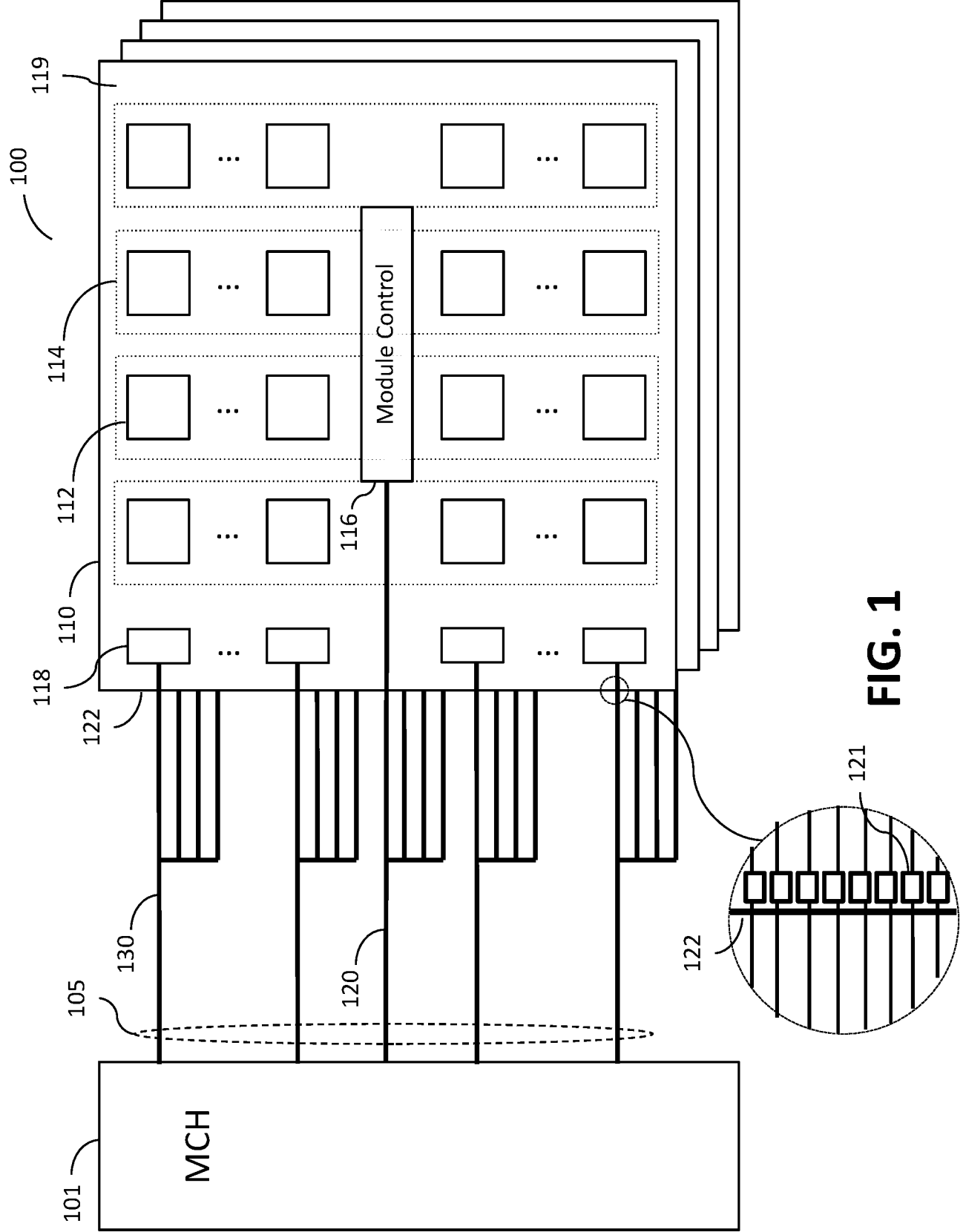


FIG. 1

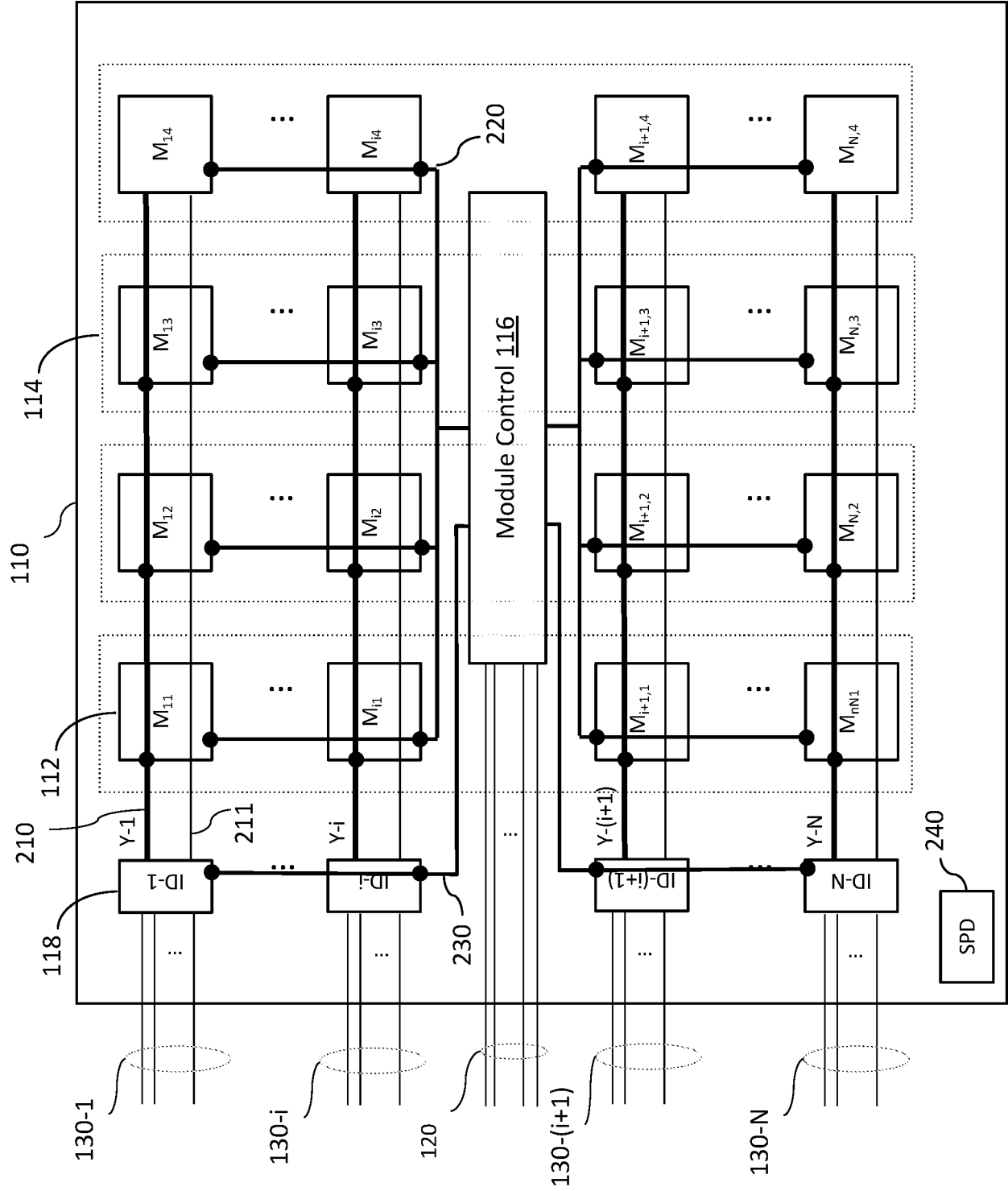


FIG. 2A

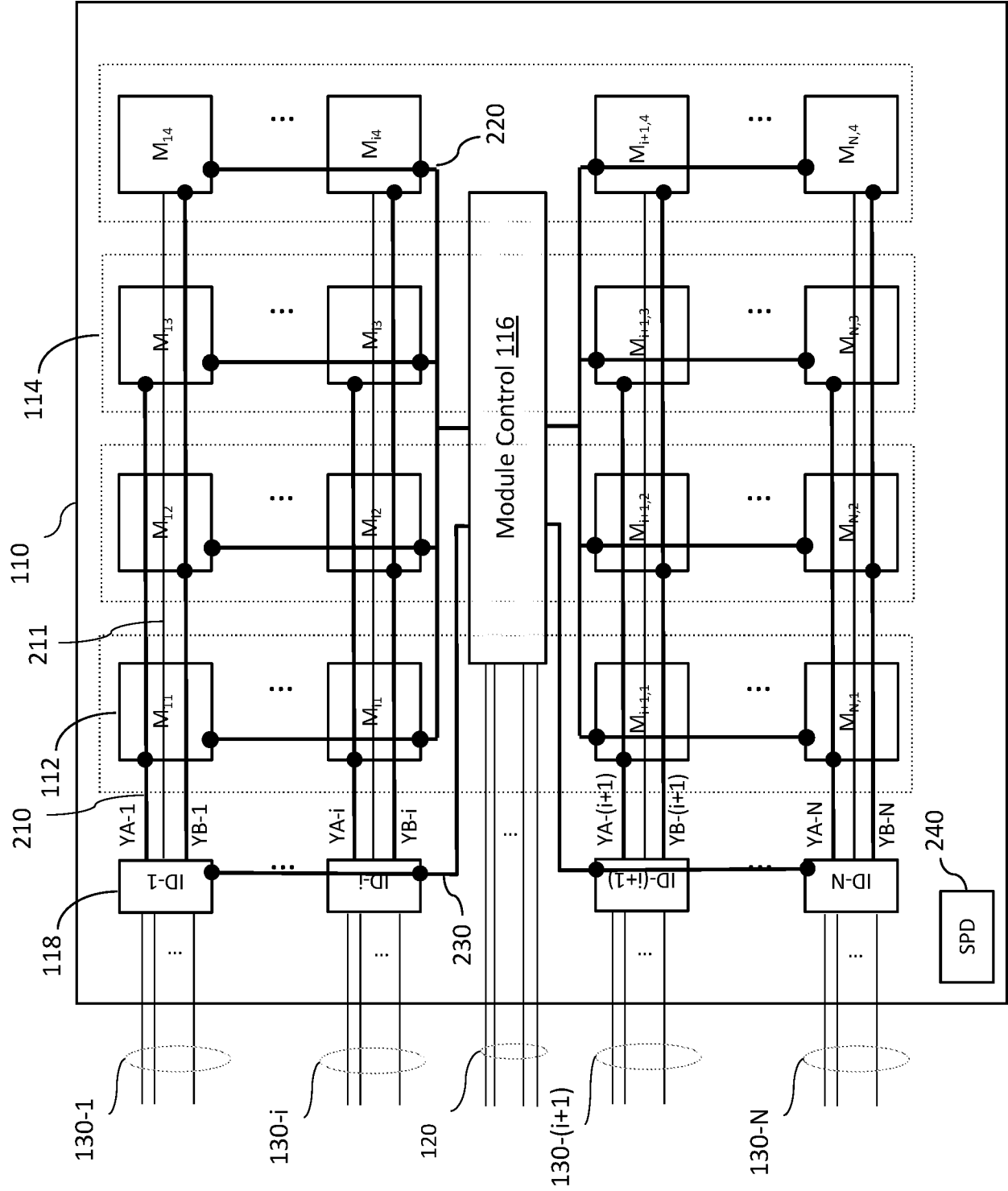


FIG. 2B

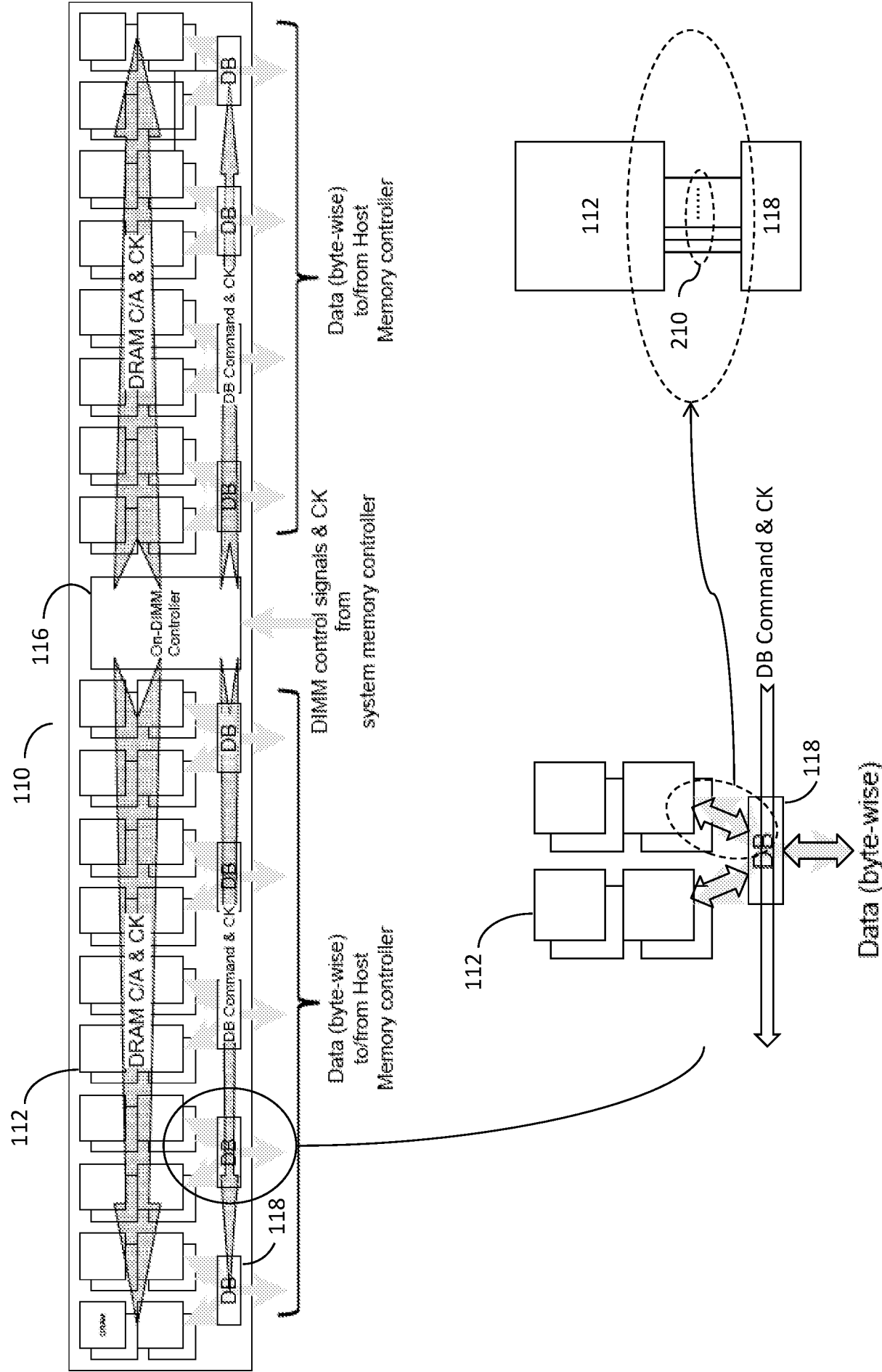


FIG. 2C

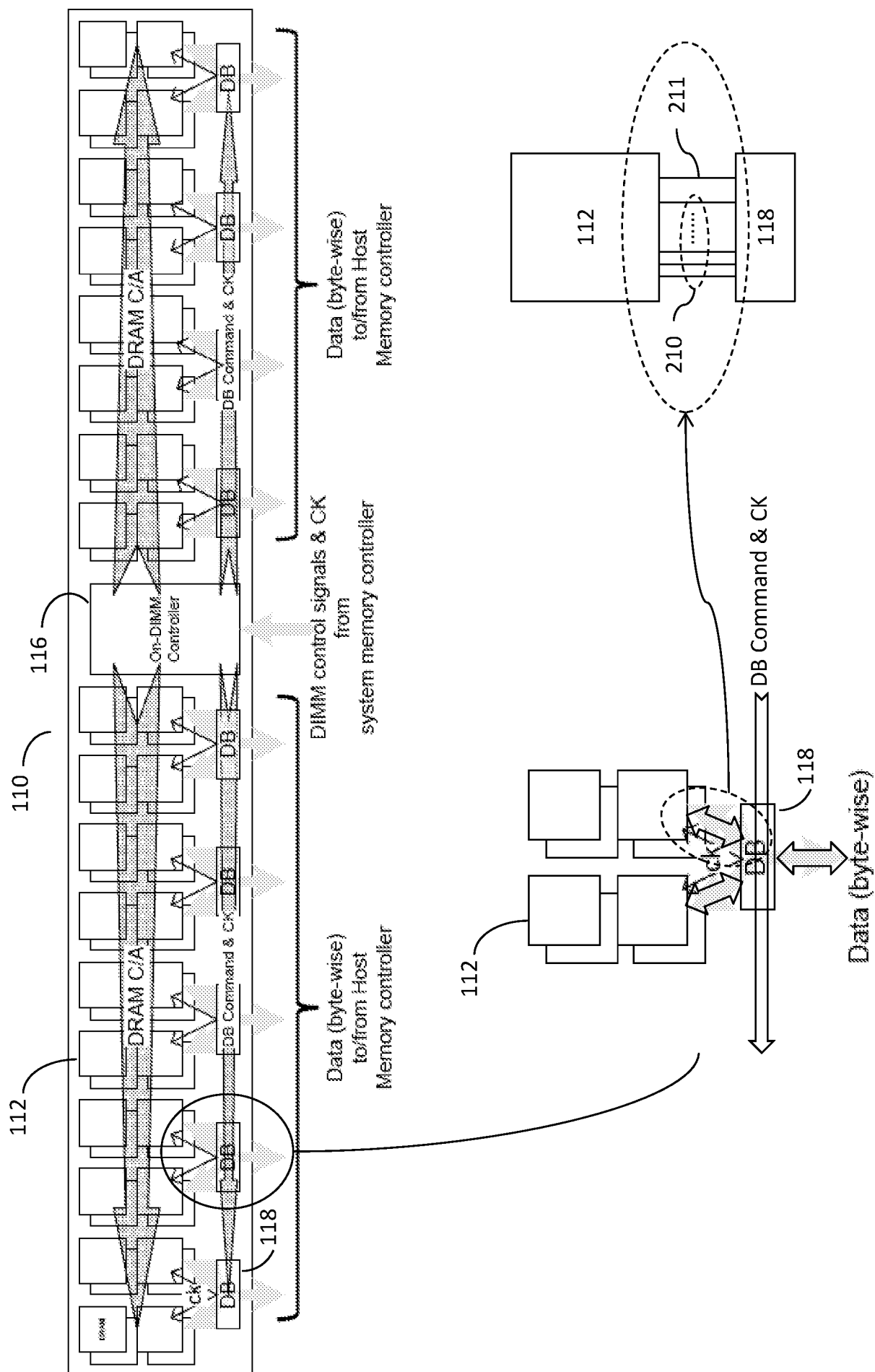


FIG. 2D

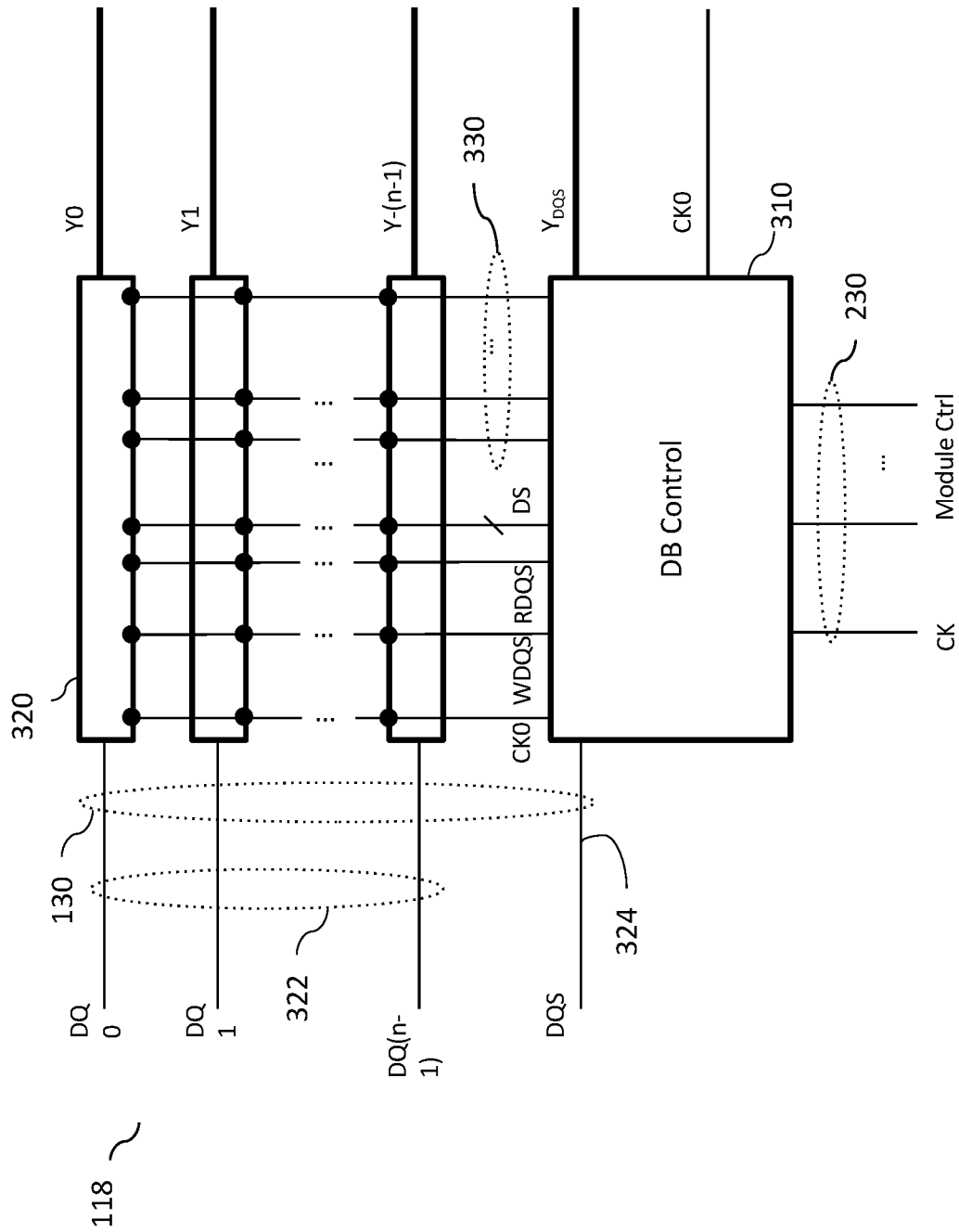


FIG. 3

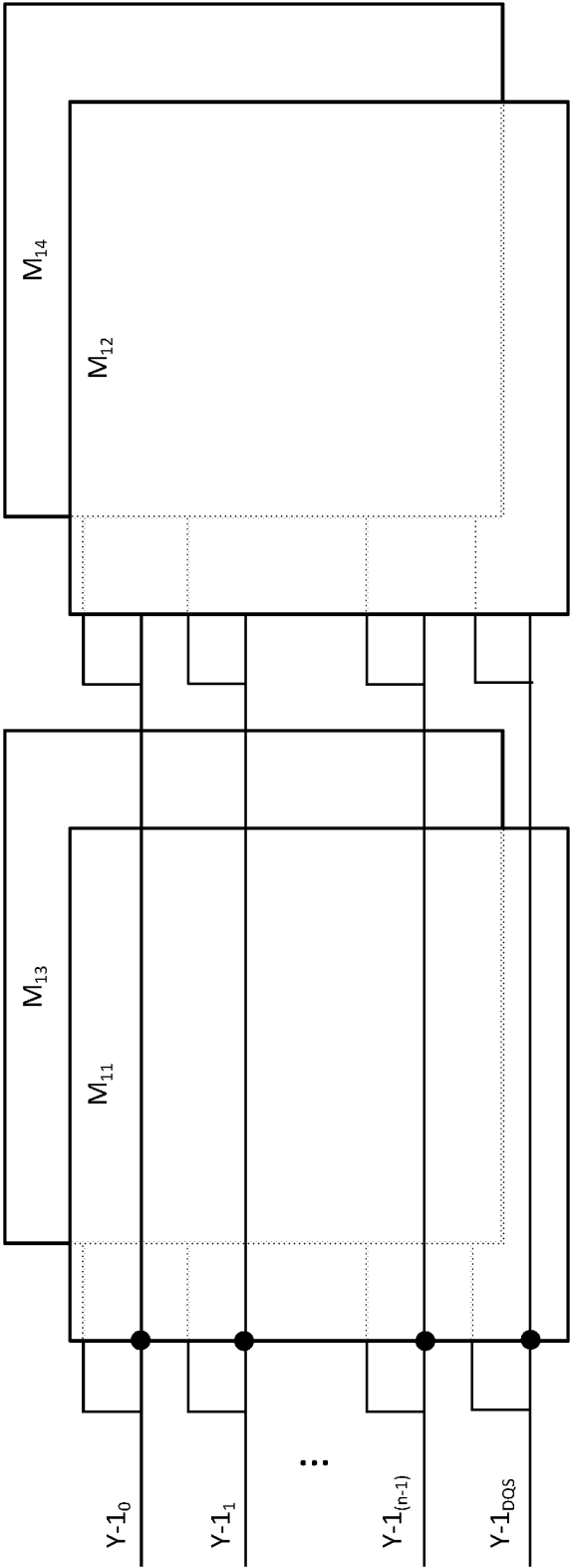


FIG. 4A

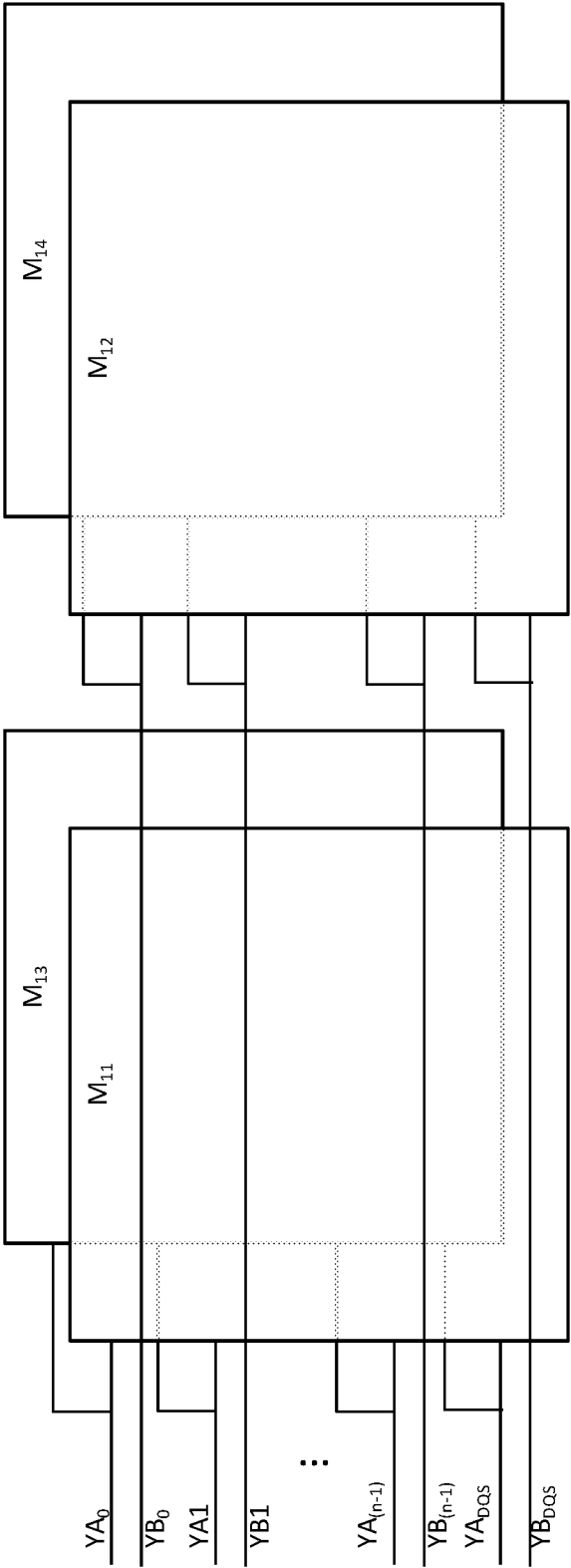


FIG. 4B

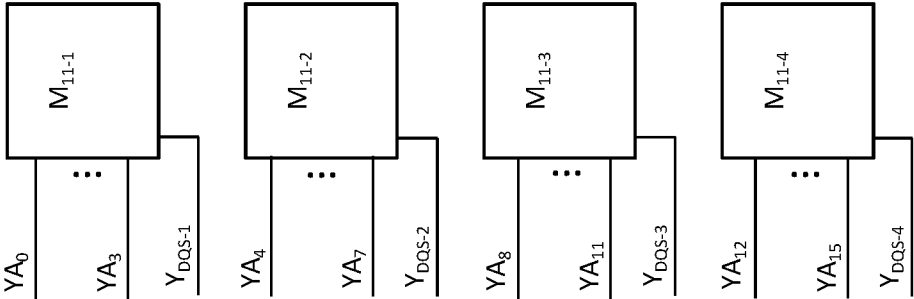


FIG. 5B

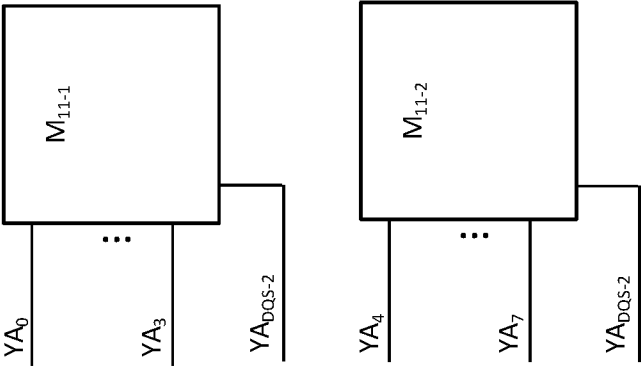


FIG. 5A

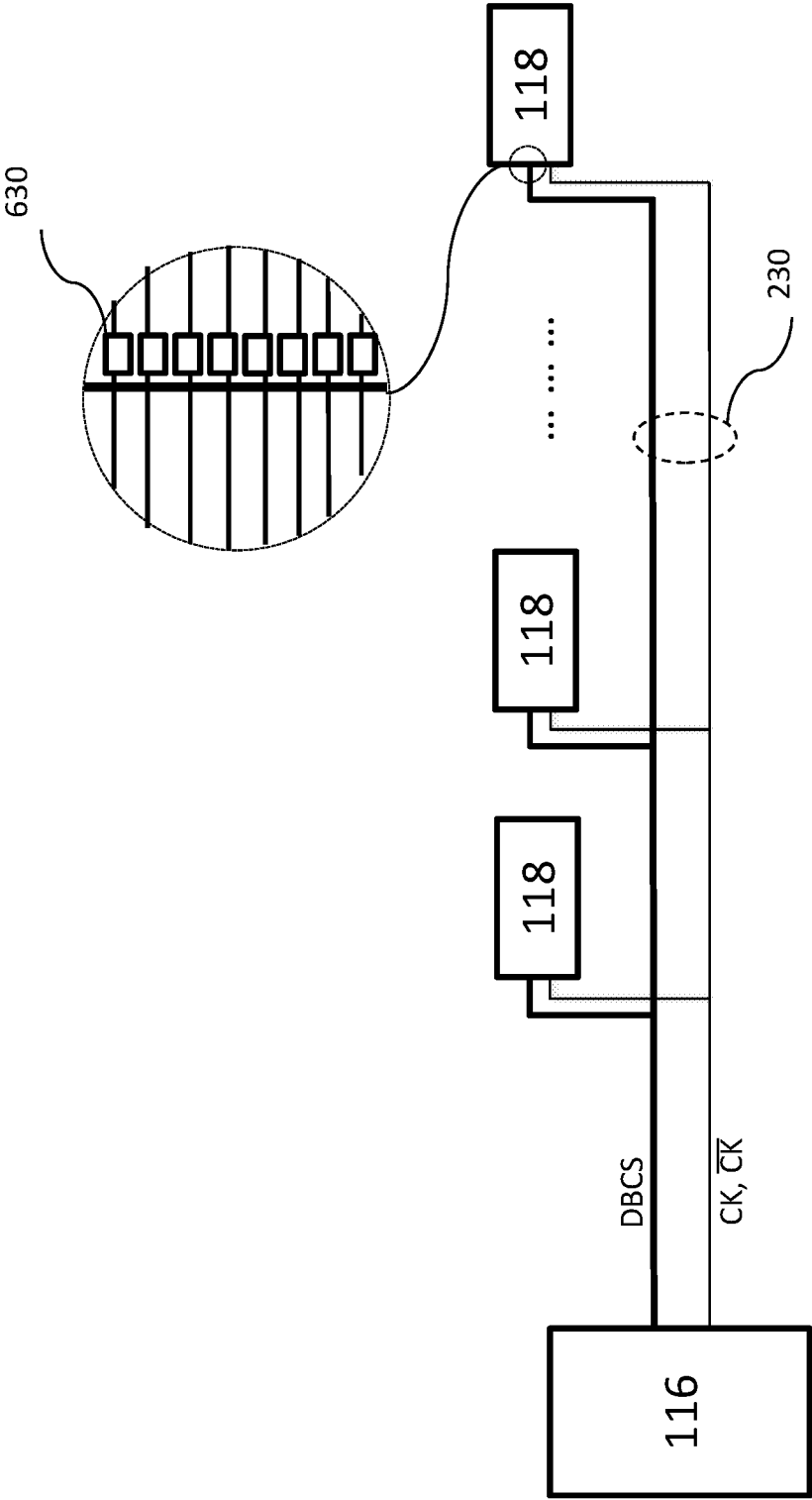


FIG. 6A

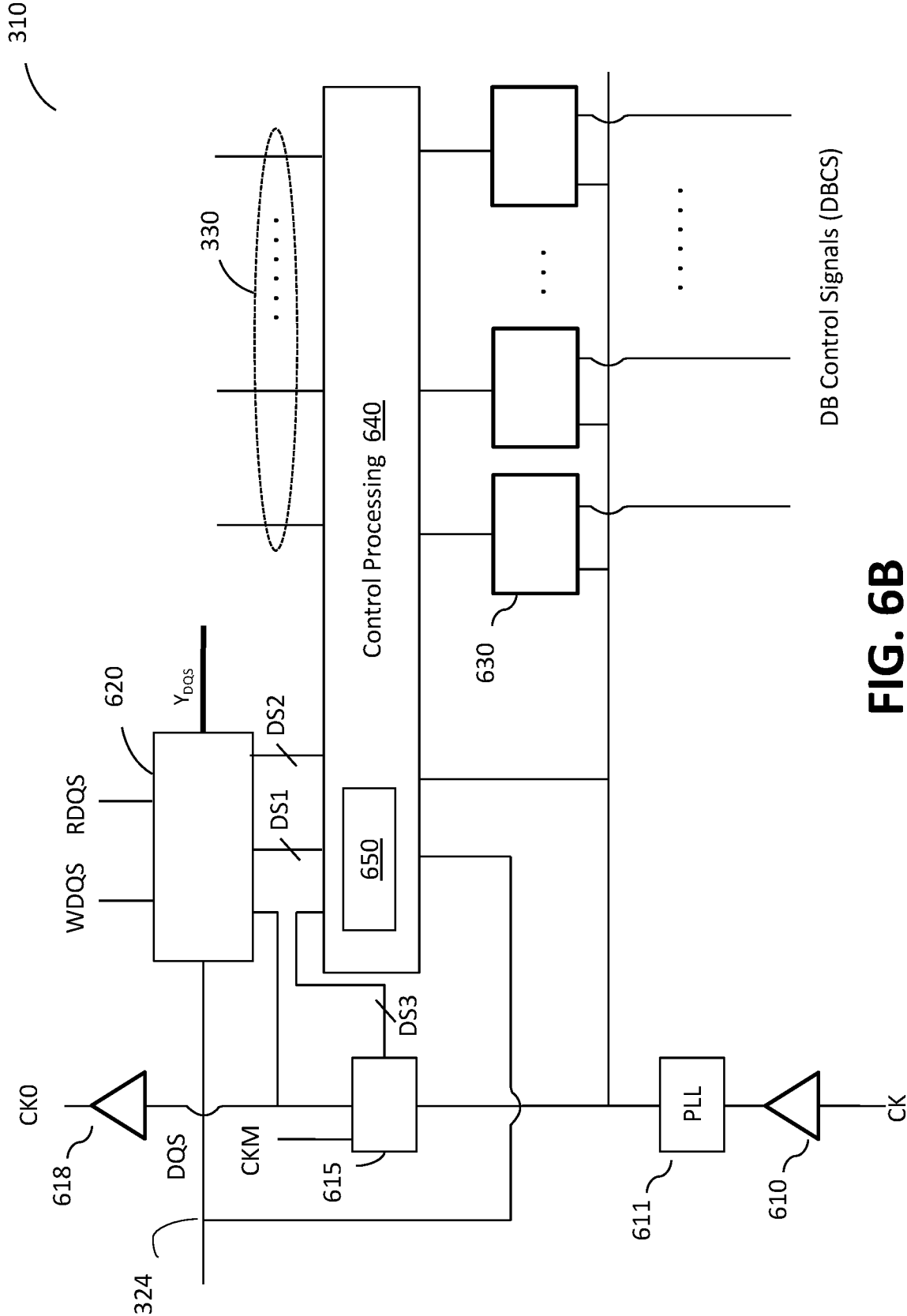


FIG. 6B

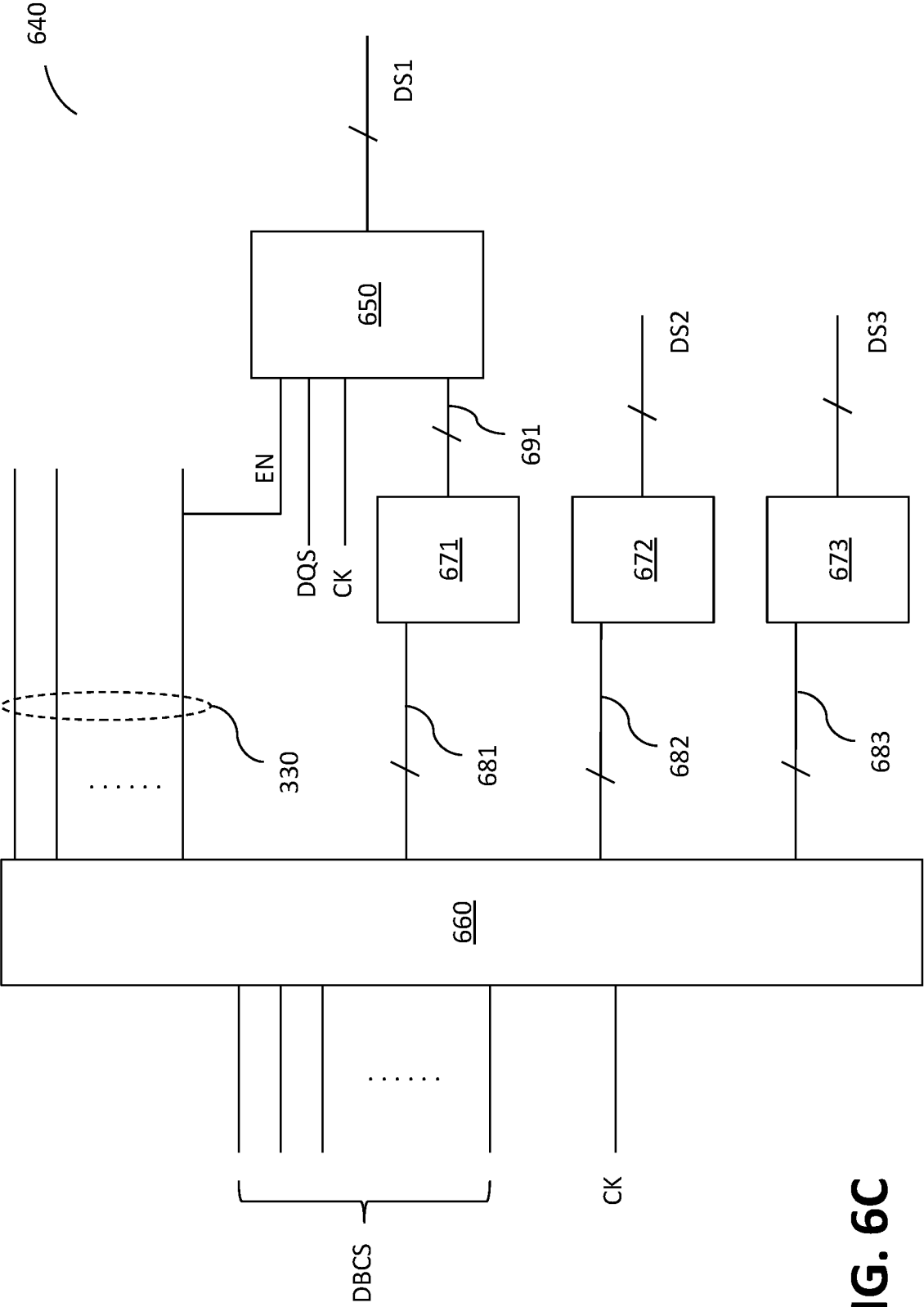


FIG. 6C

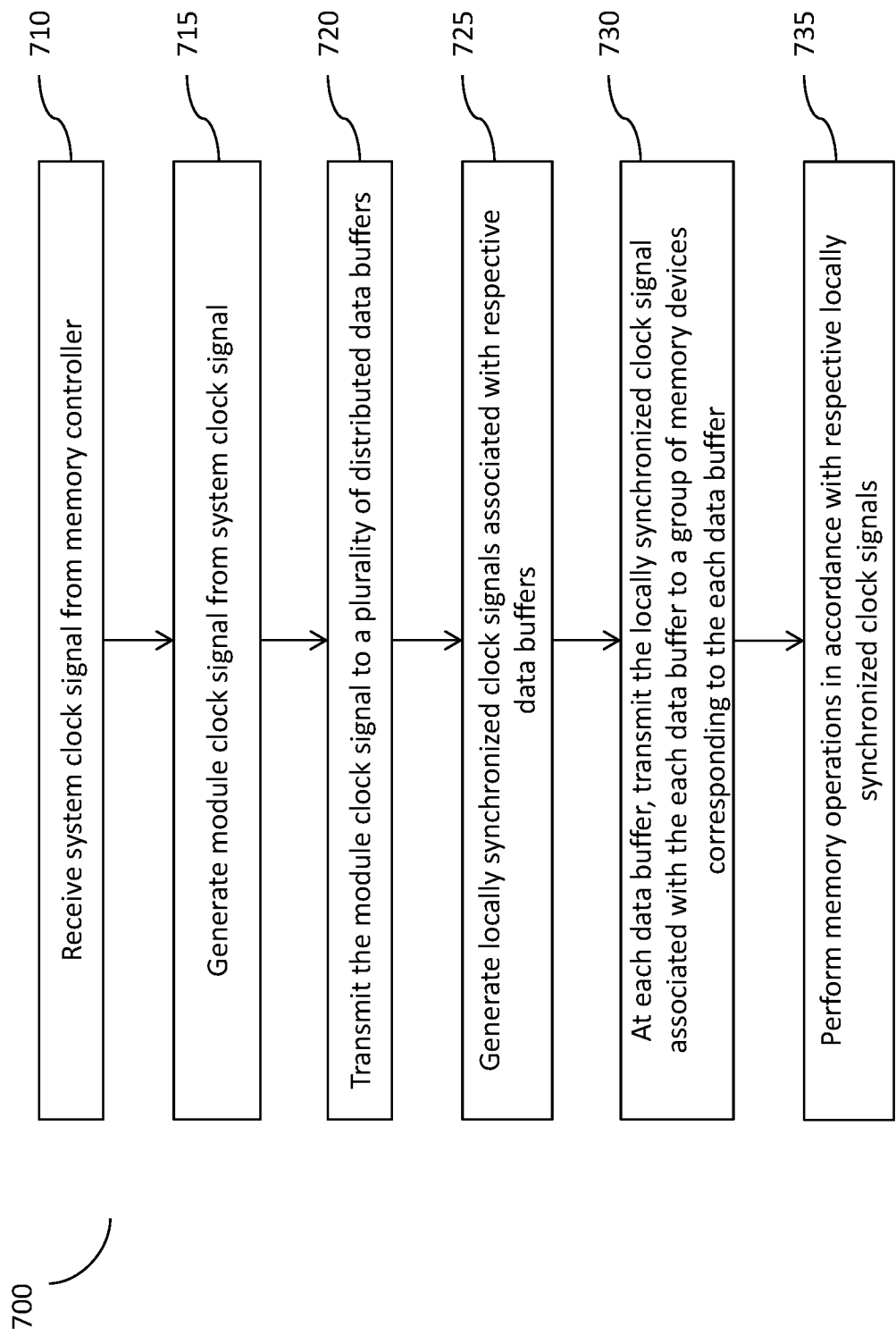


FIG. 7A

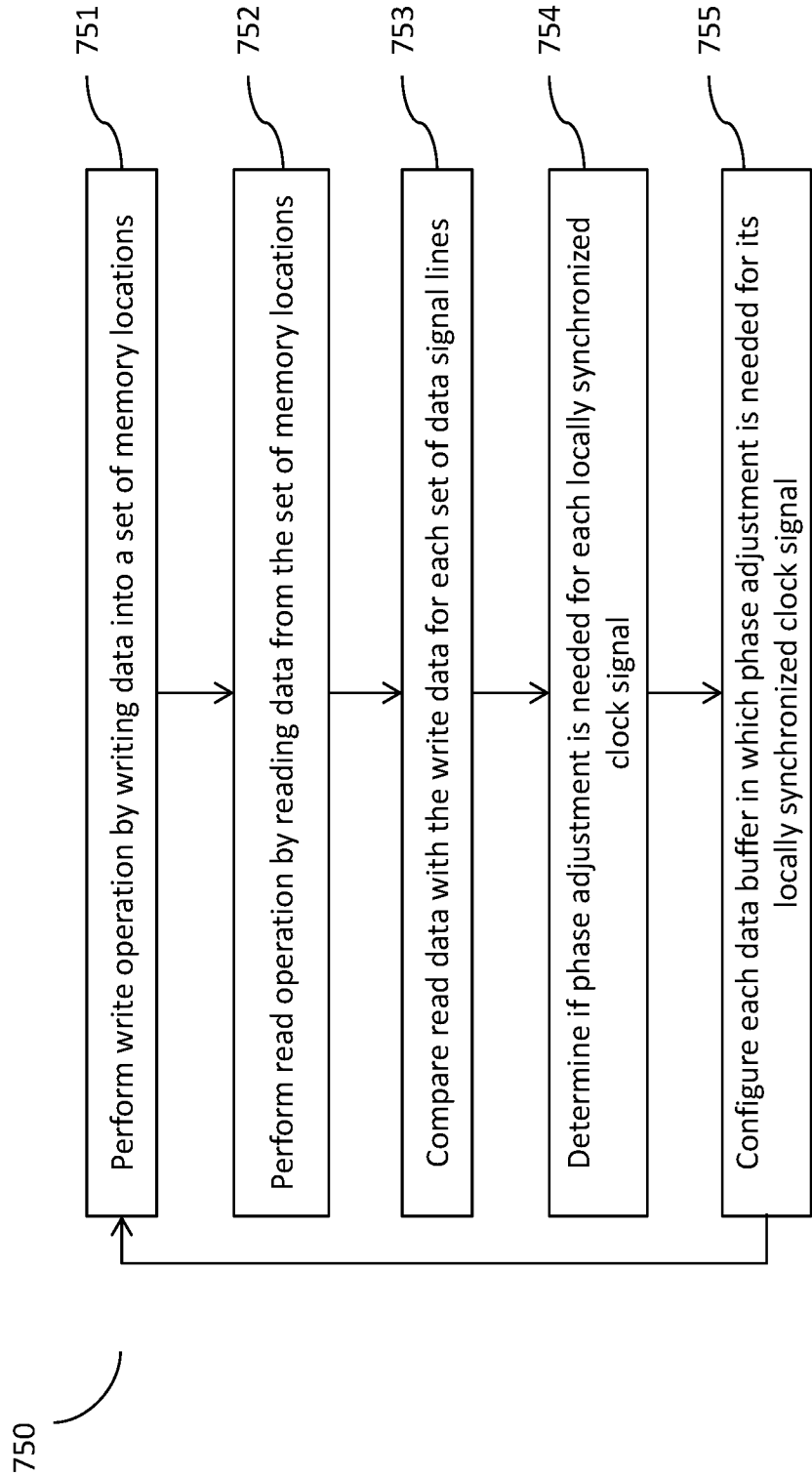


FIG. 7B

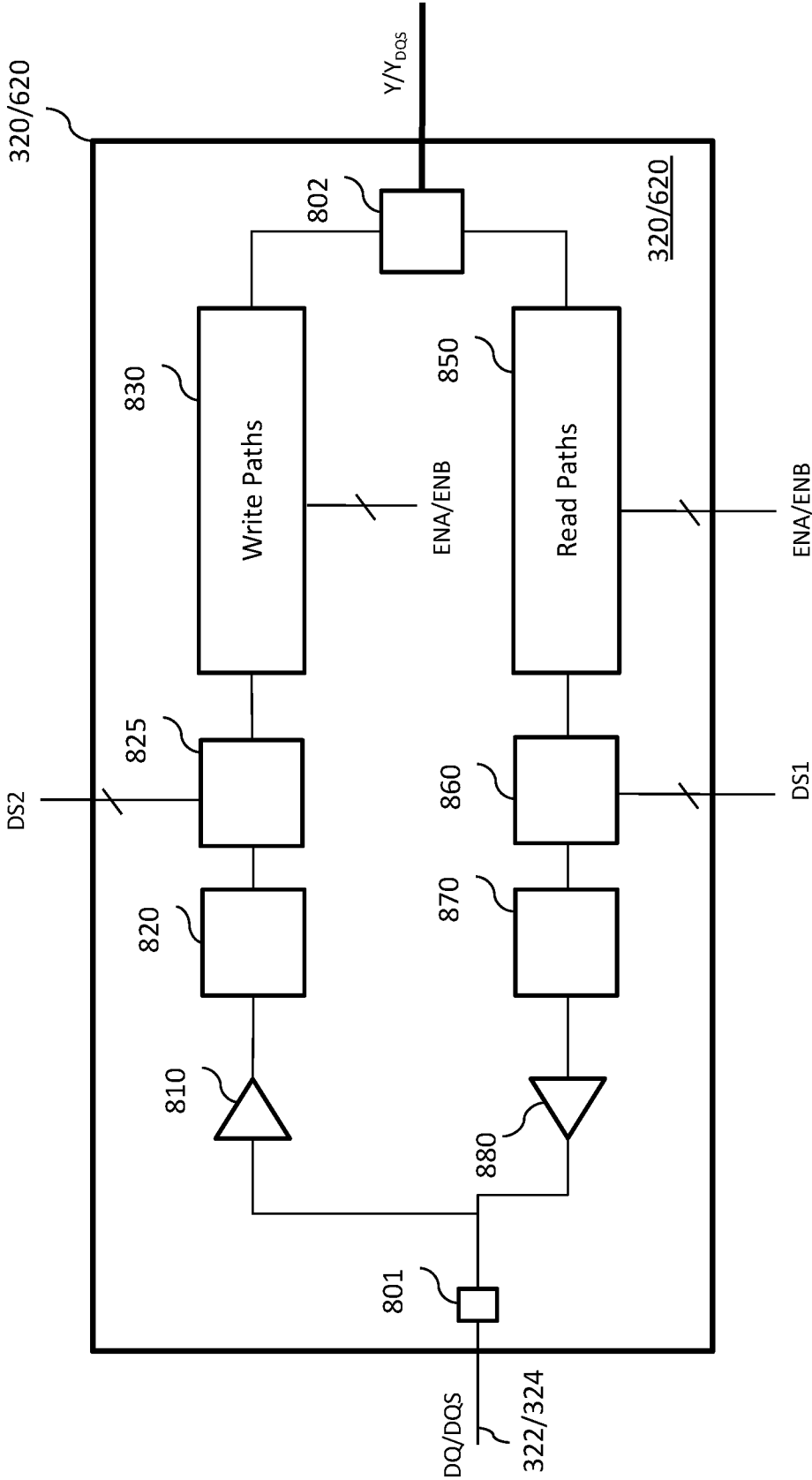


FIG. 8

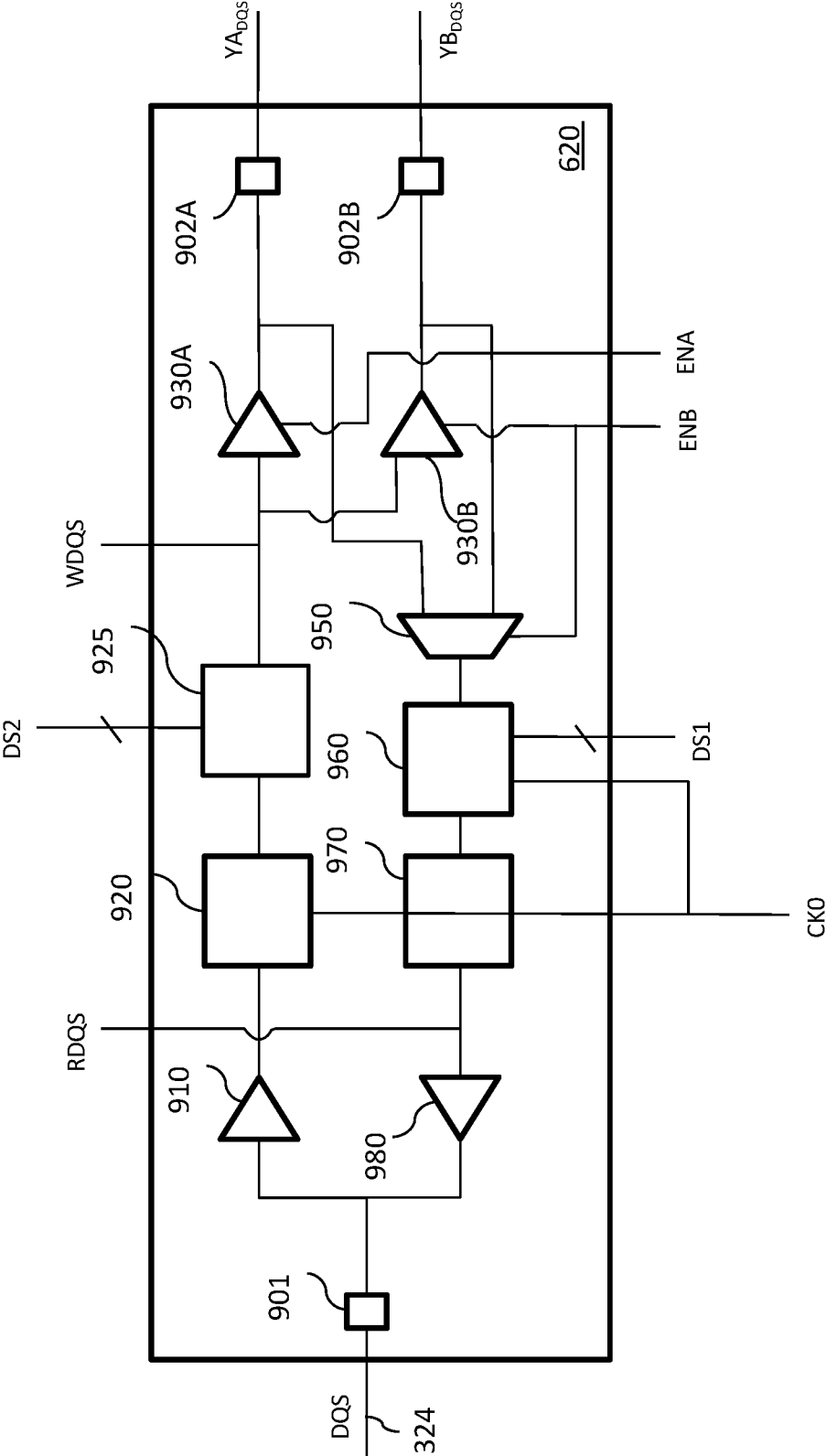


FIG. 9

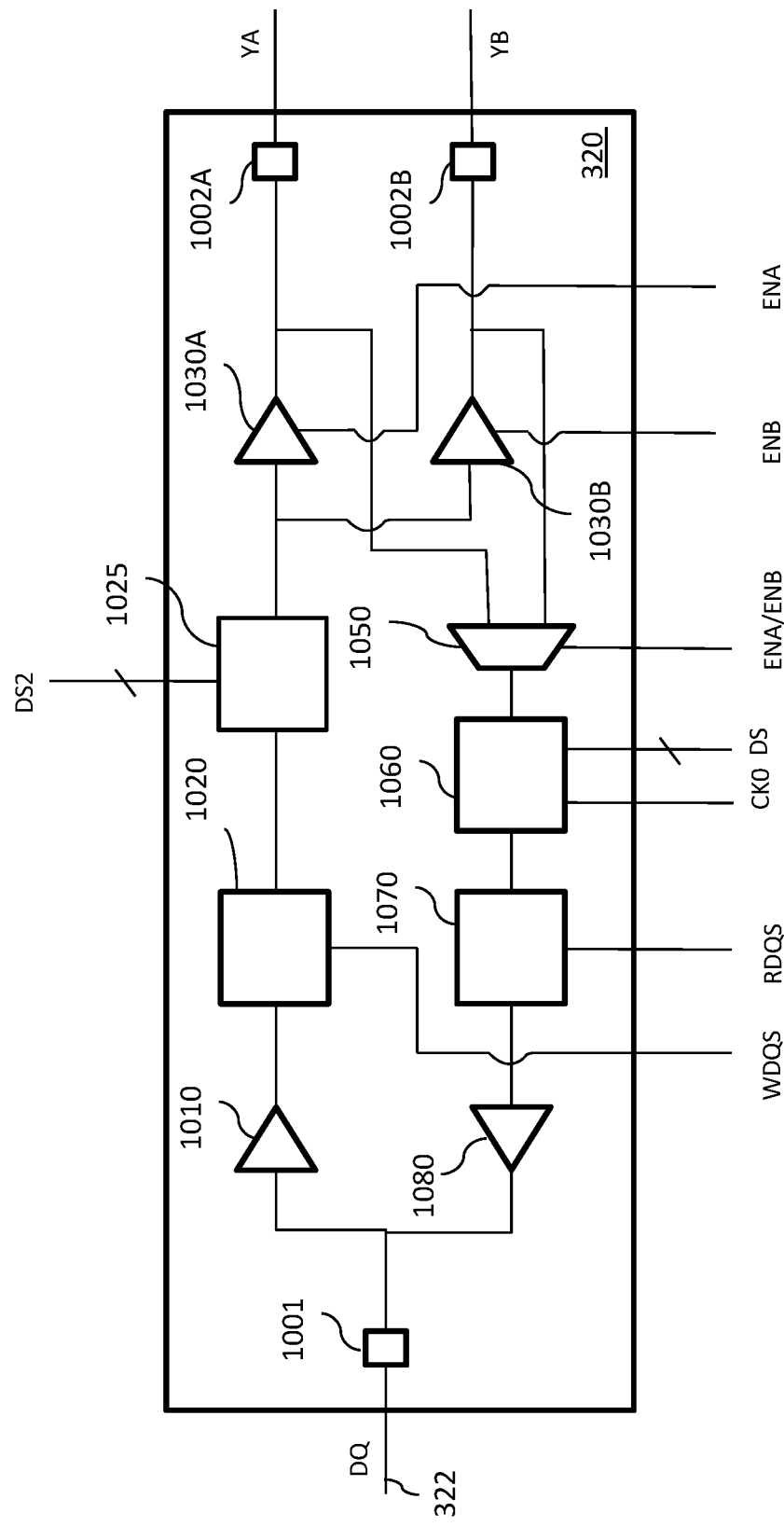


FIG. 10

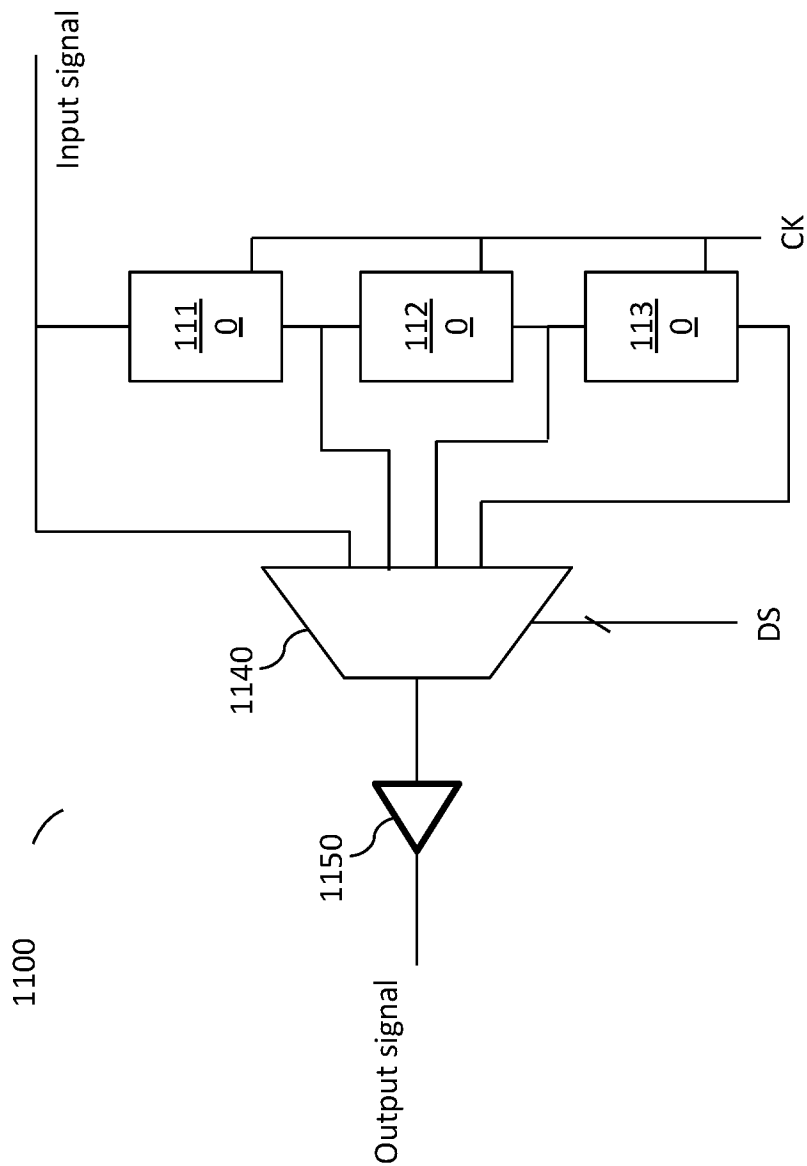


FIG. 11

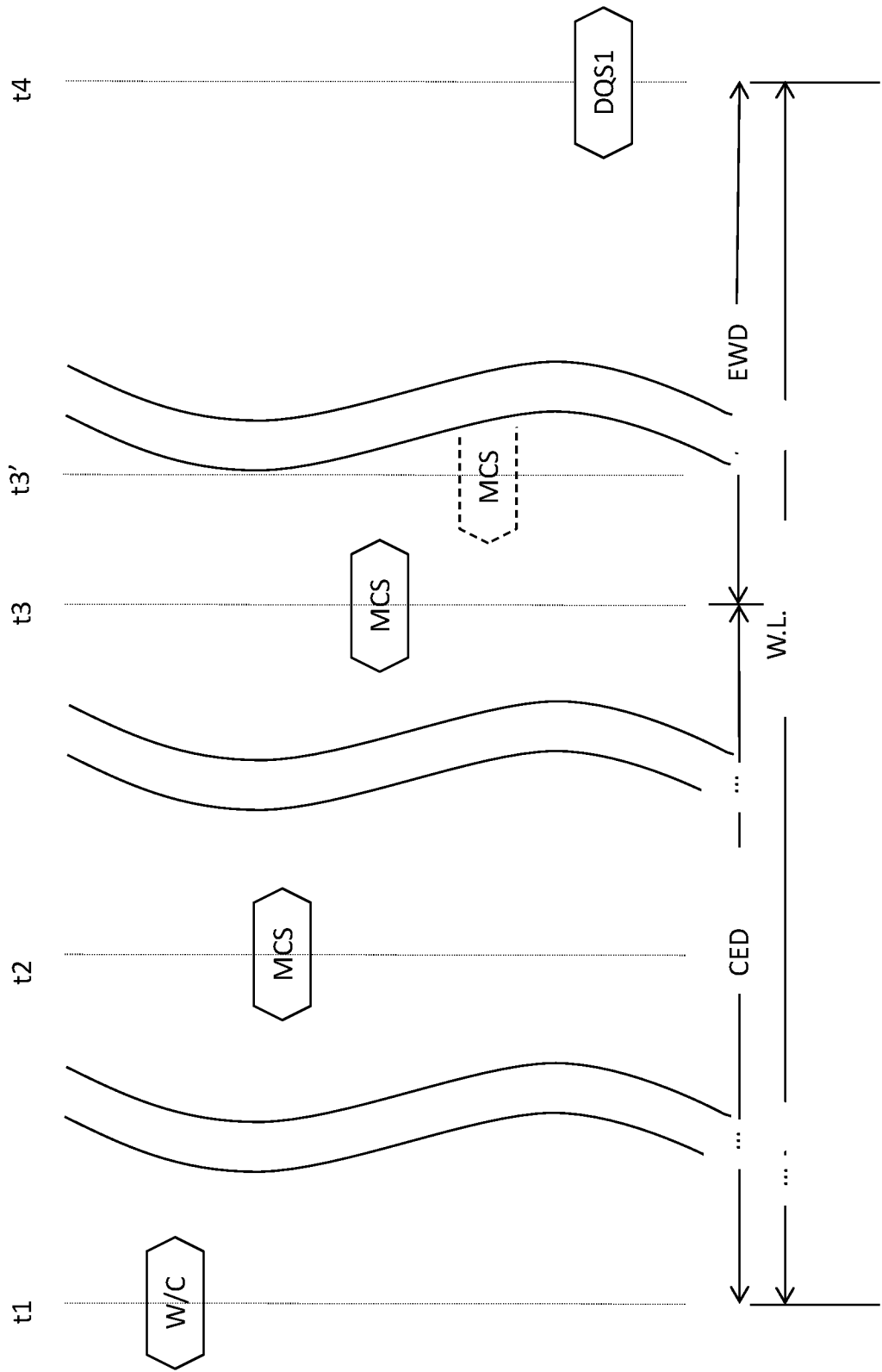


FIG. 12A

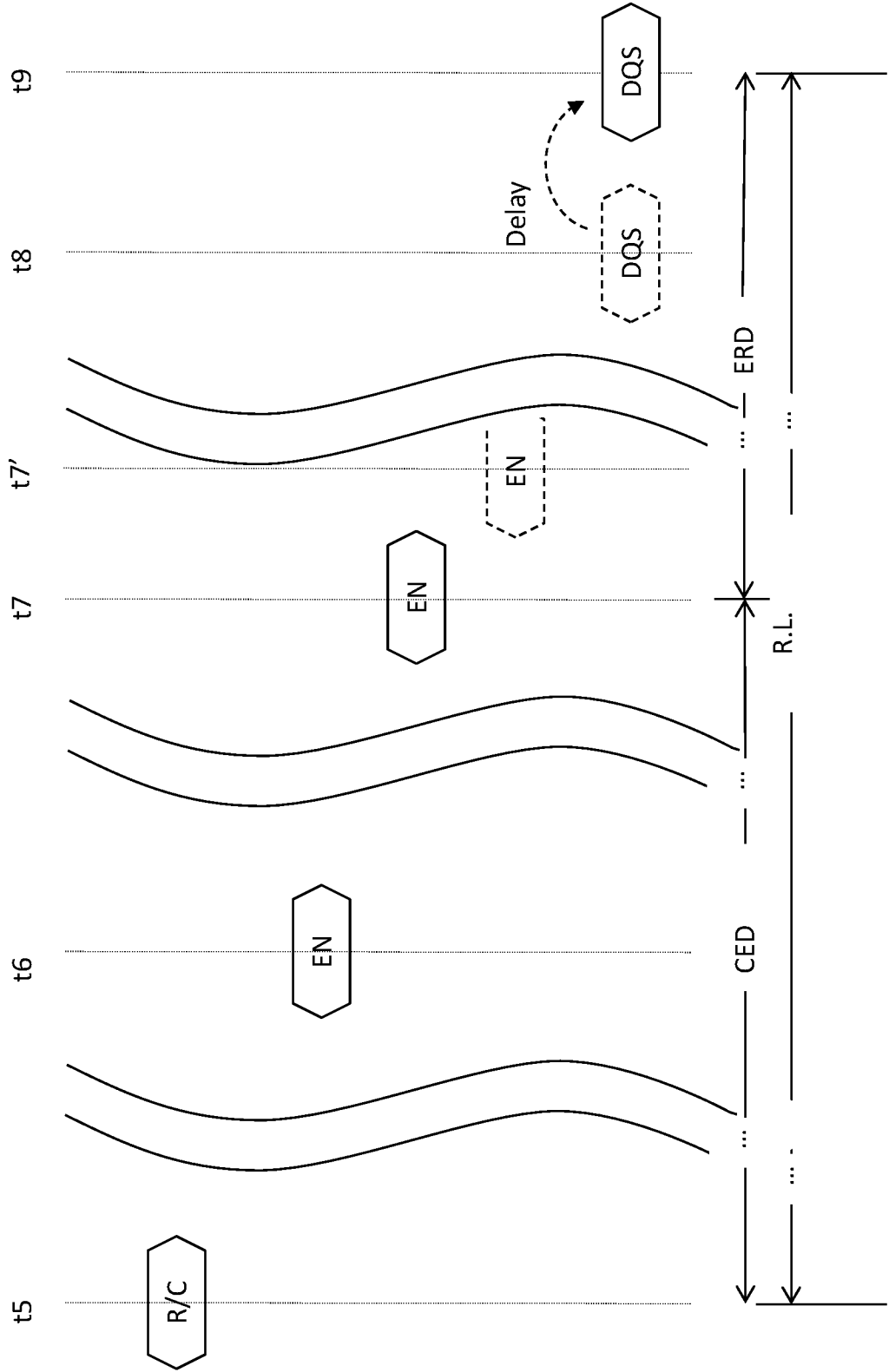


FIG. 12B

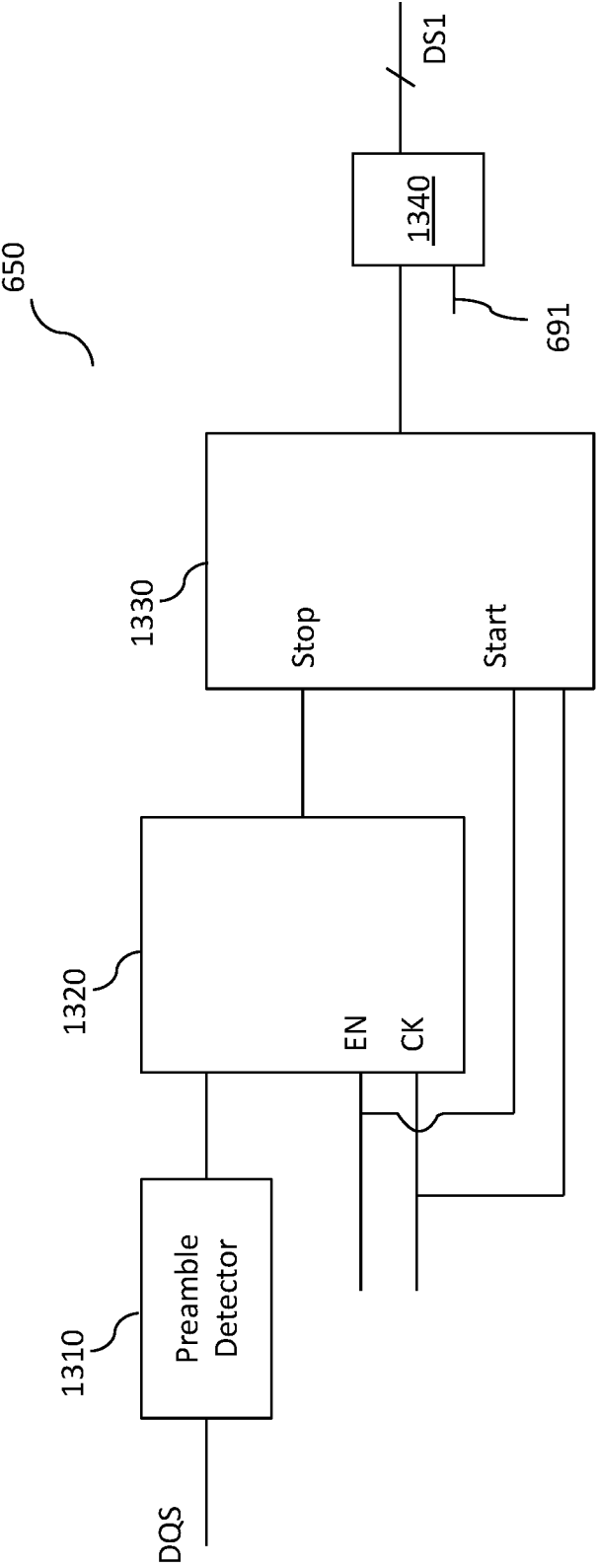


FIG. 13

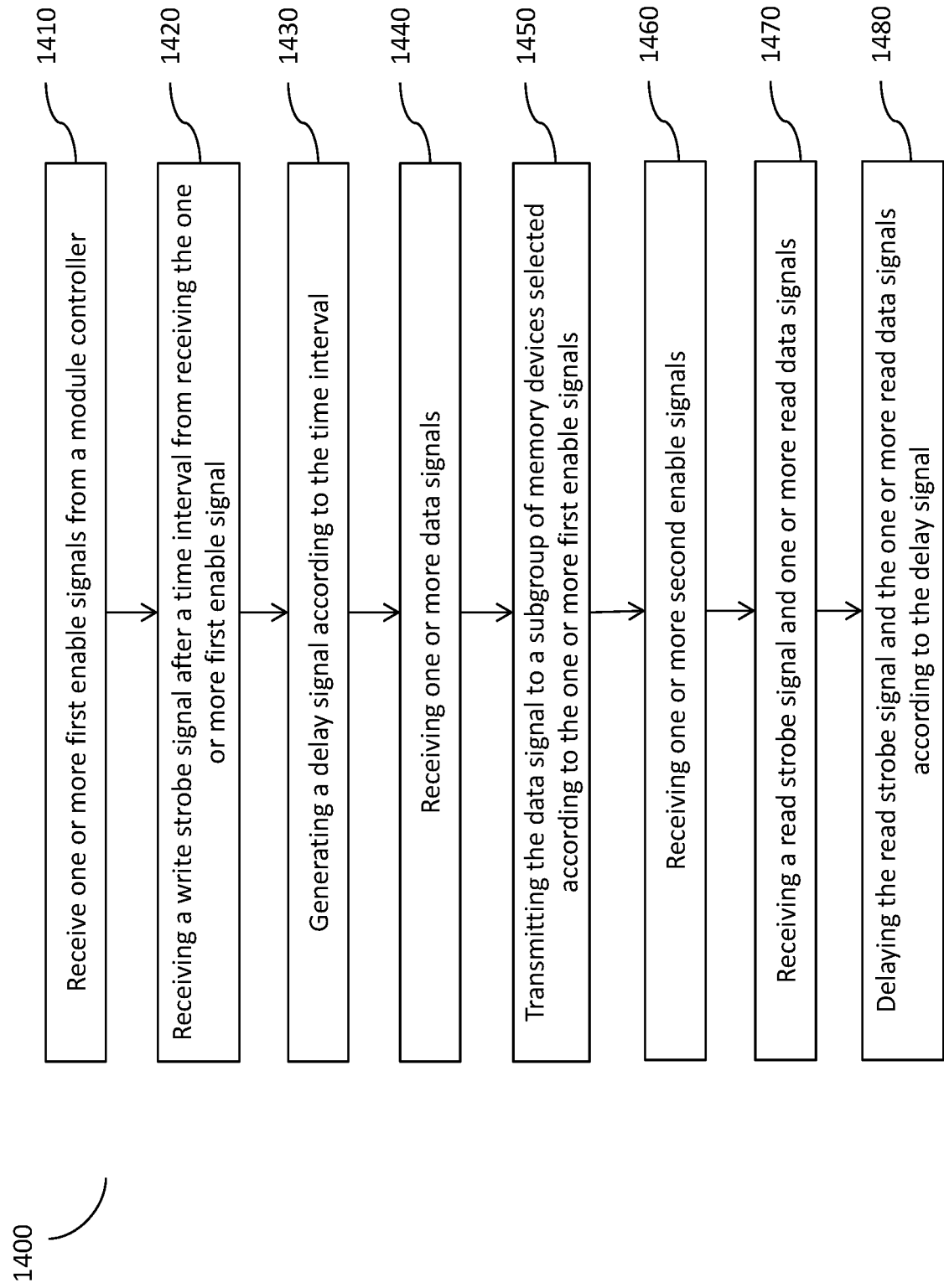


FIG. 14

GENERAL POWER OF ATTORNEY BY APPLICANT FOR UNITED STATES PATENT

I hereby revoke all previous powers of attorney given in the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

I hereby appoint the practitioner associated with the following Customer Number
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79141

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Please recognize or change the correspondence address for the application referenced in the attached transmittal letter to the address associated with the above-mentioned Customer Number.

I am the Applicant:

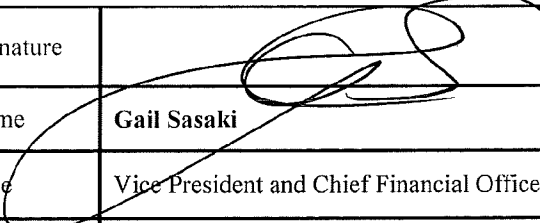
☐ Inventor or Joint Inventor

☐ Legal Representative of a Deceased or Legally Incapacitated Inventor

☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign

☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is currently being filed in this document)

SIGNATURE of Applicant for Patent

Signature		Date: January 9, 2013
Name	Gail Sasaki	
Title	Vice President and Chief Financial Officer	
Company	Netlist, Inc.	

Note: Signature - this form must be signed by the Applicant or a person authorized to act on behalf of Applicant.

Submit multiple forms for more than one signature

Total of ___ forms are submitted.

UTILITY PATENT APPLICATION TRANSMITTAL Only for new nonprovisional applications under 37 CFR § 1.53(b))		Attorney Docket No. NL073-20US04	
		First Inventor Hyun Lee	
		Title <i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	
		Electronically filed	November 1, 2024
APPLICATION ELEMENTS See MPEP Chapter 600 concerning utility patent application contents.		Address to: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450	
1. ☐ Fee Transmittal Form (with duplicate for fee processing) 2. ☐ Applicant claims Small Entity status, see 37 C.F.R. § 1.27 3. ☐ Applicant certifies Micro Entity status, see 37 C.F.R. § 1.29 4. ☒ Specification [Total Pages 32] 5. ☒ Drawing(s) (35 USC § 113) [Total Sheets 22] 6. ☒ Oath and Declaration [Total Pages 2] a. ☐ Newly executed (original or copy) b. ☒ Copy from a prior application (37 CFR § 1.63(d)) 7. ☒ Application Data Sheet, see 37 C.F.R. § 1.76 8. ☐ CD-Rom or CD-R in duplicate, large table or Computer Program (Appendix) ☐ Landscape Table on CD 9. ☐ Nucleotide and/or Amino Acid Sequence Submission (if applicable, all necessary) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statement verifying identity of above copies		ACCOMPANYING APPLICATION PARTS 10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee: Netlist, Inc. 11. ☒ 37 CFR § 3.73(c) Statement ☒ Power of Attorney 12. ☐ English Translation Document (if applicable) 13. ☐ Information Disclosure Statement and PTO-1449 ☐ Copies of citations attached 14. ☐ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP 503) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Non-Publication Request under 35 U.S.C. § 122 (b)(2)(B)(i) 18. ☐ Other:	
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19. CORRESPONDENCE ADDRESS: Customer Number 79141			
Signature /Jamie J. Zheng/		Date November 1, 2024	
Name (Print/Type) Jamie J. Zheng		Registration No. (Attorney/Agent)	51,167

**COMBINED DECLARATION AND ASSIGNMENT
FOR UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐ The attached application, or

☒ United States application or PCT international application
number 14/445,035 filed on July 28, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."

ASSIGNMENT

In consideration of the sum of One Dollar (\$1.00) and other good and valuable consideration paid, I hereby sell and assign to

Netlist, Inc.

(Name of Assignee)

of 175 Technology Dr., Suite #150, Irvine, CA 92618

(Address of Assignee)

(hereinafter designated as the Assignee) my entire right, title and interest for the United States as defined in 35 USC 100, for any invention set forth in the above-identified application.

I agree to execute all papers necessary in connection with this application for patent in the USPTO for the invention, and any continuation, divisional or reissue applications thereof and also to execute separate assignments in connection with such applications as the Assignee may deem necessary or expedient.

I agree to execute all papers necessary in connection with any post-grant proceeding which may occur in connection with this application or continuation, divisional or reissue thereof and to cooperate with the Assignee in every way possible in obtaining evidence and going forward with such post-grant proceeding.

I agree to execute all papers and documents and perform any act which may be necessary in connection with claims or provisions of the International Convention for Protection of Industrial Property or similar agreements.

I agree to perform all affirmative acts which may be necessary to obtain a grant of a valid United States patent to the

Assignee.

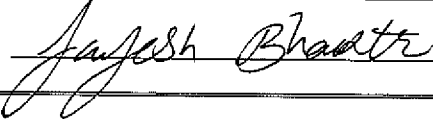
I hereby authorize and request the USPTO to issue any and all Letters Patents of the United States resulting from the application or any continuation, divisional or reissue applications thereof to the Assignee, as Assignee of my entire interest, and covenant that I have not executed, and will not execute, any agreement in conflict herewith.

First Inventor (Legal Name): Hyun Lee

Signature: 

Date: 3/31/2016

First Inventor (Legal Name): Jayesh R. Bhakta

Signature: 

Date: 3/31/2016

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application a continuation of U.S. Patent Application Ser. No. 18/059,958, filed on November 29, 2022, entitled “Memory Module with Static and Dynamic Timing Adjustment in Data Buffers,” which is a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021, entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory

module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules.

The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer

circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical

communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to

latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer

control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory

devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected

subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such

data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the

memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very

limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the

clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not

be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or

ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines $Y-1_0$, $Y-1_1$, ..., $Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup

in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YA-1₀, YA-1₁, ..., YA-1_n and module strobe line YA-1_{DQS} and memory devices M₁₂ and M₁₄, which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YB-1₀, YB-1₁, ..., YB-1_n and module strobe line YB-1_{DQS}. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M₁₁₋₁ and M₁₁₋₂ may be used in place of the memory device M₁₁. Each of the two memory devices M₁₁₋₁ and M₁₁₋₂ has a data width of 4, and together they act like a memory device M₁₁ of a data width of 8. Thus, memory device M₁₁₋₁ is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory circuit M₁₁₋₂ is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M₁₁₋₁ to M₁₁₋₄ may be used as the memory device M₁₁. Each of the four memory devices M₁₁₋₁ to M₁₁₋₄ has a data width of 4, and together they act like a memory device M₁₁ of a data width of 16. Thus, memory device M₁₁₋₁ is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory device M₁₁₋₂ is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain

embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118 based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines $Y_{A_{DQS}}$ and $Y_{B_{DQS}}$. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read

paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 681 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 691 to the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the data path enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock

signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment.

As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS} , or YA/YA_{DQS} and YB/YB_{DQS} . The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825 controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830 that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line YA_{DQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line YB_{DQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line YA_{DQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay

circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line YB_{DQS}. The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line YA_{DQ}, a third DQ pin 1002B that is coupled to a corresponding module DQ line YB_{DQ}. The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line YA_{DQ}. The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second

write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line YB_{DQ}. The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having

programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different

cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the

one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of

memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide connections between the memory module and the system bus;

memory devices mounted on the PCB and organized in a plurality of groups, a respective group of the memory devices corresponding to a respective set of the plurality of sets of data/strobe signal lines;

circuits mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines;

generate a module clock signal and module C/A signals in response to the system clock and input C/A signals;

generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and

output the module C/A signals to the memory devices; and

output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the memory devices, wherein each memory device in the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines.

2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.

3. The memory module of claim 2, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in a plurality of groups, and circuits coupled to the memory devices. The circuits including circuitry configurable to receive from the memory controller a system clock and input control and address (C/A) signals, and to generate a module clock signal and module C/A signals in response to the system clock and input C/A signals. The circuits further include circuitry configurable to generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and to output the plurality of local clock signals to respective groups of the memory devices. A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices.

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For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US04

STATEMENT UNDER 37 C.F.R. § 3.73(c)

Commissioner for Patents
P.O. Box 1450
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CONFIRMATION NO. 7292

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FILING RECEIPT



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Date Mailed: 11/13/2024

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which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.
Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/935,410**

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Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/935,410

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	3 minus 20 =	*			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY		
		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
AMENDMENT A	Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
					TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		
AMENDMENT B	Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
					TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/935,410	11/01/2024	Hyun Lee	NL073-20US04

CONFIRMATION NO. 7292

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

FORMALITIES LETTER



OC00000080490596

Date Mailed: 11/13/2024

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

An application number and filing date have been accorded to the above-identified nonprovisional application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee has not been received.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) has not been received.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$320** basic filing fee.
- **\$160** surcharge.
- **\$700** search fee.
- **\$800** examination fee.
- **\$(0)** previous unapplied payment amount.
- **\$1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/ederido/



United States Patent and Trademark Office

Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

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A handwritten signature in black ink, reading "Kathi Vidal".

Kathi Vidal

Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

A handwritten signature in black ink, reading "Derrick Brent".

Derrick Brent

Deputy Under Secretary of Commerce for Intellectual Property and
Deputy Director of the United States Patent and Trademark Office



UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/935,410

RECEIPT DATE / TIME
02/13/2025 11:41:16 PM Z ET

ATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Jamie Zheng

PATENT CENTER # 69181915

FILING DATE 11/01/2024

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun Lee

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 0

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
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Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage

submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

UNITED STATES
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
18/935,410RECEIPT DATE / TIME
02/13/2025 11:41:16 PM Z ETATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Jamie Zheng

PATENT CENTER # 69181915

AUTHORIZED BY -

CUSTOMER # 79141

FILING DATE 11/01/2024

CORRESPONDENCE
ADDRESS -FIRST NAMED
INVENTOR Hyun Lee

Payment Information

PAYMENT METHOD
CARD / 2172PAYMENT TRANSACTION ID
E20252CN41316803PAYMENT AUTHORIZED BY
Jamie Zheng

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1251	EXTENSION FOR RESPONSE WITHIN FIRST MONTH, EXCEPT PROVISIONAL APPLICATIONS	235.00	1	235.00
TOTAL AMOUNT:				\$235.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C.

371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/935,410	11/01/2024	Hyun Lee	NL073-20US04

CONFIRMATION NO. 7292

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

FORMALITIES LETTER



OC00000080490596

Date Mailed: 11/13/2024

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

An application number and filing date have been accorded to the above-identified nonprovisional application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee has not been received.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) has not been received.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$320** basic filing fee.
- **\$160** surcharge.
- **\$700** search fee.
- **\$800** examination fee.
- **\$(0)** previous unapplied payment amount.
- **\$1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/ederido/

Electronically filed February 13, 2025

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	7292
Serial No.:	18/935,410	Group Art Unit:	
Filed:	November 1, 2024	Examiner:	
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>		
		Attorney Docket No.:	NL073-20US04

**RESPONSE TO THE NOTICE TO FILE MISSING PARTS
OF NONPROVISIONAL APPLICATION**

Mail Stop: MISSING PARTS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Notice to File Missing Parts of Nonprovisional Application mailed November 13, 2024 (the "Notice"), a copy of which is enclosed, submitted herewith are the statutory basic filing, search and examination fees. The surcharge for the late payment of the statutory basic filing fee is also submitted herewith.

Applicant hereby petitions for a one-month extension of time to respond to the Notice.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL066-13US03).

Respectfully submitted,

Date: February 13, 2025

/ Jamie J. Zheng /
Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51,167
(Reg. No.)

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/935,410RECEIPT DATE / TIME
02/13/2025 11:38:57 PM Z ETATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7292	FILED BY	Jamie Zheng
PATENT CENTER #	69181888	FILING DATE	11/01/2024
CUSTOMER #	79141	FIRST NAMED INVENTOR	Hyun Lee
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073-20US04_MP_Response.pdf	1	Applicant Response to Pre-Exam Formalities Notice	69 KB
NTC_MISS_PRT.pdf	2	Applicant Response to Pre-Exam Formalities Notice	69 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
NL073-20US04_MP_Response.pdf	460F5D89453268EEFEC8A6F12C627C1FC0D81EE88A55FEA4 CA8A83E5CF7CBC781857FA23965F7AEED8A7DAF656200CC6 163E9DEB5B9EF11EB6A3705C3246C42A
NTC_MISS_PRT.pdf	FCAC27CEC7B971A86C6C14CF26CAAF18BFF6FD98B5667945

BCE8E26DC2E3EC70236C248880AD7B1A96943D9A1D065F2C
B4180FE7E3A8A1D13E63AFB5A32F2B33

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
18/935,410RECEIPT DATE / TIME
02/13/2025 11:38:57 PM Z ETATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Jamie Zheng

PATENT CENTER # 69181888

AUTHORIZED BY -

CUSTOMER # 79141

FILING DATE 11/01/2024

CORRESPONDENCE
ADDRESS -FIRST NAMED
INVENTOR Hyun Lee

Payment Information

PAYMENT METHOD
CARD / 2172PAYMENT TRANSACTION ID
E20252CN39276801PAYMENT AUTHORIZED BY
Jamie Zheng

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T))	350.00	1	350.00
1111	UTILITY PATENT APPL. SEARCH FEE	770.00	1	770.00
1311	PATENT APPL. EXAMINATION FEE	880.00	1	880.00
1051	SURCHARGE- LATE FILING FEE, SEARCH FEE, EXAMINATION FEE, INVENTOR'S OATH OR DECLARATION, OR APPLICATION FILED WITHOUT AT LEAST ONE CLAIM OR BY REFERENCE	170.00	1	170.00
TOTAL				\$2,170.00

AMOUNT:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/935,410

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	350
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	770
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	880
TOTAL CLAIMS (37 CFR 1.16(i))	3 minus 20 =	* 0			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	2000

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	* Minus **		=	x =		OR	x =	
Independent (37 CFR 1.16(h))	* Minus ***		=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	* Minus **		=	x =		OR	x =	
Independent (37 CFR 1.16(h))	* Minus ***		=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/935,410	11/01/2024		2170	NL073-20US04	3	1

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

CONFIRMATION NO. 7292

UPDATED FILING RECEIPT



0000000084760847

Date Mailed: 02/24/2025

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Applications for which benefit is claimed

This application is a CON of 18/059,958 11/29/2022 PAT 12,135,644
which is a CON of 17/141,978 01/05/2021 PAT 11,513,955
which is a CON of 16/432,700 06/05/2019 PAT 10,884,923
which is a CON of 14/445,035 07/28/2014 PAT 10,324,841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - **None.**
Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/935,410**

Projected Publication Date: 06/05/2025

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/935,410	11/01/2024	Hyun Lee	NL073-20US04

CONFIRMATION NO. 7292

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

PUBLICATION NOTICE



OC000000089740009

Date Mailed: 06/05/2025

Title:MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Publication No.US-2025-0181504-A1

Publication Date:06/05/2025

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be viewed using the USPTO's publicly available Searchable Databases via the Patent Public Search tool at www.uspto.gov. The direct link to access the Patent Public Search tool is currently <https://ppubs.uspto.gov/pubwebapp/static/pages/ppubsbasic.html>.

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PATENT APPLICATION FEE DETERMINATION RECORD				Application or Docket Number 18/935,410		Filing Date 11/01/2024		☐ To be Mailed	
Substitute for Form PTO-875									
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
FOR		(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A	N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A	N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A	N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL					
APPLICATION AS AMENDED - PART II									
AMENDMENT	11/06/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$200 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$600 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/LAMONT FLETCHER/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
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Electronically filed November 6, 2025

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	7292
Serial No.:	18/935,410	Group Art Unit:	2132
Filed:	November 1, 2024	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US04

PRELIMINARY AMENDMENT

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

Applicant files herewith a preliminary amendment for the above identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US04).

Amendments to the Specification

Please amend the Title as follows:

MEMORY MODULE WITH LOCAL CLOCK SIGNALS SYNCHRONIZATION

Please amend paragraph [0001] as follows:

[0001] The present application is a continuation of U.S. Patent Application Ser. No. 18/059,958, filed on November 29, 2022 (now US Patent No. 12,135,644), entitled “Memory Module with Static and Dynamic Timing Adjustment in Data Buffers,” which is a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021 (now US Patent No. 11,513,955), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, ~~a set of~~ control/address (C/A) signal lines and ~~a plurality of sets of~~ data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;

memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding ~~to a respective subset of the plurality of sets of data/strobe signal lines;~~

~~circuits/circuitry~~ mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the ~~set of~~ C/A signal lines;

generate ~~a module clock signal and~~ module C/A signals in response to the ~~system clock and input C/A signals;~~

generate a plurality of local clocks ~~signals~~ corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks ~~signals~~ having respective phase relationships with the ~~module system clock signal, the~~ respective phase relationships being programmable independently of each other; and

output the module C/A signals to the memory devices; and

output the plurality of local clocks ~~signals~~ to ~~respective groups of the~~ memory devices, wherein a respective local clock ~~signal~~ having a respective programmable phase relationship with the ~~module system clock signal~~ is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein ~~each memory device in the corresponding group of the~~

memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding subset of the plurality of sets of data/strobe signal lines in accordance with the respective local clock.

2. (Currently Amended) The memory module of claim 1, wherein the ~~circuits~~ circuitry includes logic and configuration registers programmable by the logic to control the respective ~~programmable~~ phase relationships.

3. (Currently Amended) The memory module of claim 2, wherein:
the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform ~~a set of operations and the logic is further configurable~~ one or more operations to program the respective configuration registers ~~based on information derived from the set of operations.~~

4. (New) The memory module of claim 3, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

5. (New) The memory module of claim 2, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the plurality of local clocks, programmable delay circuitry to add respective delays to respective ones of the plurality of local clocks, and clock driver circuitry to drive the plurality of local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

6. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines,

control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising:

- a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;

- memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices;

- circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to:

 - receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;

 - generate module C/A signals in response to the input C/A signals;

 - generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and

 - output the module C/A signals to the memory devices; and

 - output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines.

7. (New) The memory module of claim 6, wherein the circuitry includes logic and configuration registers programmable by the logic, the registers including one or more

first registers to control the first phase shift and one or more second registers to control the second phase shift.

8. (New) The memory module of claim 7, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform one or more operations to program the configuration registers.

9. (New) The memory module of claim 8, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

10. (New) The memory module of claim 7, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the local clocks, programmable delay circuitry to add respective delays to respective ones of the local clocks, and clock driver circuitry to drive the local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

11. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding, respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines;

memory devices mounted on the PCB;

a module control device mounted on the PCB and configurable to receive input control and address (C/A) signals for a memory write operation to write data to the

memory module, and to output module C/A signals and module control signals in response the input C/A signals;

a plurality of data buffers mounted on the PCB and coupled to the module control device, wherein:

each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a respective section of the memory devices;

the plurality of data buffers include at least m data buffers;

the m data buffers are configurable to receive data associated with the write operation;

the data associated with the write operation has a bit width of N and includes m n-bit-wide sections, $N = mxn$, wherein n and m are integers greater than 1;

each respective data buffer of the m data buffers is configurable to receive the module control signals and to receive a respective n-bit-wide section of the data associated with the write operation in response to the module control signals;

the each respective data buffer of the m data buffers is further configurable to transmit the respective n-bit-wide section of the data associated with the write operation to at least a first branch of the respective section of the memory devices;

the each respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and

at least one first memory devices in the first branch of the respective section of the memory devices is configurable to receive at least some of the module C/A signals and the respective local clock, and to receive and store the respective n-bit-wide section of the data associated with the write operation in response to the at least some of the registered C/A signals and in accordance with the respective local clock.

12. (New) The memory module of claim 11, wherein:

the module control device is further configurable to receive a system clock and to output a module clock; and

the each respective data buffer of the m data buffers is further configured to receive the module clock and to generate the respective local clock from the module clock, the respective local clock having a respective phase shift from the module clock.

13. (New) The memory module of claim 12, wherein the each respective data buffer includes logic and respective configuration registers programmable by the logic, and wherein the respective phase shift is dependent on one or more register settings in the respective configuration registers.

14. (New) The memory module of claim 13, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the write operation is performed in the normal operation mode; and

the respective configuration registers are programmed in one or more operations during the configuration mode.

15. (New) The memory module of claim 14, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

16. (New) The memory module of claim 12, wherein:

the m data buffers include a first data buffer and a second data buffer;

the first data buffer is configurable to generate a first local clock from the module clock in response to the module control signals and to transmit the first local clock to at least a first branch of a first section of memory devices;

the second data buffer is configurable to generate a second local clock from the module clock in response to the module control signals and to transmit the second local clock to at least a first branch of a second section of memory devices;

the first local clock has a first phase shift from the module clock;

the second local clock has a second phase shift from the module clock; and

the first shift and the second phase shift are programmable independently of each other.

17. (New) The memory module of claim 11, wherein:

the each respective data buffer of the m data buffers includes a first n -bit-wide write path coupled to the first branch of the respective section of the memory device and not to a second branch of the respective section of the memory device, and a second n -bit-wide write path coupled to the second branch of the respective section of the memory device and not to the first branch of the respective section of the memory device; and

the each respective data buffer of the m data buffers is configurable to transmit the respective n -bit-wide section of the data associated with the write operation to the first branch of the respective section of the memory device via the first n -bit-wide write path and not to any memory device in the second branch of the respective section of the memory device via the second n -bit-wide write path.

18. (New) The memory module of claim 17, wherein:

the each respective data buffer of the m data buffers further includes input buffers configurable to receive the respective n -bit-wide section of the data associated with the write operation;

selection circuitry coupled to the input buffers; and

logic coupled to the selection circuitry and configurable to control the selection circuitry to select the first n -bit-wide write path to transmit the respective n -bit-wide section of the data associated with the write operation in response to the module control signals.

19. (New) The memory module of claim 17, wherein the at least one first memory device includes one or more dynamic random access memory (DRAM) device each having a bit width of n/k , where k is a number of DRAM devices in the at least one first memory device, and k is 1, 2, or 4.

20. (New) The memory module of claim 19, wherein $n = 8$ or 16.

REMARKS

In this preliminary amendment, claims 1-3 are amended, and new claims 4-20 are presented.

Support for the amendments can be found throughout the specification and drawings of the application as filed. No new matter has been added.

After entry of this amendment, the pending claims are: claims 1-20.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: November 6, 2025

/Jamie J. Zheng/

Jamie J. Zheng

Customer No. 79141

51,167

(Reg. No.)

Electronically filed November 6, 2025

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	7292
Serial No.:	18/935,410	Group Art Unit:	2132
Filed:	November 1, 2024	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US04

PRELIMINARY AMENDMENT

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

Applicant files herewith a preliminary amendment for the above identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US04).

Amendments to the Specification

Please amend the Title as follows:

MEMORY MODULE WITH LOCAL CLOCK SIGNALS SYNCHRONIZATION

Please amend paragraph [0001] as follows:

[0001] The present application is a continuation of U.S. Patent Application Ser. No. 18/059,958, filed on November 29, 2022 (now US Patent No. 12,135,644), entitled “Memory Module with Static and Dynamic Timing Adjustment in Data Buffers,” which is a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021 (now US Patent No. 11,513,955), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, ~~a set of~~ control/address (C/A) signal lines and ~~a plurality of sets of~~ data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;

memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding to a respective subset of the plurality of ~~sets of~~ data/strobe signal lines;

~~circuits~~ circuitry mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the ~~set of~~ C/A signal lines;

generate ~~a module clock signal and~~ module C/A signals in response to the ~~system clock and~~ input C/A signals;

generate a plurality of local clocks ~~signals~~ corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks ~~signals~~ having respective phase relationships with the ~~module system clock signal~~, the respective phase relationships being programmable independently of each other;

and

output the module C/A signals to the memory devices; and

output the plurality of local clocks ~~signals~~ to ~~respective groups of the~~ memory devices, wherein a respective local clock ~~signal~~ having a respective programmable phase relationship with the ~~module system clock signal~~ is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein ~~each memory device in~~ the corresponding group of the

memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding subset of the plurality of sets of data/strobe signal lines in accordance with the respective local clock.

2. (Currently Amended) The memory module of claim 1, wherein the ~~circuits~~ circuitry includes logic and configuration registers programmable by the logic to control the respective ~~programmable~~ phase relationships.

3. (Currently Amended) The memory module of claim 2, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform ~~a set of operations and the logic is further configurable~~ one or more operations to program the respective configuration registers ~~based on information derived from the set of operations.~~

4. (New) The memory module of claim 3, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

5. (New) The memory module of claim 2, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the plurality of local clocks, programmable delay circuitry to add respective delays to respective ones of the plurality of local clocks, and clock driver circuitry to drive the plurality of local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

6. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines,

control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising:

- a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;

- memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices;

- circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to:

 - receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;

 - generate module C/A signals in response to the input C/A signals;

 - generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and

 - output the module C/A signals to the memory devices; and

 - output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines.

7. (New) The memory module of claim 6, wherein the circuitry includes logic and configuration registers programmable by the logic, the registers including one or more

first registers to control the first phase shift and one or more second registers to control the second phase shift.

8. (New) The memory module of claim 7, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform one or more operations to program the configuration registers.

9. (New) The memory module of claim 8, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

10. (New) The memory module of claim 7, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the local clocks, programmable delay circuitry to add respective delays to respective ones of the local clocks, and clock driver circuitry to drive the local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

11. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding, respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines;

memory devices mounted on the PCB;

a module control device mounted on the PCB and configurable to receive input control and address (C/A) signals for a memory write operation to write data to the

memory module, and to output module C/A signals and module control signals in response the input C/A signals;

a plurality of data buffers mounted on the PCB and coupled to the module control device, wherein:

each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a respective section of the memory devices;

the plurality of data buffers include at least m data buffers;

the m data buffers are configurable to receive data associated with the write operation;

the data associated with the write operation has a bit width of N and includes m n -bit-wide sections, $N = mxn$, wherein n and m are integers greater than 1;

each respective data buffer of the m data buffers is configurable to receive the module control signals and to receive a respective n -bit-wide section of the data associated with the write operation in response to the module control signals;

the each respective data buffer of the m data buffers is further configurable to transmit the respective n -bit-wide section of the data associated with the write operation to at least a first branch of the respective section of the memory devices;

the each respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and

at least one first memory devices in the first branch of the respective section of the memory devices is configurable to receive at least some of the module C/A signals and the respective local clock, and to receive and store the respective n -bit-wide section of the data associated with the write operation in response to the at least some of the registered C/A signals and in accordance with the respective local clock.

12. (New) The memory module of claim 11, wherein:

the module control device is further configurable to receive a system clock and to output a module clock; and

the each respective data buffer of the m data buffers is further configured to receive the module clock and to generate the respective local clock from the module clock, the respective local clock having a respective phase shift from the module clock.

13. (New) The memory module of claim 12, wherein the each respective data buffer includes logic and respective configuration registers programmable by the logic, and wherein the respective phase shift is dependent on one or more register settings in the respective configuration registers.

14. (New) The memory module of claim 13, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the write operation is performed in the normal operation mode; and

the respective configuration registers are programmed in one or more operations during the configuration mode.

15. (New) The memory module of claim 14, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

16. (New) The memory module of claim 12, wherein:

the m data buffers include a first data buffer and a second data buffer;

the first data buffer is configurable to generate a first local clock from the module clock in response to the module control signals and to transmit the first local clock to at least a first branch of a first section of memory devices;

the second data buffer is configurable to generate a second local clock from the module clock in response to the module control signals and to transmit the second local clock to at least a first branch of a second section of memory devices;

the first local clock has a first phase shift from the module clock;

the second local clock has a second phase shift from the module clock; and

the first shift and the second phase shift are programmable independently of each other.

17. (New) The memory module of claim 11, wherein:

the each respective data buffer of the m data buffers includes a first n -bit-wide write path coupled to the first branch of the respective section of the memory device and not to a second branch of the respective section of the memory device, and a second n -bit-wide write path coupled to the second branch of the respective section of the memory device and not to the first branch of the respective section of the memory device; and

the each respective data buffer of the m data buffers is configurable to transmit the respective n -bit-wide section of the data associated with the write operation to the first branch of the respective section of the memory device via the first n -bit-wide write path and not to any memory device in the second branch of the respective section of the memory device via the second n -bit-wide write path.

18. (New) The memory module of claim 17, wherein:

the each respective data buffer of the m data buffers further includes input buffers configurable to receive the respective n -bit-wide section of the data associated with the write operation;

selection circuitry coupled to the input buffers; and

logic coupled to the selection circuitry and configurable to control the selection circuitry to select the first n -bit-wide write path to transmit the respective n -bit-wide section of the data associated with the write operation in response to the module control signals.

19. (New) The memory module of claim 17, wherein the at least one first memory device includes one or more dynamic random access memory (DRAM) device each having a bit width of n/k , where k is a number of DRAM devices in the at least one first memory device, and k is 1, 2, or 4.

20. (New) The memory module of claim 19, wherein $n = 8$ or 16.

REMARKS

In this preliminary amendment, claims 1-3 are amended, and new claims 4-20 are presented.

Support for the amendments can be found throughout the specification and drawings of the application as filed. No new matter has been added.

After entry of this amendment, the pending claims are: claims 1-20.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: November 6, 2025

/Jamie J. Zheng/

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www.uspto.gov**ELECTRONIC ACKNOWLEDGEMENT RECEIPT**APPLICATION #
18/935,410RECEIPT DATE / TIME
11/06/2025 08:52:21 PM Z ETATTORNEY DOCKET #
NL073-20US04**Title of Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application InformationAPPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Jamie Zheng

PATENT CENTER # 73077246

FILING DATE 11/01/2024

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun LeeCORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents**TOTAL DOCUMENTS: 4**

DOCUMENT		PAGES	DESCRIPTION	SIZE (KB)
NL073-20US04_Preliminary_Amendment.pdf		10	-	146 KB
NL073-20US04_Preliminary_Amendment-A.PE.pdf	(1-1)	1	Preliminary Amendment	100 KB
NL073-20US04_Preliminary_Amendment-SPEC.pdf	(2-2)	1	Specification	93 KB
NL073-20US04_Preliminary_Amendment-CLM.pdf	(3-9)	7	Claims	128 KB
NL073-20US04_Preliminary_Amendment	(10-10)	1	Applicant Arguments/Remarks Made in an Amendment	93 KB

dment-REM.pdf

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
NL073- 20US04_Preliminary_Amendm ent.pdf	3768755807AFCD943EF59799EAFDE8410D0556CD448E18224 58CE99F44C451D1803859883256A04A7764048331A5D8D26B5 4FC771C6739C7514887E333C56E4A
NL073- 20US04_Preliminary_Amendm ent-A.PE.pdf	895ED0CD9E8A3B16C307AB0B94845EF7210171B80FF9C64D1 AFB88FA66B67F6132F6E369B185832EE4DF3EE5216414BCD6 3A32EBB8BFD36B4D2BEAAE58401659
NL073- 20US04_Preliminary_Amendm ent-SPEC.pdf	CF5C6ED59D47A48C60D12CEDD3B5F854EAACB4B97A1953A F5431C5402DA5489381CC320D5761AC466A735676B8E6EFEC D09F366390F7BC2ADC396191EFC56004
NL073- 20US04_Preliminary_Amendm ent-CLM.pdf	F539FB87240B1DAC155E0EE74002F555E6EB49F6734164360 D0D8D79A032A62F6AB0071C815585530269F2B01787B080487 54A3E12DCA76C7AFD8F96D61A0697
NL073- 20US04_Preliminary_Amendm ent-REM.pdf	5659669E3AD4CFC9E2E3D536043966900F6737C447F8307F66 0AEBBAB8C1E624A9DE7F49E4918FF6241A40503AEC194EE6 1013E5500194DB9B96381BA0E3628A

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	11/01/2024	Hyun Lee	NL073-20US04	7292
79141	7590	12/04/2025	EXAMINER	
USCH Law, PC 3524 Waverley St. Palo Alto, CA 94306			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			12/04/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

cofficeaction@appcoll.com
patents@usclaw.com

Office Action Summary

Application No.

18/935,410

Applicant(s)

Lee et al.

Examiner

ZUBAIR AHMED

Art Unit

2132

AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 2 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 06 November 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☐ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☐ Claim(s) ____ is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☒ Claim(s) 1-20 are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 01 November 2024 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to preliminary amendment filed on 11/06/2025. New claims 4-20 were added in this amendment. Claims 1-20 are pending.

Election/Restrictions

Restriction to one of the following inventions is required under 35 U.S.C. 121:

- I. Group I includes claims 1-10. Claims 1-5 are drawn to an apparatus "A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data/strobe signal lines; circuitry mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines; generate module C/A signals in response to the input C/A signals; generate a plurality of local clocks corresponding, respectively, to the

plurality of groups of the memory devices, the plurality of local clocksu having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and output the module C/A signals to the memory devices; and output the plurality of local clocksu to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding subset of data/strobe signal lines in accordance with the respective local clock” and claims 6-10 are drawn to an apparatus “A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices; circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A

signal lines; generate module C/A signals in response to the input C/A signals; generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and output the module C/A signals to the memory devices; and output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines” ***classified in CPC G11C 5/04 (Supports for storage elements {e.g., memory modules} Mounting or fixing of storage elements on such supports), G11C 7/1066 (Output synchronization), G11C 7/1093 (Input synchronization), G11C 29/023 (in clock generator or timing circuitry), G11C 29/028 (with adaption or trimming of parameters),***

G06F 1/10 (Distribution of clock signals {e.g., skew}), G06F 13/4243 (with synchronous protocol), G06F 13/4256 (using a clocked protocol).

- II. Claims 11-20 are drawn to an apparatus “A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding, respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines; memory devices mounted on the PCB; a module control device mounted on the PCB and configurable to receive input control and address (C/A) signals for a memory write operation to write data to the memory module, and to output module C/A signals and module control signals in response the input C/A signals; a plurality of data buffers mounted on the PCB and coupled to the module control device, wherein: each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a respective section of the memory devices; the plurality of data buffers include at least m data buffers; the m data buffers are configurable to receive data associated with the write operation; the data associated with the write operation has a bit width of N and includes m n-bit-wide sections, $N = mxn$, wherein n and

m are integers greater than 1; each respective data buffer of the m data buffers is configurable to receive the module control signals and to receive a respective n-bit-wide section of the data associated with the write operation in response to the module control signals; the each respective data buffer of the m data buffers is further configurable to transmit the respective n-bit-wide section of the data associated with the write operation to at least a first branch of the respective section of the memory devices; the each respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and at least one first memory devices in the first branch of the respective section of the memory devices is configurable to receive at least some of the module C/A signals and the respective local clock, and to receive and store the respective n-bit-wide section of the data associated with the write operation in response to the at least some of the registered C/A signals and in accordance with the respective local clock” ***classified in CPC G06F 12/0802 (Addressing of a memory level in which the access to the desired data or data block requires associative addressing means, e.g., caches), G11C 5/04 (Supports for storage elements {e.g., memory modules} Mounting or fixing of storage elements on such supports), G11C 7/1066 (Output synchronization), G11C 7/1093 (Input synchronization), G11C 29/023 (in clock generator or timing circuitry), G11C 29/028 (with adaption or***

trimming of parameters), G06F 1/10 (Distribution of clock signals {e.g., skew}), G06F 13/4243 (with synchronous protocol), G06F 13/4256 (using a clocked protocol).

The inventions are distinct, each from each other because of the following reasons:

Inventions Group I and Group II are related as subcombinations disclosed as usable together in a single combination. The subcombinations are distinct from each other if they are shown to be separately usable.

In the instant case, invention Group I has separate utility such as “A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data/strobe signal lines; circuitry mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines; generate module C/A signals in response to the input C/A signals; generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the

plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and output the module C/A signals to the memory devices; and output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data/strobe signals with the memory controller via a corresponding subset of data/strobe signal lines in accordance with the respective local clock” and “A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices; circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines; generate module C/A signals in response to the input C/A signals; generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and output the module

C/A signals to the memory devices; and output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines” ***classified in CPC G11C 5/04 (Supports for storage elements {e.g., memory modules} Mounting or fixing of storage elements on such supports), G11C 7/1066 (Output synchronization), G11C 7/1093 (Input synchronization), G11C 29/023 (in clock generator or timing circuitry), G11C 29/028 (with adaption or trimming of parameters), G06F 1/10 (Distribution of clock signals {e.g., skew}), G06F 13/4243 (with synchronous protocol), G06F 13/4256 (using a clocked protocol).***

Invention group II has separate utility such as “A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding,

respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines; memory devices mounted on the PCB; a module control device mounted on the PCB and configurable to receive input control and address (C/A) signals for a memory write operation to write data to the memory module, and to output module C/A signals and module control signals in response the input C/A signals; a plurality of data buffers mounted on the PCB and coupled to the module control device, wherein: each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a respective section of the memory devices; the plurality of data buffers include at least m data buffers; the m data buffers are configurable to receive data associated with the write operation; the data associated with the write operation has a bit width of N and includes m n -bit-wide sections, $N = mxn$, wherein n and m are integers greater than 1; each respective data buffer of the m data buffers is configurable to receive the module control signals and to receive a respective n -bit-wide section of the data associated with the write operation in response to the module control signals; the each respective data buffer of the m data buffers is further configurable to transmit the respective n -bit-wide section of the data associated with the write operation to at least a first branch of the respective section of the memory devices; the each respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and at least one first memory devices in the first branch of the respective section of the memory devices is configurable to receive at least some of the module C/A signals and the respective local clock, and to receive and store the respective n -bit-wide section of

the data associated with the write operation in response to the at least some of the registered C/A signals and in accordance with the respective local clock” ***classified in CPC G06F 12/0802 (Addressing of a memory level in which the access to the desired data or data block requires associative addressing means, e.g., caches), G11C 5/04 (Supports for storage elements {e.g., memory modules} Mounting or fixing of storage elements on such supports), G11C 7/1066 (Output synchronization), G11C 7/1093 (Input synchronization), G11C 29/023 (in clock generator or timing circuitry), G11C 29/028 (with adaption or trimming of parameters), G06F 1/10 (Distribution of clock signals {e.g., skew}), G06F 13/4243 (with synchronous protocol), G06F 13/4256 (using a clocked protocol).***

- A. These inventions are separately usable and have acquired a separate status in the art as shown by their different classification and their recognized divergent subject matter.
- B. The search required for one Group is not required for the other Group (i.e., the search for Group II is not required for Group I).

For the reasons above, restriction for examination purposes as indicated is proper.

Applicant is advised that the reply to this requirement to be complete **must** include **(i) an election of an invention to be examined** even though the requirement may be traversed (37 CFR 1.143) and **(ii) identification of the claims encompassing the elected invention.**

The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other invention.

Applicant is requested to formally cancel the non-elected claims.

Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be amended in compliance with 37 CFR 1.48(b) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. Any amendment of inventorship must be accompanied by a request under 37 CFR 1.48(b) and by the fee required under 37 CFR 1.17(i).

Applicant is also advised that the response must be submitted to the Office within 60 calendar days.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, HOSAIN T. ALAM can be reached at (571) 272-3978. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Application/Control Number: 18/935,410
Art Unit: 2132

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/ZUBAIR AHMED/
Examiner, Art Unit 2132

/HOSAIN T ALAM/
Supervisory Patent Examiner, Art Unit 2132

Electronically filed February 4, 2026

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	7292
Serial No.:	18/935,410	Group Art Unit:	2132
Filed:	November 1, 2024	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US04

RESPONSE TO RESTRICTION REQUIREMENT

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This amendment is submitted in response to the Restriction Requirement dated December 4, 2025.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US04).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and ~~data/strobe~~ data signal lines, the memory module comprising:

- a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;

- memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the ~~data/strobe~~ data signal lines;

- circuitry mounted on the PCB and configurable to:

- receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;

- generate module C/A signals in response to the input C/A signals;

- generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and

- output the module C/A signals to the memory devices; and

- output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating ~~data/strobe~~ data signals with the memory controller via a corresponding subset of ~~data/strobe~~ data signal lines in accordance with the respective local clock.

2. (Previously Presented) The memory module of claim 1, wherein the circuitry includes logic and configuration registers programmable by the logic to control the respective phase relationships.
3. (Currently Amended) The memory module of claim 2, wherein:
 - the memory module is operable in at least a normal operation mode and a configuration mode;
 - the memory module in the normal operation mode is configurable to output or receive ~~data/strobe~~ data signals via the ~~data/strobe~~ data signal lines in response to memory read or write commands received via the C/A signal lines; and
 - the memory module in the configuration mode is configurable to perform one or more operations to program the respective configuration registers.
4. (Previously Presented) The memory module of claim 3, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.
5. (Previously Presented) The memory module of claim 2, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the plurality of local clocks, programmable delay circuitry to add respective delays to respective ones of the plurality of local clocks, and clock driver circuitry to drive the plurality of local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.
6. (Previously Presented) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;
 - memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices;

circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to:

- receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;

- generate module C/A signals in response to the input C/A signals;

- generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and

- output the module C/A signals to the memory devices; and

- output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines.

7. (Previously Presented) The memory module of claim 6, wherein the circuitry includes logic and configuration registers programmable by the logic, the registers including one or more first registers to control the first phase shift and one or more second registers to control the second phase shift.

8. (Previously Presented) The memory module of claim 7, wherein:

- the memory module is operable in at least a normal operation mode and a configuration mode;

- the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform one or more operations to program the configuration registers.

9. (Previously Presented) The memory module of claim 8, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

10. (Previously Presented) The memory module of claim 7, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the local clocks, programmable delay circuitry to add respective delays to respective ones of the local clocks, and clock driver circuitry to drive the local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

11. (Currently Amended) ~~A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising~~
The memory module of claim 1, wherein:

~~a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding, respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines;~~

the input C/A signals include first input C/A signals for a memory write operation to write first data to the memory module~~memory devices mounted on the PCB;~~

the circuitry includes a module control device mounted on the PCB and configurable to receive the input control and address (C/A) C/A signals via the C/A connectors for a memory write operation to write data to the memory module, and to output first module C/A signals and first module control signals for the write operation in response the first input C/A signals;

the circuitry further includes a plurality of data buffers mounted on the PCB and coupled to the module control device,~~wherein;~~

each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a ~~respective section~~ corresponding group of the memory devices;

~~the plurality of data buffers include at least m data buffers;~~

~~the m data buffers are configurable to receive data associated with the write operation;~~

~~the first data associated with the write operation~~ has a bit width of N and includes m n-bit-wide sections corresponding, respectively, to m subsets of the data signal lines and to m data buffers of the plurality of data buffers, $N = mxn$, wherein n and m are integers greater than 1;

~~[[each]] a respective data buffer of the m data buffers is configurable to receive the first module control signals and to receive a respective corresponding n-bit-wide section of the first data associated with the write operation in response to the first module control signals;~~

~~the [[each]] respective data buffer of the m data buffers is further configurable to transmit the respective corresponding n-bit-wide section of the first data associated with the write operation to at least a first branch of memory devices in the respective section-group of the memory devices coupled to the respective data buffer;~~

~~the [[each]] respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and~~

~~at least one first memory device in the first branch of the respective section of the memory devices is configurable to receive at least some of the first module C/A signals and the respective local clock, and to receive and store the respective corresponding n-bit-wide section of the first data associated with the write operation in response to the at least some of the registered first module C/A signals and in accordance with the respective local clock.~~

12. (Currently Amended) The memory module of claim 11, wherein:

the module control device is further configurable to receive ~~[[a]]~~the system clock and to output a module clock; and

the each respective data buffer of the m data buffers is further configured to receive the module clock and to generate the respective local clock from the module clock, the respective local clock having a respective phase shift from the module clock.

13. (Previously Presented) The memory module of claim 12, wherein the each respective data buffer includes logic and respective configuration registers programmable by the logic, and wherein the respective phase shift is dependent on one or more register settings in the respective configuration registers.

14. (Previously Presented) The memory module of claim 13, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the write operation is performed in the normal operation mode; and

the respective configuration registers are programmed in one or more operations during the configuration mode.

15. (Previously Presented) The memory module of claim 14, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

16. (Previously Presented) The memory module of claim 12, wherein:

the m data buffers include a first data buffer and a second data buffer;

the first data buffer is configurable to generate a first local clock from the module clock in response to the module control signals and to transmit the first local clock to at least a first branch of a first section of memory devices;

the second data buffer is configurable to generate a second local clock from the module clock in response to the module control signals and to transmit the second local clock to at least a first branch of a second section of memory devices;

the first local clock has a first phase shift from the module clock;

the second local clock has a second phase shift from the module clock; and

the first shift and the second phase shift are programmable independently of each other.

17. (Previously Presented) The memory module of claim 11, wherein:

the each respective data buffer of the m data buffers includes a first n -bit-wide write path coupled to the first branch of the respective section of the memory device and not to a second branch of the respective section of the memory device, and a second n -bit-wide write path coupled to the second branch of the respective section of the memory device and not to the first branch of the respective section of the memory device; and

the each respective data buffer of the m data buffers is configurable to transmit the respective n -bit-wide section of the data associated with the write operation to the first branch of the respective section of the memory device via the first n -bit-wide write path and not to any memory device in the second branch of the respective section of the memory device via the second n -bit-wide write path.

18. (Previously Presented) The memory module of claim 17, wherein:

the each respective data buffer of the m data buffers further includes input buffers configurable to receive the respective n -bit-wide section of the data associated with the write operation;

selection circuitry coupled to the input buffers; and

logic coupled to the selection circuitry and configurable to control the selection circuitry to select the first n -bit-wide write path to transmit the respective n -bit-wide section of the data associated with the write operation in response to the module control signals.

19. (Previously Presented) The memory module of claim 17, wherein the at least one first memory device includes one or more dynamic random access memory (DRAM) device each having a bit width of n/k , where k is a number of DRAM devices in the at least one first memory device, and k is 1, 2, or 4.

20. (Previously Presented) The memory module of claim 19, wherein $n = 8$ or 16.

REMARKS

This amendment is in response to the Restriction Requirement dated December 4, 2025. According to the Restriction Requirement, restriction to one of the following inventions is required under 35 U.S.C. § 121:

- Group I: Claims 1-10, drawn to an apparatus as claimed; and
- Group II: Claims 11-20, drawn to an apparatus as claimed.

In response, Applicant hereby elects Group I (claims 1-10) for examination in the current application. The election is made without traverse.

Claim 11 has been amended to depend from claim 1. Thus, claim 11 as amended now belongs to Invention Group I. Claims 12-20 depend from claim 11 and thus also belong to Invention Group I.

Applicant reserves the right to pursue the non-elected claims through filing of divisional applications. Favorable consideration of the elected claims is respectfully requested.

After entry of this amendment, the pending claims are: claims 1-20.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: February 4, 2026

/Jamie J. Zheng/

Jamie J. Zheng

Customer No. 79141

51,167

(Reg. No.)



UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/935,410

RECEIPT DATE / TIME
02/05/2026 02:48:19 AM Z ET

ATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7292	FILED BY	Jamie Zheng
PATENT CENTER #	74339719	FILING DATE	11/01/2024
CUSTOMER #	79141	FIRST NAMED INVENTOR	Hyun Lee
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 4

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Certificate_Transmission_sb0092.pdf	2	Transmittal Letter	880 KB
NL073-20US04_Amendment_RR.pdf	9	-	142 KB
NL073-20US04_Amendment_RR-ELC..pdf (1-1)	1	Response to Election / Restriction Filed	96 KB
NL073-20US04_Amendment_RR-CLM.pdf (2-8)	7	Claims	123 KB
NL073-20US04_Amendment_RR-REM.pdf (9-9)	1	Applicant Arguments/Remarks Made in an Amendment	95 KB

Digest

DOCUMENT

MESSAGE DIGEST(SHA-512)

Certificate_Transmission_sb0092.pdf

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20US04_Amendment_RR-
REM.pdf

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/Jamie J. Zheng/

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US Pat. Appl. No. 18/935,410 - Response to Restriction Requirement

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Electronically filed February 4, 2026

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	7292
Serial No.:	18/935,410	Group Art Unit:	2132
Filed:	November 1, 2024	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US04

RESPONSE TO RESTRICTION REQUIREMENT

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This amendment is submitted in response to the Restriction Requirement dated December 4, 2025.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US04).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and ~~data/strobe~~ data signal lines, the memory module comprising:

- a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;
- memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the ~~data/strobe~~ data signal lines;
- circuitry mounted on the PCB and configurable to:
 - receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;
 - generate module C/A signals in response to the input C/A signals;
 - generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and
 - output the module C/A signals to the memory devices; and
 - output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating ~~data/strobe~~ data signals with the memory controller via a corresponding subset of ~~data/strobe~~ data signal lines in accordance with the respective local clock.

2. (Previously Presented) The memory module of claim 1, wherein the circuitry includes logic and configuration registers programmable by the logic to control the respective phase relationships.
3. (Currently Amended) The memory module of claim 2, wherein:
 - the memory module is operable in at least a normal operation mode and a configuration mode;
 - the memory module in the normal operation mode is configurable to output or receive ~~data/strobe data~~ signals via the ~~data/strobe data~~ signal lines in response to memory read or write commands received via the C/A signal lines; and
 - the memory module in the configuration mode is configurable to perform one or more operations to program the respective configuration registers.
4. (Previously Presented) The memory module of claim 3, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.
5. (Previously Presented) The memory module of claim 2, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the plurality of local clocks, programmable delay circuitry to add respective delays to respective ones of the plurality of local clocks, and clock driver circuitry to drive the plurality of local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.
6. (Previously Presented) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;
 - memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices;

circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to:

- receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;

- generate module C/A signals in response to the input C/A signals;

- generate local clocks from the system clock, the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock, the first phase shift and the second phase shift being programmable independently of each other; and

- output the module C/A signals to the memory devices; and

- output the local clocks to the memory devices, wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines.

7. (Previously Presented) The memory module of claim 6, wherein the circuitry includes logic and configuration registers programmable by the logic, the registers including one or more first registers to control the first phase shift and one or more second registers to control the second phase shift.

8. (Previously Presented) The memory module of claim 7, wherein:

- the memory module is operable in at least a normal operation mode and a configuration mode;

- the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform one or more operations to program the configuration registers.

9. (Previously Presented) The memory module of claim 8, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

10. (Previously Presented) The memory module of claim 7, wherein the circuitry further includes phase-locked loop circuit (PLL) circuitry to provide phase locking between the system clock and the local clocks, programmable delay circuitry to add respective delays to respective ones of the local clocks, and clock driver circuitry to drive the local clocks to the memory devices, the respective delays being controlled by respective registers that are programmable independently of each other.

11. (Currently Amended) ~~A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising~~The memory module of claim 1, wherein:

~~a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus, the connectors including C/A connectors corresponding, respectively, to the C/A signal lines, and data connectors corresponding, respectively, to the data signal lines;~~

the input C/A signals include first input C/A signals for a memory write operation to write first data to the memory module~~memory devices mounted on the PCB;~~

the circuitry includes a module control device mounted on the PCB and configurable to receive the input control and address (C/A)C/A signals via the C/A connectors for a memory write operation to write data to the memory module, and to output first module C/A signals and first module control signals for the write operation in response the first input C/A signals;

the circuitry further includes a plurality of data buffers mounted on the PCB and coupled to the module control device,~~wherein;~~

each data buffer of the plurality of data buffers is coupled between a respective subset of the data connectors and a ~~respective section corresponding~~ group of the memory devices;

~~the plurality of data buffers include at least m data buffers;~~

~~the m data buffers are configurable to receive data associated with the write operation;~~

~~the first data associated with the write operation~~ has a bit width of N and includes m n-bit-wide sections corresponding, respectively, to m subsets of the data signal lines and to m data buffers of the plurality of data buffers, $N = mxn$, wherein n and m are integers greater than 1;

~~[[each]] a respective data buffer of the m data buffers is configurable to receive the first module control signals and to receive a respective corresponding n-bit-wide section of the first data associated with the write operation in response to the first module control signals;~~

~~the [[each]] respective data buffer of the m data buffers is further configurable to transmit the respective corresponding n-bit-wide section of the first data associated with the write operation to at least a first branch of memory devices in the respective section group of the memory devices coupled to the respective data buffer;~~

~~the [[each]] respective data buffer is further configurable to generate a respective local clock in response to the module control signals and to transmit the respective local clock to at least the first branch of the respective section of the memory devices; and~~

~~at least one first memory device in the first branch of the respective section of the memory devices is configurable to receive at least some of the first module C/A signals and the respective local clock, and to receive and store the respective corresponding n-bit-wide section of the first data associated with the write operation in response to the at least some of the registered first module C/A signals and in accordance with the respective local clock.~~

12. (Currently Amended) The memory module of claim 11, wherein:

the module control device is further configurable to receive ~~[[a]]~~the system clock and to output a module clock; and

the each respective data buffer of the m data buffers is further configured to receive the module clock and to generate the respective local clock from the module clock, the respective local clock having a respective phase shift from the module clock.

13. (Previously Presented) The memory module of claim 12, wherein the each respective data buffer includes logic and respective configuration registers programmable by the logic, and wherein the respective phase shift is dependent on one or more register settings in the respective configuration registers.

14. (Previously Presented) The memory module of claim 13, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the write operation is performed in the normal operation mode; and

the respective configuration registers are programmed in one or more operations during the configuration mode.

15. (Previously Presented) The memory module of claim 14, wherein the memory module is configurable to perform the one or more operations in response to mode register command signals received from the memory controller.

16. (Previously Presented) The memory module of claim 12, wherein:

the m data buffers include a first data buffer and a second data buffer;

the first data buffer is configurable to generate a first local clock from the module clock in response to the module control signals and to transmit the first local clock to at least a first branch of a first section of memory devices;

the second data buffer is configurable to generate a second local clock from the module clock in response to the module control signals and to transmit the second local clock to at least a first branch of a second section of memory devices;

the first local clock has a first phase shift from the module clock;

the second local clock has a second phase shift from the module clock; and

the first shift and the second phase shift are programmable independently of each other.

17. (Previously Presented) The memory module of claim 11, wherein:

the each respective data buffer of the m data buffers includes a first n -bit-wide write path coupled to the first branch of the respective section of the memory device and not to a second branch of the respective section of the memory device, and a second n -bit-wide write path coupled to the second branch of the respective section of the memory device and not to the first branch of the respective section of the memory device; and

the each respective data buffer of the m data buffers is configurable to transmit the respective n -bit-wide section of the data associated with the write operation to the first branch of the respective section of the memory device via the first n -bit-wide write path and not to any memory device in the second branch of the respective section of the memory device via the second n -bit-wide write path.

18. (Previously Presented) The memory module of claim 17, wherein:

the each respective data buffer of the m data buffers further includes input buffers configurable to receive the respective n -bit-wide section of the data associated with the write operation;

selection circuitry coupled to the input buffers; and

logic coupled to the selection circuitry and configurable to control the selection circuitry to select the first n -bit-wide write path to transmit the respective n -bit-wide section of the data associated with the write operation in response to the module control signals.

19. (Previously Presented) The memory module of claim 17, wherein the at least one first memory device includes one or more dynamic random access memory (DRAM) device each having a bit width of n/k , where k is a number of DRAM devices in the at least one first memory device, and k is 1, 2, or 4.

20. (Previously Presented) The memory module of claim 19, wherein $n = 8$ or 16.

REMARKS

This amendment is in response to the Restriction Requirement dated December 4, 2025. According to the Restriction Requirement, restriction to one of the following inventions is required under 35 U.S.C. § 121:

- Group I: Claims 1-10, drawn to an apparatus as claimed; and
- Group II: Claims 11-20, drawn to an apparatus as claimed.

In response, Applicant hereby elects Group I (claims 1-10) for examination in the current application. The election is made without traverse.

Claim 11 has been amended to depend from claim 1. Thus, claim 11 as amended now belongs to Invention Group I. Claims 12-20 depend from claim 11 and thus also belong to Invention Group I.

Applicant reserves the right to pursue the non-elected claims through filing of divisional applications. Favorable consideration of the elected claims is respectfully requested.

After entry of this amendment, the pending claims are: claims 1-20.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: February 4, 2026

/Jamie J. Zheng/

Jamie J. Zheng

Customer No. 79141

51,167

(Reg. No.)

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 18/935,410		Filing Date 11/01/2024		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	02/05/2026	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0			x \$200 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0			x \$600 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								SLIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/LISA R WRIGHT/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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79141 7590 04/08/2026
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER
2132	

DATE MAILED: 04/08/2026

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	11/01/2024	Hyun Lee	NL073-20US04	7292

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	07/08/2026

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

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P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

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Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	11/01/2024	Hyun Lee	NL073-20US04	7292

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	07/08/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	11/01/2024	Hyun Lee	NL073-20US04	7292
79141	7590	04/08/2026	EXAMINER	
USCH Law, PC 3524 Waverley St. Palo Alto, CA 94306			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
DATE MAILED: 04/08/2026				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 18/935,410	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 02/05/2026.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material _____. | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/HOSAIN T ALAM/
Supervisory Patent Examiner, Art Unit 2132

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to amendment filed on 02/05/2026 in response to a restriction requirement mailed on 12/04/2025. The restriction required election from one of two groups: Group I comprising claims 1-10, and Group II comprising claims 11-20. Applicant elected Group I for examination without traverse. Further, Applicant amended independent claim 11 to depend from claim 1 of invention Group I. Claims 12-20 depend from claim 11. Thus, claims 11-20 also belong to Group I. Accordingly, claims 1-20 have been examined.

Allowable Subject Matter

Claims 1-20 are allowed.

Reasons for Allowance

The following is an examiner's statement of reasons for allowance.

Any comments considered necessary by applicants must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

After careful considerations, examination and search of the claimed invention, the closest prior art of record does not teach or anticipate the claimed feature of claim 1 "generate a plurality of local clocks corresponding, respectively, to the plurality of

groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other” and “output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data/strobe data signal lines in accordance with the respective local clock” in combination with the overall claimed limitations when interpreted in light of the specification.

Three prior art references were considered by the Examiner. The references are Matsui US 2004/0105292 (“Matsui”), Sutardja et al. US 2007/0168781 (“Sutardja”), and Hampel et al. US 2010/0115191 (“Hampel”). Matsui and Sutardja are prior art references with respect to the parent application 14/445,035. The Hampel reference was obtained in an updated search. The three references are believed to be the closest prior art.

Regarding the Matsui-Sutardja combination, the Patent Trial and Appeal Board (PTAB) rendered a decision with respect to the parent application 14/445,035 on 09/27/2018. The PTAB decided that the Examiner did not sufficiently provide “articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007). See page 6 of the PTAB decision.

In addition to the decision rendered by PTAB, the Matsui-Sutardja combination does not teach “output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data/strobe data signal lines in accordance with the respective local clock”.

Therefore, Matsui alone or in any combination with Sutardja do not teach all of the limitations of independent claim 1.

The Examiner selected Hampel as a substitute for Sutardja. Hampel teaches “Clock circuit 270 includes one or more clock alignment circuits for phase or delay adjusting internal clock signals with respect to an external clock. Clock alignment circuits may utilize an external clock from an existing clock generator, such as clock source 130, or an internal clock generator to provide an internal clock, to generate internal synchronizing clock signals having a predetermined temporal relationship.” Paragraph [0036] of Hampel. That is, Hampel teaches “generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other” required by independent claim 1. However, Hampel alone or in any combination with Matsui do not teach “output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the

system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data/strobe data signal lines in accordance with the respective local clock” required by claim 1.

Therefore, claim 1 is allowable.

Claims 2-20 directly or indirectly depend from claim 1 and these claims are also allowable by virtue of their dependency to claim 1.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, HOSAIN T. ALAM can be reached at (571) 272-3978. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is

available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/HOSAIN T ALAM/
Supervisory Patent Examiner, Art Unit 2132

<i>Notice of References Cited</i>	Application/Control No. 18/935,410		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20040105292-A1	2004-06-03	Matsui, Yoshinori	G11C7/1093	365/189.05
*	B	US-20070168781-A1	2007-07-19	Sutardja; Sehat	G11C29/76	714/718
*	C	US-20100115191-A1	2010-05-06	Hampel; Craig	G11C29/32	711/170
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title, Date, Publisher, Edition or Volume, Pertinent Pages
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	11/25/2025	03/09/2026							
1	1	÷	=							
2	2	÷	=							
3	3	÷	=							
4	4	÷	=							
5	5	÷	=							
6	6	÷	=							
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19	19	÷	=							
20	20	÷	=							

Search Notes 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802, 1/10, 13/4243, 13/4256	03/09/2026	ZA
G11C 5/04, 7/1066, 7/1093, 29/023, 29/028	03/09/2026	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	03/09/2026	ZA
711	119	03/09/2026	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E search	03/09/2026	ZA
Inventor search	03/09/2026	ZA
Assignee search	03/09/2026	ZA
NPL search	03/09/2026	ZA
This application is allowed in view of the PTAB decision regarding the parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/12/0802	03/09/2026	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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Issue Classification 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	09 March 2026 (Date)	Total Claims Allowed: 20	
/HOSAIN T ALAM/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	13 March 2026 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 0802
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	120	119				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	09 March 2026 (Date)	Total Claims Allowed: 20	
/HOSAIN T ALAM/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	13 March 2026 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/935,410	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10	19	19										
2	2	11	11	20	20										
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6	6	15	15												
7	7	16	16												
8	8	17	17												
9	9	18	18												

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	09 March 2026 (Date)	Total Claims Allowed: 20	
/HOSAIN T ALAM/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	13 March 2026 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

Bibliographic Data

Application No: 18/935,410

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
11/01/2024	711	2132	NL073-20US04
RULE			

APPLICANTS

NETLIST, INC., Irvine, CA, UNITED STATES

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 18059958 11/29/2022 PAT 12135644

18059958 is a CON of 17141978 01/05/2021 PAT 11513955

17141978 is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

11/12/2024

STATE OR COUNTRY

UNITED STATES

ADDRESS

USCH Law, PC

3524 Waverley St.

Palo Alto, CA 94306

UNITED STATES

FILING FEE RECEIVED

\$2,170

PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	34930	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028 OR G06F1/10 OR G06F13/4243 OR G06F13/4256).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:15 PM
N2	38	N1 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:15 PM
N3	18	N1 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:19 PM
N4	17	N1 AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:20 PM
N5	1688	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:21 PM
N6	1	N5 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:21 PM
N7	1	N5 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:22 PM

N8	2	simmm) N5 AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:22 PM
N9	10	((lee near2 hyun).in. OR (bhakta near2 jayesh).in.) AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:22 PM
N10	6	netlist AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2026/03/09 05:23 PM

PE2E SEARCH - Search History (Similarity Search)

No Similarity Results available



Search History

Report Run Date: 2026-03-09 UTC
Searches: 1
Markers: 0

Search #	Timestamp	Results	Page
1	2026-03-09 21:06 UTC	60,731,913	3



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Luxembourg Applications | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Patents | Israel Applications | Jordan Patents | Jordan Applications | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Saudi Arabia Applications | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Patents | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IEEE Xplore Publications: Ericsson Technology Review | IEEE Xplore Publications: Manning Books | IEEE Xplore Publications: Packt Books | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2026-03-09 21:06 UTC | 60,731,913 results from

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Text

A memory module operable in a computer system having a memory controller and a system bus, the system bus

including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising: a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines; circuitry mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines; generate module C/A signals in response to the input C/A signals; generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and output the module C/A signals to the memory devices; and output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data/strobe data signal lines in accordance with the respective local clock.

Concept Modifiers: None

Filters: None

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("20250181504").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2025/11/04 01:47 PM
L2	1	("12135644").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2025/11/24 10:43 AM
L3	1	("11513955").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2025/11/24 10:53 AM
L4	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2025/11/24 10:55 AM
L5	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2025/11/24 10:57 AM
L6	69281	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028 OR G06F1/10 OR G06F13/4243 OR G06F13/4256).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:09 PM
L7	17	L6 AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:10 PM
L8	81	L6 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:18 PM
L9	71	(generat\$3 WITH local near3 clocks SAME phase WITH (system	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2026/03/09 03:27 PM

		OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	SG, TW, WO); EPO; JPO; IBM_TDB)				
L10	32	L6 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:34 PM
L11	61	L6 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:40 PM
L12	35	L6 AND ((memory OR dram) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 03:55 PM
L13	43	L6 AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:03 PM
L14	206	((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks) AND controller	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:08 PM
L15	87	((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks) AND (memory near2 controller)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:20 PM
L16	6	((lee near2 hyun).in. OR (bhakta near2 jayesh).in.) AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:25 PM
L17	14	((lee near2 hyun).in. OR (bhakta near2	(US-PGPUB; USPAT; USOCR; FIT (AU, CN,	OR	ON	ON	2026/03/09 04:27 PM

L18	19	jayesh).in.) AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:30 PM
L19	6	((lee near2 hyun).in. OR (bhakta near2 jayesh).in.) AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:34 PM
L20	23	netlist AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:35 PM
L21	27	netlist AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:40 PM
L22	1699	(711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:45 PM
L23	1	L22 AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:45 PM
L24	1	L22 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:45 PM
L25	1	L22 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:46 PM

L26	1	near2 clock) AND ((memory near2 module) OR dimm OR simm) L22 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm OR memory)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:46 PM
L27	2	L22 AND ((memory OR dram) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:46 PM
L28	2	L22 AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 04:48 PM
L29	49	(Artificial Intelligence) Similar to: 18/935,410 with 0 CPC Selections and 0 Text Selections Text: (CN-110428855-B OR US-20210271593-A1 OR US-20140040568-A1 OR US-20170337125-A1 OR US-20110016269-A1 OR DE-60308183-T2 OR US-20170097904-A1 OR EP-2313891-B1 OR US-20040260864-A1 OR US-8788747-B2 OR US-9542343-B2 OR US-20100115191-A1 OR WO-2008121559-A1 OR WO-2013115783-A1 OR KR-102653147-B1 OR US-8553470-B2 OR US-20130086449-A1 OR KR-20210013387-A OR US-11461246-B2 OR US-20210248089-A1 OR US-11416398-B2 OR KR-20170031825-A OR US-20170206165-A1 OR KR-20080104184-A	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO))	OR	ON	ON	2026/03/09 04:49 PM

L30	50	OR US-20200089626-A1 OR US-7389387-B2 OR US-20150186267-A1 OR US-11003596-B2 OR US-20140325315-A1 OR US-6446184-B2 OR US-20210255969-A1 OR KR-102194003-B1 OR KR-102633091-B1 OR US-20190266106-A1 OR US-20190294548-A1 OR US-6202138-B1 OR TW-I246648-B OR CN-112306760-A OR KR-101752117-B1 OR US-20120066442-A1 OR WO-2010148359-A1 OR US-20140281200-A1 OR KR-20210154277-A OR US-9390783-B1 OR US-9003153-B2 OR US-8725983-B2 OR WO-03083662-A2 OR WO-2013147886-A1 OR CN-109388517-A OR KR-100633828-B1).did. (Artificial Intelligence) Similar to: 18/935,410 with 4 CPC Selections and 0 Text Selections Text: (CN-110428855-B OR US-20050166006-A1 OR US-20050071542-A1 OR US-20170109067-A1 OR JP-2025522175-A OR KR-20210154277-A OR JP-7553477-B2 OR CN-113853592-B OR JP-2026012814-A OR US-11741020-B2 OR US-20200371938-A1 OR JP-2024174950-A OR US-20230004500-A1 OR US-12141073-B1 OR US-20200371939-A1 OR US-20250036573-A1 OR US-12147353-B2 OR US-20250028645-A1 OR US-	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO))	OR	ON	ON	2026/03/09 04:55 PM
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L31	49	20230236974-A1 OR US-12105640-B2 OR US-20230281126-A1 OR US-12072814-B2 OR EP-4432288-A2 OR JP-7762781-B2 OR US- 20250139019-A1 OR US-20240370380-A1 OR CN-121478680-A OR US-12541468-B2 OR US-11640357-B2 OR US-20240232100- A1 OR US- 20200371946-A1 OR US-11940929-B2 OR US-20200371956-A1 OR US-12393521-B2 OR US-20250348438- A1 OR US- 20250225083-A1 OR CN-113853592-A OR EP-3977299-B1 OR JP- 6986369-B2 OR US- 11693791-B2 OR US- 20230333991-A1 OR US-11693790-B2 OR US-20240020242-A1 OR US-20200371921- A1 OR US- 20260037451-A1 OR US-20230401162-A1 OR EP-4636601-A2 OR JP-2024167392-A OR JP-7551659-B2 OR US- 20200371964-A1).did. (Artificial Intelligence) Similar to: 18/935,410 with 9 CPC Selections and 0 Text Selections Text: (CN-110428855-B OR DE-102018120862-A1 OR US-7814382-B2 OR US-20140289575-A1 OR KR-20180048071-A OR US-20070168812- A1 OR KR- 20170139438-A OR CN-107481746-A OR CN-107481746-B OR US-20070168811-A1 OR KR-102772896-B1 OR US-7823030-B2 OR JP-6986369-B2 OR KR- 20210154277-A OR JP- 2017220237-A OR TW-	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO))	OR	ON	ON	2026/03/09 05:00 PM
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L32	49	201743219-A OR US-10810144-B2 OR CN-114968857-B OR EP-1997111-B1 OR US-20210210126-A1 OR US-20070214335-A1 OR US-11127443-B2 OR US-11574665-B2 OR US-8488405-B2 OR US-7613883-B2 OR KR-102491651-B1 OR US-20250217070-A1 OR CN-106997324-A OR US-20170168931-A1 OR KR-20170070920-A OR US-20130021858-A1 OR US-9971697-B2 OR US-20190220222-A1 OR CN-106997324-B OR CN-109313918-B OR JP-2025522175-A OR US-11914888-B2 OR US-20230009384-A1 OR KR-20010007493-A OR KR-100375188-B1 OR WO-2018044391-A1 OR US-20070097753-A1 OR US-11302369-B2 OR US-20220084568-A1 OR JP-4842765-B2 OR JP-2007128639-A OR KR-102514772-B1 OR JP-7500365-B2 OR US-6347055-B1 OR US-20170109067-A1).did. (Artificial Intelligence) Similar to: 18/935,410 with 9 CPC Selections and 2 Text Selections Text: (CN-110428855-B OR DE-102018120862-A1 OR US-7814382-B2 OR US-20140289575-A1 OR JP-6986369-B2 OR KR-20170139438-A OR JP-2017220237-A OR TW-201743219-A OR KR-20210154277-A OR US-20070168812-A1 OR CN-107481746-A OR US-10810144-B2 OR US-20070168811-	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO))	OR	ON	ON	2026/03/09 05:05 PM
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		A1 OR CN-107481746-B OR US-7823030-B2 OR KR-102491651-B1 OR CN-114968857-B OR US-20170168931-A1 OR KR-20170070920-A OR CN-106997324-A OR US-9971697-B2 OR CN-106997324-B OR KR-20180048071-A OR US-20210271593-A1 OR US-20170206028-A1 OR KR-102772896-B1 OR US-20140325315-A1 OR US-10019367-B2 OR CN-102812518-B OR KR-102633091-B1 OR KR-20180109071-A OR US-20170206036-A1 OR WO-2017127386-A1 OR US-20170337125-A1 OR EP-3532933-B1 OR US-9891864-B2 OR US-20210210126-A1 OR US-20140040568-A1 OR WO-2018080783-A1 OR US-20050270891-A1 OR US-11127443-B2 OR TW-201732831-A OR WO-2011094437-A2 OR US-10847196-B2 OR CN-102812518-A OR US-12213548-B2 OR US-20210349658-A1 OR US-20160216912-A1 OR WO-2008121559-A1 OR US-11574665-B2).did.					
L33	0	L32 AND (generat\$3 WITH local near3 clocks SAME phase WITH relationship WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:12 PM
L34	1	L32 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:12 PM
L35	1	L32 AND (generat\$3	(US-PGPUB; USPAT;	OR	ON	ON	2026/03/09

		WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)				05:13 PM
L36	1	L32 AND (generat\$3 WITH local near3 clocks SAME phase WITH (system OR board OR module) near2 clock) AND ((memory near2 module) OR dimm OR simm)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:14 PM
L37	1	L32 AND ((memory OR dram) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:14 PM
L38	1	L32 AND ((memory OR dram OR sram OR chip) WITH (rank\$3 OR group\$3) WITH local near2 clocks)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:14 PM
L39	2	("20100115191").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2026/03/09 05:27 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Similarity Search)

Ref #	Sort and Retrieval Information	Text Selections	CPC Selections
L29	Sort results by: Similarity to Application # 18/935,410 Retrieve results by: Application # 18/935,410	No selections	No selections
L30	Sort results by: Similarity to Application # 18/935,410 Retrieve results by: Application # 18/935,410	No selections	(G06F12/0802) Addressing of a memory level in which the access to the desired data or data block requires associative addressing means, e.g. caches (G06F1/10) Distribution of clock signals, e.g. skew

L31	Sort results by: Similarity to Application # 18/935,410 Retrieve results by: Application # 18/935,410	No selections	(G06F13/4243) Bus transfer protocol, e.g. handshake; Synchronisation on a parallel bus being a memory bus with synchronous protocol (G06F13/4256) Bus transfer protocol, e.g. handshake; Synchronisation on a daisy chain bus using a clocked protocol (G06F12/0802) Addressing of a memory level in which the access to the desired data or data block requires associative addressing means, e.g. caches (G06F1/10) Distribution of clock signals, e.g. skew (G06F13/4243) Bus transfer protocol, e.g. handshake; Synchronisation on a parallel bus being a memory bus with synchronous protocol (G06F13/4256) Bus transfer protocol, e.g. handshake; Synchronisation on a daisy chain bus using a clocked protocol (G11C5/04) Supports for storage elements, e.g. memory modules; Mounting or fixing of storage elements on such supports (G11C7/1066) Output synchronization (G11C7/1093) Input synchronization (G11C29/023) Detection or location of defective auxiliary circuits, e.g. defective refresh counters in clock generator or timing circuitry (G11C29/028) Detection or location of defective auxiliary circuits, e.g. defective refresh counters with adaption or trimming of parameters
L32	Sort results by: Similarity to Application # 18/935,410 Retrieve results by: Application # 18/935,410	(Claims) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobedata signal lines, the memory module comprising: circuitry mounted on the PCB and configurable to: receive from the memory	(G06F12/0802) Addressing of a memory level in which the access to the desired data or data block requires associative addressing means, e.g. caches (G06F1/10) Distribution of clock signals, e.g. skew (G06F13/4243) Bus transfer protocol, e.g. handshake; Synchronisation on a parallel bus being a memory bus with synchronous protocol (G06F13/4256) Bus transfer protocol, e.g. handshake; Synchronisation on a daisy chain

		controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines; generate module C/A signals in response to the input C/A signals; generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices, the plurality of local clocks having respective phase relationships with the system clock, the respective phase relationships being programmable independently of each other; and output the module C/A signals to the memory devices; and output the plurality of local clocks to the memory devices, wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices, wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data/strobedata signals with the memory controller via a corresponding subset of data/strobe data signal lines in accordance with the respective local clock. wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data/strobedata signal lines; (Claims) a printed circuit board (PCB) having	bus using a clocked protocol (G11C5/04) Supports for storage elements, e.g. memory modules; Mounting or fixing of storage elements on such supports (G11C7/1066) Output synchronization (G11C7/1093) Input synchronization (G11C29/023) Detection or location of defective auxiliary circuits, e.g. defective refresh counters in clock generator or timing circuitry (G11C29/028) Detection or location of defective auxiliary circuits, e.g. defective refresh counters with adaption or trimming of parameters
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		connectors configured to provide electrical connections between the memory module and the system bus; memory devices mounted on the PCB and organized in a plurality of groups,	
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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/935,410	11/01/2024	2132	2170	NL073-20US04	3	1

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

CONFIRMATION NO. 7292
CORRECTED FILING RECEIPT



0000000104646135

Date Mailed: 04/17/2026

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Applications for which benefit is claimed

This application is a CON of 18/059,958 11/29/2022 PAT 12,135,644
which is a CON of 17/141,978 01/05/2021 PAT 11,513,955
which is a CON of 16/432,700 06/05/2019 PAT 10,884,923
which is a CON of 14/445,035 07/28/2014 PAT 10,324,841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - **None.** Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 11/12/2024

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/935,410**

Projected Publication Date: Not Applicable.

Non-Publication Request: No

Early Publication Request: No
Title

MEMORY MODULE WITH LOCAL CLOCK SIGNALS

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop

technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 6 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

79141

7590

04/08/2026

USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	11/01/2024	Hyun Lee	NL073-20US04	7292

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	07/08/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 USCH Law, PC

2

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

Netlist, Inc.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

6. Courtesy Ceremonial Copy

☒ Applicant requests a courtesy ceremonial copy of the electronic patent grant. Applicant understands that the electronic patent grant is the official statutory patent grant.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature 

Date 5/11/2026

Typed or printed name Francisco A. Rubio-Campos

Registration No. 45,358



UNITED STATES
PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/935,410

RECEIPT DATE / TIME
05/11/2026 05:11:39 PM Z ET

ATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL CLOCK SIGNALS

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	7292	FILED BY	Erin Hong
PATENT CENTER #	76111654	FILING DATE	11/01/2024
CUSTOMER #	79141	FIRST NAMED INVENTOR	Hyun Lee
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Francisco Rubio-Campos

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
USPTO 18-935.410 Application Issue Fee_Form Part B - Transmittal 2026-05-11.pdf	1	Issue Fee Payment (PTO-85B)	222 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
USPTO 18-935.410 Application Issue Fee_Form Part B - Transmittal 2026-05-11.pdf	6B2BBA4C9B881FB75B4DA0C959638A384CDE8B5C413AC79D 866BA6AEE675E2183AF7C0A45029E0510AE5BFE3BFF5A7B92 340D93B2A5CC329DB5F9B936F4BE9C7

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
18/935,410

RECEIPT DATE / TIME
05/11/2026 05:11:39 PM Z ET

ATTORNEY DOCKET #
NL073-20US04

Title of Invention

MEMORY MODULE WITH LOCAL CLOCK SIGNALS

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 7292

FILED BY Erin Hong

PATENT CENTER # 76111654

AUTHORIZED BY Francisco Rubio-Campos

CUSTOMER # 79141

FILING DATE 11/01/2024

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Hyun Lee

Payment Information

PAYMENT METHOD
CARD / 7003

PAYMENT TRANSACTION ID
E20265AH13187844

PAYMENT AUTHORIZED BY
Erin Hong

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
18/935,410	11/01/2024	789669

The following fields have been changed to Customer Number 207611 on 05/11/2026 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
18/935,410	11/01/2024	789669

The following fields have been changed to Customer Number 207611 on 05/11/2026 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/935,410	07/07/2026	12675407	NL073-20US04	7292

207611 7590 06/24/2026
Netlist, Inc.
111 Academy
Irvine, CA 92617

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

The USPTO will automatically mail a courtesy ceremonial copy of the electronic patent grant to the correspondence address of record if the issue fee was paid before March 9, 2026. If the issue fee was paid on or after March 9, 2026, a courtesy ceremonial copy will be mailed if the applicant opted-in to receiving it. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Hyun Lee, Ladera Ranch, CA;
Jayesh Bhakta, Cerritos, CA;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

NETLIST, INC., Irvine, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).